

Chapter 1

Strategies for solving problems

Physics involves a great deal of problem solving. Whether you are doing cutting-edge research or reading a book on a well-known subject, you are going to need to solve some problems. In the latter case (the presently relevant one, given what is in your hand right now), it is fairly safe to say that the true test of understanding something is the ability to solve problems on it. Reading about a topic is often a necessary step in the learning process, but it is by no means a sufficient one. The more important step is spending as much time as possible solving problems (which is inevitably an active task) beyond the time you spend reading (which is generally a more passive task). I have therefore included a very large number of problems/exercises in this book.

However, if I'm going to throw all these problems at you, I should at least give you some general strategies for solving them. These strategies are the subject of the present chapter. They are things you should always keep in the back of your mind when tackling a problem. Of course, they are generally not sufficient by themselves; you won't get too far without understanding the physical concepts behind the subject at hand. But when you add these strategies to your physical understanding, they can make your life a lot easier.

1.1 General strategies

There are a number of general strategies you should invoke without hesitation when solving a problem. They are:

1. **Draw a diagram, if appropriate.**

In the diagram, be sure to label clearly all the relevant quantities (forces, lengths, masses, etc.). Diagrams are absolutely critical in certain types of problems. For example, in problems involving “free-body” diagrams (discussed in Chapter 3) or relativistic kinematics (discussed in Chapter 11), drawing a diagram can change a hopelessly complicated problem into a near-trivial one. And even in cases where diagrams aren't this crucial, they're invariably very helpful. A picture is definitely worth a thousand words (and even a few more, if you label things!).

2. Write down what you know, and what you are trying to find.

In a simple problem, you may just do this in your head without realizing it. But in more difficult problems, it is very useful to explicitly write things out. For example, if there are three unknowns that you're trying to find, but you've written down only two facts, then you know there must be another fact you're missing (assuming that the problem is in fact solvable), so you can go searching for it. It might be a conservation law, or an $F = ma$ equation, etc.

3. Solve things symbolically.

If you are solving a problem where the given quantities are specified numerically, you should immediately change the numbers to letters and solve the problem in terms of the letters. After you obtain an answer in terms of the letters, you can plug in the actual numerical values to obtain a numerical answer. There are many advantages to using letters:

- **IT'S QUICKER.** It's much easier to multiply a g by an ℓ by writing them down on a piece of paper next to each other, than it is to multiply them together on a calculator. And with the latter strategy, you'd undoubtedly have to pick up your calculator at least a few times during the course of a problem.
- **YOU'RE LESS LIKELY TO MAKE A MISTAKE.** It's very easy to mistype an 8 for a 9 in a calculator, but you're probably not going to miswrite a q for a g on a piece of paper. But if you do, you'll quickly realize that it should be a g . You certainly won't just give up on the problem and deem it unsolvable because no one gave you the value of q !
- **YOU CAN DO THE PROBLEM ONCE AND FOR ALL.** If someone comes along and says, oops, the value of ℓ is actually 2.4 m instead of 2.3 m, then you won't have to do the whole problem again. You can simply plug the new value of ℓ into your final symbolic answer.
- **YOU CAN SEE THE GENERAL DEPENDENCE OF YOUR ANSWER ON THE VARIOUS GIVEN QUANTITIES.** For example, you can see that it grows with quantities a and b , decreases with c , and doesn't depend on d . There is much, much more information contained in a symbolic answer than in a numerical one. And besides, symbolic answers nearly always look nice and pretty.
- **YOU CAN CHECK UNITS AND SPECIAL CASES.** These checks go hand-in-hand with the previous "general dependence" advantage. But since they're so important, we'll postpone their discussion and devote Sections 1.2 and 1.3 to them.

Having said all this, it should be noted that there are occasionally times when things get a bit messy when working with letters. For example, solving a system of three equations in three unknowns might be rather cumbersome unless you plug in the actual numbers. But in the vast majority of problems, it is highly advantageous to work entirely with letters.

4. Consider units/dimensions.

This is extremely important. See Section 1.2 for a detailed discussion.

5. Check limiting/special cases.

This is also extremely important. See Section 1.3 for a detailed discussion.

6. Check order of magnitude if you end up getting a numerical answer.

If you end up with an actual numerical answer to a problem, be sure to do a sanity check to see if the number is reasonable. If you've calculated the distance along the ground that a car skids before it comes to rest, and if you've gotten an answer of a kilometer or a millimeter, then you know you've probably done something wrong. Errors of this sort often come from forgetting some powers of 10 (say, when converting kilometers to meters) or from multiplying something instead of dividing (although you should be able to catch this by checking your units, too).

You will inevitably encounter problems, physics ones or otherwise, where you don't end up obtaining a rigorous answer, either because the calculation is intractable, or because you just don't feel like doing it. But in these cases it's usually still possible to make an educated guess, to the nearest power of 10. For example, if you walk past a building and happen to wonder how many bricks are in it, or what the labor cost was in constructing it, then you can probably give a reasonable answer without doing any severe computations. The physicist Enrico Fermi was known for his ability to estimate things quickly and produce order-of-magnitude guesses with only minimal calculation. Hence, a problem where the goal is to simply obtain the nearest power-of-10 estimate is known as a "Fermi problem." Of course, sometimes in life you need to know things to better accuracy than the nearest power of 10 . . .

How Fermi could estimate things!

Like the well-known Olympic ten rings,

And the one hundred states,

And weeks with ten dates,

And birds that all fly with one . . . wings.

In the following two sections, we'll discuss the very important strategies of checking units and special cases. Then in Section 1.4 we'll discuss the technique of solving problems numerically, which is what you need to do when you end up with a set of equations you can't figure out how to solve. Section 1.4 isn't quite analogous to Sections 1.2 and 1.3, in that these first two are relevant to basically any problem you'll ever do, whereas solving equations numerically is something you'll do only for occasional problems. But it's nevertheless something that every physics student should know.

In all three of these sections, we'll invoke various results derived later in the book. For the present purposes, the derivations of these results are completely irrelevant, so don't worry at all about the physics behind them – there will be

plenty of opportunity for that later on! The main point here is to learn what to do with the result of a problem once you've obtained it.

1.2 Units, dimensional analysis

The units, or dimensions, of a quantity are the powers of mass, length, and time associated with it. For example, the units of a speed are length per time. The consideration of units offers two main benefits. First, looking at units before you start a problem can tell you roughly what the answer has to look like, up to numerical factors. Second, checking units at the end of a calculation (which is something you should *always* do) can tell you if your answer has a chance at being correct. It won't tell you that your answer is definitely correct, but it might tell you that your answer is definitely incorrect. For example, if your goal in a problem is to find a length, and if you end up with a mass, then you know it's time to look back over your work.

“Your units are wrong!” cried the teacher.

“Your church weighs six joules – what a feature!

And the people inside

Are four hours wide,

And eight gauss away from the preacher!”

In practice, the second of the above two benefits is what you will generally make use of. But let's do a few examples relating to the first benefit, because these can be a little more exciting. To solve the three examples below exactly, we would need to invoke results derived in later chapters. But let's just see how far we can get by using only dimensional analysis. We'll use the “[]” notation for units, and we'll let M stand for mass, L for length, and T for time. For example, we'll write a speed as $[v] = L/T$ and the gravitational constant as $[G] = L^3/(MT^2)$ (you can figure this out by noting that Gm_1m_2/r^2 has the dimensions of force, which in turn has dimensions ML/T^2 , from $F = ma$). Alternatively, you can just use the mks units, kg, m, s, instead of M , L , T , respectively.¹

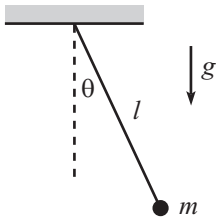


Fig. 1.1

Example (Pendulum): A mass m hangs from a massless string of length ℓ (see Fig. 1.1) and swings back and forth in the plane of the paper. The acceleration due to gravity is g . What can we say about the frequency of oscillations?

Solution: The only dimensionful quantities given in the problem are $[m] = M$, $[\ell] = L$, and $[g] = L/T^2$. But there is one more quantity, the maximum angle θ_0 , which is dimensionless (and easy to forget). Our goal is to find the frequency, which

¹ When you check units at the end of a calculation, you will invariably be working with the kg,m,s notation. So that notation will inevitably get used more. But I'll use the M , L , T notation here, because I think it's a little more instructive. At any rate, just remember that the letter m (or M) stands for “meter” in one case, and “mass” in the other.

has units of $1/T$. The only combination of our given dimensionful quantities that has units of $1/T$ is $\sqrt{g/\ell}$. But we can't rule out any θ_0 dependence, so the most general possible form of the frequency is²

$$\omega = f(\theta_0) \sqrt{\frac{g}{\ell}}, \quad (1.1)$$

where f is a dimensionless function of the dimensionless variable θ_0 .

REMARKS:

1. It just so happens that for small oscillations, $f(\theta_0)$ is essentially equal to 1, so the frequency is essentially equal to $\sqrt{g/\ell}$. But there is no way to show this by using only dimensional analysis; you actually have to solve the problem for real. For larger values of θ_0 , the higher-order terms in the expansion of f become important. Exercise 4.23 deals with the leading correction, and the answer turns out to be $f(\theta_0) = 1 - \theta_0^2/16 + \dots$.
2. Since there is only one mass in the problem, there is no way that the frequency (with units of $1/T$) can depend on $[m] = M$. If it did, there would be nothing to cancel the units of mass and produce a pure inverse-time.
3. We claimed above that the only combination of our given dimensionful quantities that has units of $1/T$ is $\sqrt{g/\ell}$. This is easy to see here, but in more complicated problems where the correct combination isn't so obvious, the following method will always work. Write down a general product of the given dimensionful quantities raised to arbitrary powers ($m^a \ell^b g^c$ in this problem), and then write out the units of this product in terms of a , b , and c . If we want to obtain units of $1/T$ here, then we need

$$M^a L^b \left(\frac{L}{T^2} \right)^c = \frac{1}{T}. \quad (1.2)$$

Matching up the powers of the three kinds of units on each side of this equation gives

$$M : a = 0, \quad L : b + c = 0, \quad T : -2c = -1. \quad (1.3)$$

The solution to this system of equations is $a = 0$, $b = -1/2$, and $c = 1/2$, so we have reproduced the $\sqrt{g/\ell}$ result. ♣

What can we say about the total energy of the pendulum (with the potential energy measured relative to the lowest point)? We'll talk about energy in Chapter 5, but the only thing we need to know here is that energy has units of ML^2/T^2 . The only combination of the given dimensionful constants of this form is $mg\ell$. But again, we can't rule out any θ_0 dependence, so the energy must take the form $f(\theta_0)mg\ell$, where f is some function. That's as far as we can go with dimensional analysis. However, if we actually invoke a little physics, we can say that the total energy equals the potential energy at the highest point, which is $mg\ell(1 - \cos \theta_0)$. Using the Taylor expansion for $\cos \theta$ (see Appendix A for a discussion of Taylor series), we see that $f(\theta_0) = \theta_0^2/2 - \theta_0^4/24 + \dots$. So in contrast with the frequency result above, the maximum angle θ_0 plays a critical role in the energy.

² We'll measure frequency here in radians per second, denoted by ω . So we're actually talking about the "angular frequency." Just divide by 2π (which doesn't affect the units) to obtain the "regular" frequency in cycles per second (hertz), usually denoted by ν . We'll talk at great length about oscillations in Chapter 4.

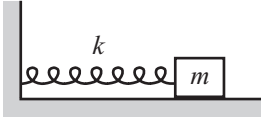


Fig. 1.2

Example (Spring): A spring with spring constant k has a mass m on its end (see Fig. 1.2). The spring force is $F(x) = -kx$, where x is the displacement from the equilibrium position. What can we say about the frequency of oscillations?

Solution: The only dimensionful quantities in this problem are $[m] = M$, $[k] = M/T^2$ (obtained by noting that kx has the dimensions of force), and the maximum displacement from the equilibrium, $[x_0] = L$. (There is also the equilibrium length, but the force doesn't depend on this, so there is no way it can come into the answer.) Our goal is to find the frequency, which has units of $1/T$. The only combination of our given dimensionful quantities with these units is

$$\omega = C\sqrt{\frac{k}{m}}, \quad (1.4)$$

where C is a dimensionless number. It just so happens that C is equal to 1 (assuming that we're measuring ω in radians per second), but there is no way to show this by using only dimensional analysis. Note that, in contrast with the pendulum above, the frequency cannot have any dependence on the maximum displacement.

What can we say about the total energy of the spring? Energy has units of ML^2/T^2 , and the only combination of the given dimensionful constants of this form is Bkx_0^2 , where B is a dimensionless number. It turns out that $B = 1/2$, so the total energy equals $kx_0^2/2$.

REMARK: A real spring doesn't have a perfectly parabolic potential (that is, a perfectly linear force), so the force actually looks something like $F(x) = -kx + bx^2 + \dots$. If we truncate the series at the second term, then we have one more dimensionful quantity to work with, $[b] = M/LT^2$. To form a quantity with the dimensions of frequency, $1/T$, we need x_0 and b to appear in the combination x_0b , because this is the only way to get rid of the L . You can then see (by using the strategy of writing out a general product of the variables, discussed in the third remark in the pendulum example above) that the frequency must be of the form $f(x_0b/k)\sqrt{k/m}$, where f is some function. We can therefore have x_0 dependence in this case. This answer must reduce to $C\sqrt{k/m}$ for $b = 0$. Hence, f must be of the form $f(y) = C + c_1y + c_2y^2 + \dots$ ♣

Example (Low-orbit satellite): A satellite of mass m travels in a circular orbit just above the earth's surface. What can we say about its speed?

Solution: The only dimensionful quantities in the problem are $[m] = M$, $[g] = L/T^2$, and the radius of the earth $[R] = L$.³ Our goal is to find the speed, which has units of L/T . The only combination of our dimensionful quantities with these units is

$$v = C\sqrt{gR}. \quad (1.5)$$

It turns out that $C = 1$.

³ You might argue that the mass of the earth, M_E , and Newton's gravitational constant, G , should be also included here, because Newton's gravitational force law for a particle on the surface of the earth is $F = GM_Em/R^2$. But since this force can be written as $m(GM_E/R^2) \equiv mg$, we can absorb the effects of M_E and G into g .

1.3 Approximations, limiting cases

As with units, the consideration of limiting cases (or perhaps we should say special cases) offers two main benefits. First, it can help you get started on a problem. If you're having trouble figuring out how a given system behaves, then you can imagine making, for example, a certain length become very large or very small, and then you can see what happens to the behavior. Having convinced yourself that the length actually affects the system in extreme cases (or perhaps you will discover that the length doesn't affect things at all), it will then be easier to understand how it affects the system in general, which will then make it easier to write down the relevant quantitative equations (conservation laws, $F = ma$ equations, etc.), which will allow you to fully solve the problem. In short, modifying the various parameters and observing the effects on the system can lead to an enormous amount of information.

Second, as with checking units, checking limiting cases (or special cases) is something you should *always* do at the end of a calculation. But as with checking units, it won't tell you that your answer is definitely correct, but it might tell you that your answer is definitely incorrect. It is generally true that your intuition about limiting cases is much better than your intuition about generic values of the parameters. You should use this fact to your advantage.

Let's do a few examples relating to the second benefit. The initial expressions given in each example below are taken from various examples throughout the book, so just accept them for now. For the most part, I'll repeat here what I'll say later on when we work through the problems for real. A tool that comes up often in checking limiting cases is the Taylor series approximations; the series for many functions are given in Appendix A.

Example (Dropped ball): A beach ball is dropped from rest at height h . Assume that the drag force from the air takes the form $F_d = -\alpha v$. We'll find in Section 3.3 that the ball's velocity and position are given by

$$v(t) = -\frac{g}{\alpha} (1 - e^{-\alpha t}), \quad \text{and} \quad y(t) = h - \frac{g}{\alpha} \left(t - \frac{1}{\alpha} (1 - e^{-\alpha t}) \right). \quad (1.6)$$

These expressions are a bit complicated, so for all you know, I could have made a typo in writing them down. Or worse, I could have completely botched the solution. So let's look at some limiting cases. If these limiting cases yield expected results, then we can feel a little more confident that the answers are actually correct.

If t is very small (more precisely, if $\alpha t \ll 1$; see the discussion following this example), then we can use the Taylor series, $e^{-x} \approx 1 - x + x^2/2$, to make approximations

to leading order in αt . The $v(t)$ in Eq. (1.6) becomes

$$\begin{aligned} v(t) &= -\frac{g}{\alpha} \left(1 - \left(1 - \alpha t + \frac{(\alpha t)^2}{2} - \dots \right) \right) \\ &\approx -gt, \end{aligned} \quad (1.7)$$

plus terms of higher order in αt . This answer is expected, because the drag force is negligible at the start, so we essentially have a freely falling body with acceleration g downward. For small t , Eq. (1.6) also gives

$$\begin{aligned} y(t) &= h - \frac{g}{\alpha} \left[t - \frac{1}{\alpha} \left(1 - \left(1 - \alpha t + \frac{(\alpha t)^2}{2} - \dots \right) \right) \right] \\ &\approx h - \frac{gt^2}{2}, \end{aligned} \quad (1.8)$$

plus terms of higher order in αt . Again, this answer is expected, because we essentially have a freely falling body at the start, so the distance fallen is the standard $gt^2/2$.

We can also look at large t (or rather, large αt). In this case, $e^{-\alpha t}$ is essentially zero, so the $v(t)$ in Eq. (1.6) becomes (there's no need for a Taylor series in this case)

$$v(t) \approx -\frac{g}{\alpha}. \quad (1.9)$$

This is the “terminal velocity.” Its value makes sense, because it is the velocity for which the total force, $-mg - m\alpha v$, vanishes. For large t , Eq. (1.6) also gives

$$y(t) \approx h - \frac{gt}{\alpha} + \frac{g}{\alpha^2}. \quad (1.10)$$

Apparently for large t , g/α^2 is the distance (and this does indeed have units of length, because α has units of T^{-1} , because $m\alpha v$ has units of force) that our ball lags behind another ball that started out already at the terminal velocity, $-g/\alpha$.

Whenever you derive approximate answers as we just did, you gain something and you lose something. You lose some truth, of course, because your new answer is technically not correct. But you gain some aesthetics. Your new answer is invariably much cleaner (sometimes involving only one term), and this makes it a lot easier to see what's going on.

In the above example, it actually makes no sense to look at the limit where t is small or large, because t has dimensions. Is a year a large or small time? How about a hundredth of a second? There is no way to answer this without knowing what problem you're dealing with. A year is short on the time scale of galactic evolution, but a hundredth of a second is long on the time scale of a nuclear process. It makes sense only to look at the limit of a small (or large) *dimensionless* quantity. In the above example, this quantity is αt . The given constant α has units of T^{-1} , so $1/\alpha$ sets a typical time scale for the system. It

therefore makes sense to look at the limit where $t \ll 1/\alpha$ (that is, $\alpha t \ll 1$), or where $t \gg 1/\alpha$ (that is, $\alpha t \gg 1$). In the limit of a small dimensionless quantity, a Taylor series can be used to expand an answer in powers of the small quantity, as we did above. We sometimes get sloppy and say things like, “In the limit of small t .” But you know that we really mean, “In the limit of some small dimensionless quantity that has a t in the numerator,” or, “In the limit where t is much smaller than a certain quantity that has the dimensions of time.”

REMARK: As mentioned above, checking special cases tells you that either (1) your answer is consistent with your intuition, or (2) it’s wrong. It never tells you that it’s definitely correct. This is the same as what happens with the scientific method. In the real world, everything comes down to experiment. If you have a theory that you think is correct, then you need to check that its predictions are consistent with experiments. The specific experiments you do are the analog of the special cases you check after solving a problem; these two things represent what you know is true. If the results of the experiments are inconsistent with your theory, then you need to go back and fix your theory, just as you would need to go back and fix your answer. If, on the other hand, the results are consistent, then although this is good, the only thing it really tells you is that your theory *might* be correct. And considering the way things usually turn out, the odds are that it’s not actually correct, but rather the limiting case of a more correct theory (just as Newtonian physics is a limiting case of relativistic physics, which is a limiting case of quantum field theory, etc.). That’s how physics works. You can’t prove anything, so you learn to settle for the things you can’t disprove.

Consider, when seeking gestalts,
The theories that physics exalts.
It’s not that they’re known
To be written in stone.
It’s just that we can’t say they’re false. ♣

When making approximations, how do you know how many terms in the Taylor series to keep? In the example above, we used $e^{-x} \approx 1 - x + x^2/2$. But why did we stop at the x^2 term? The honest (but slightly facetious) answer is, “Because I had already done this problem before writing it up, so I knew how many terms to keep.” But the more informative (although perhaps no more helpful) answer is that before you do the calculation, there’s really no way of knowing how many terms to keep. So you should just keep a few and see what happens. If everything ends up canceling out, then this tells you that you need to repeat the calculation with another term in the series. For example, in Eq. (1.8), if we had stopped the Taylor series at $e^{-x} \approx 1 - x$, then we would have obtained $y(t) = h - 0$, which isn’t very useful, since the general goal is to get the leading-order behavior in the parameter we’re looking at (which is t here). So in this case we’d know we’d have to go back and include the $x^2/2$ term in the series. If we were doing a problem in which there was still no t (or whatever variable) dependence at that order, then we’d have to go back and include the $-x^3/6$ term in the series. Of course, you could just play it safe and keep terms up to, say, fifth order. But that’s invariably a poor strategy, because you’ll probably never in your life have to go out that far in a series. So just start with one or two terms and see what it gives you. Note that in Eq. (1.7), we actually didn’t need the

second-order term, so we in fact could have gotten by with only $e^{-x} \approx 1 - x$. But having the extra term here didn't end up causing much heartache.

After you make an approximation, how do you know if it's a "good" one? Well, just as it makes no sense to ask if a dimensionful quantity is large or small without comparing it to another quantity, it makes no sense to ask if an approximation is "good" or "bad" without stating the accuracy you want. In the above example, if you're looking at a t value for which $\alpha t \approx 1/100$, then the term we ignored in Eq. (1.7) is smaller than gt by a factor $\alpha t/2 \approx 1/200$. So the error is on the order of 1%. If this is enough accuracy for whatever purpose you have in mind, then the approximation is a good one. If not, it's a bad one, and you should add more terms in the series until you get your desired accuracy.

The results of checking limits generally fall into two categories. Most of the time you know what the result should be, so this provides a double-check on your answer. But sometimes an interesting limit pops up that you might not expect. Such is the case in the following examples.



Fig. 1.3

Example (Two masses in 1-D): A mass m with speed v approaches a stationary mass M (see Fig. 1.3). The masses bounce off each other elastically. Assume that all motion takes place in one dimension. We'll find in Section 5.6.1 that the final velocities of the particles are

$$v_m = \frac{(m - M)v}{m + M}, \quad \text{and} \quad v_M = \frac{2mv}{m + M}. \quad (1.11)$$

There are three special cases that beg to be checked:

- If $m = M$, then Eq. (1.11) tells us that m stops, and M picks up a speed v . This is fairly believable (and even more so for pool players). And it becomes quite clear once you realize that these final speeds certainly satisfy conservation of energy and momentum with the initial conditions.
- If $M \gg m$, then m bounces backward with speed $\approx v$, and M hardly moves. This makes sense, because M is basically a brick wall.
- If $m \gg M$, then m keeps plowing along at speed $\approx v$, and M picks up a speed of $\approx 2v$. This $2v$ is an unexpected and interesting result (it's easier to see if you consider what's happening in the reference frame of the heavy mass m), and it leads to some neat effects, as in Problem 5.23.

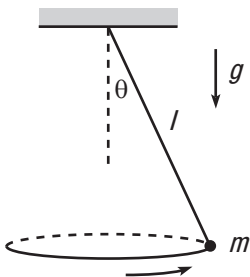


Fig. 1.4

Example (Circular pendulum): A mass hangs from a massless string of length ℓ . Conditions have been set up so that the mass swings around in a horizontal circle, with the string making a constant angle θ with the vertical (see Fig. 1.4). We'll find in Section 3.5 that the angular frequency, ω , of this motion is

$$\omega = \sqrt{\frac{g}{\ell \cos \theta}}. \quad (1.12)$$

As far as θ is concerned, there are two limits we should definitely check:

- If $\theta \rightarrow 90^\circ$, then $\omega \rightarrow \infty$. This makes sense; the mass has to spin very quickly to avoid flopping down.
- If $\theta \rightarrow 0$, then $\omega \rightarrow \sqrt{g/\ell}$, which is the same as the frequency of a standard “plane” pendulum of length ℓ (for small oscillations). This is a cool result and not at all obvious. (But once we get to $F = ma$ in Chapter 3, you can convince yourself why this is true by looking at the projection of the force on a given horizontal line.)

In the above examples, we checked limiting and special cases of answers that were correct (I hope!). This whole process is more useful (and a bit more fun, actually) when you check the limits of an answer that is *incorrect*. In this case, you gain the unequivocal information that your answer is wrong. But rather than leading you into despair, this information is in fact something you should be quite happy about, considering that the alternative is to carry on in a state of blissful ignorance. Once you know that your answer is wrong, you can go back through your work and figure out where the error is (perhaps by checking limits at various stages to narrow down where the error could be). Personally, if there’s any way I’d like to discover that my answer is garbage, this is it. At any rate, checking limiting cases can often save you a lot of trouble in the long run...

The lemmings get set for their race.
 With one step and two steps they pace.
 They take three and four,
 And then head on for more,
 Without checking the limiting case.

1.4 Solving differential equations numerically

Solving a physics problem often involves solving a differential equation. A differential equation is one that involves derivatives (usually with respect to time, in our physics problems) of the variable you’re trying to solve for. The differential equation invariably comes about from using $F = ma$, and/or $\tau = I\alpha$, or the Lagrangian technique we’ll discuss in Chapter 6. For example, consider a falling body. $F = ma$ gives $-mg = ma$, which can be written as $-g = \ddot{y}$, where a dot denotes a time derivative. This is a rather simple differential equation, and you can quickly guess that $y(t) = -gt^2/2$ is a solution. Or, more generally with the constants of integration thrown in, $y(t) = y_0 + v_0t - gt^2/2$.

However, the differential equations produced in some problems can get rather complicated, so sooner or later you will encounter one that you can’t solve exactly (either because it’s in fact impossible to solve, or because you can’t think of the

appropriate clever trick). Having resigned yourself to not getting the exact answer, you should ponder how to obtain a decent approximation to it. Fortunately, it's easy to write a short program that will give you a very good numerical answer to your problem. Given enough computer time, you can obtain any desired accuracy (assuming that the system isn't chaotic, but we won't have to worry about this for the systems we'll be dealing with).

We'll demonstrate the procedure by considering a standard problem, one that we'll solve exactly and in great depth in Chapter 4. Consider the equation,

$$\ddot{x} = -\omega^2 x. \quad (1.13)$$

This is the equation for a mass on a spring, with $\omega = \sqrt{k/m}$. We'll find in Chapter 4 that the solution can be written, among other ways, as

$$x(t) = A \cos(\omega t + \phi). \quad (1.14)$$

But let's pretend we don't know this. If someone comes along and gives us the values of $x(0)$ and $\dot{x}(0)$, then it seems that somehow we should be able to find $x(t)$ and $\dot{x}(t)$ for any later t , just by using Eq. (1.13). Basically, if we're told how the system starts, and if we know how it evolves, via Eq. (1.13), then we should know everything about it. So here's how we find $x(t)$ and $\dot{x}(t)$.

The plan is to discretize time into intervals of some small unit (call it ϵ), and to then determine what happens at each successive point in time. If we know $x(t)$ and $\dot{x}(t)$, then we can easily find (approximately) the value of x at a slightly later time, by using the definition of $\dot{x}$. Similarly, if we know $\dot{x}(t)$ and $\ddot{x}(t)$, then we can easily find (approximately) the value of $\dot{x}$ at a slightly later time, by using the definition of $\ddot{x}$. Using the definitions of the derivatives, the relations are simply

$$\begin{aligned} x(t + \epsilon) &\approx x(t) + \epsilon \dot{x}(t), \\ \dot{x}(t + \epsilon) &\approx \dot{x}(t) + \epsilon \ddot{x}(t). \end{aligned} \quad (1.15)$$

These two equations, combined with (1.13), which gives us $\ddot{x}$ in terms of x , allow us to march along in time, obtaining successive values for x , $\dot{x}$, and $\ddot{x}$.⁴

Here's what a typical program might look like.⁵ (This is a Maple program, but even if you aren't familiar with this, the general idea should be clear.) Let's say

⁴ Of course, another expression for $\ddot{x}$ is the definitional one, analogous to Eqs. (1.15), involving the third derivative. But this would then require knowledge of the third derivative, and so on with higher derivatives, and we would end up with an infinite chain of relations. An *equation of motion* such as Eq. (1.13) (which in general could be an $F = ma$, $\tau = I\alpha$, or Euler-Lagrange equation) relates $\ddot{x}$ back to x (and possibly $\dot{x}$), thereby creating an intertwined relation among x , $\dot{x}$, and $\ddot{x}$, and eliminating the need for an infinite and useless chain.

⁵ We've written the program in the most straightforward way, without any concern for efficiency, because computing time isn't an issue in this simple system. But in more complex systems that require programs for which computing time is an issue, a major part of the problem-solving process is developing a program that is as efficient as possible.

that the particle starts from rest at position $x = 2$, and let's pick $\omega^2 = 5$. We'll use the notation where $x1$ stands for $\dot{x}$, and $x2$ stands for $\ddot{x}$. And e stands for ϵ . Let's calculate x at, say, $t = 3$.

```
x:=2:          # initial position
x1:=0:         # initial velocity
e:=.01:        # small time interval
for i to 300 do # do 300 steps (ie, up to 3 seconds)
x2:=-5*x:      # the given equation
x:=x+e*x1:     # how x changes, by definition of x1
x1:=x1+e*x2:   # how x1 changes, by definition of x2
end do:        # the Maple command to stop the do loop
x;             # print the value of x
```

This procedure won't give the exact value for x , because x and $\dot{x}$ don't really change according to Eqs. (1.15). These equations are just first-order approximations to the full Taylor series with higher-order terms. Said differently, there is no way the above procedure can be exactly correct, because there are ambiguities in how the program can be written. Should line 5 come before or after line 7? That is, in determining $\dot{x}$ at time $t + \epsilon$, should we use the $\dot{x}$ at time t or $t + \epsilon$? And should line 7 come before or after line 6? The point is that for very small ϵ , the order doesn't matter much. And in the limit $\epsilon \rightarrow 0$, the order doesn't matter at all.

If we want to obtain a better approximation, we can just shorten ϵ down to 0.001 and increase the number of steps to 3000. If the result looks basically the same as with $\epsilon = 0.01$, then we know we pretty much have the right answer. In the present example, $\epsilon = 0.01$ yields $x \approx 1.965$ after 3 seconds. If we set $\epsilon = 0.001$, then we obtain $x \approx 1.836$. And if we set $\epsilon = 0.0001$, then we get $x \approx 1.823$. The correct answer must therefore be somewhere around $x = 1.82$. And indeed, if we solve the problem exactly, we obtain $x(t) = 2 \cos(\sqrt{5} t)$. Plugging in $t = 3$ gives $x \approx 1.822$.

This is a wonderful procedure, but it shouldn't be abused. It's nice to know that we can always obtain a decent numerical approximation if all else fails. But we should set our initial goal on obtaining the correct algebraic expression, because this allows us to see the overall behavior of the system. And besides, nothing beats the truth. People tend to rely a bit too much on computers and calculators nowadays, without pausing to think about what is actually going on in a problem.

The skill to do math on a page
 Has declined to the point of outrage.
 Equations quadratica
 Are solved on Math'matica,
 And on birthdays we don't know our age.

1.5 Problems

Section 1.2: Units, dimensional analysis

1.1. Escape velocity *

As given below in Exercise 1.9, show that the escape velocity from the earth is $v = \sqrt{2GM_E/R}$, up to numerical factors. You can use the fact that the form of Newton's gravitation force law implies that the acceleration (and hence overall motion) of the particle doesn't depend on its mass.

1.2. Mass in a tube *

A tube of mass M and length ℓ is free to swing around a pivot at one end. A mass m is positioned inside the (frictionless) tube at this end. The tube is held horizontal and then released (see Fig. 1.5). Let η be the fraction of the tube that the mass has traversed by the time the tube becomes vertical. Does η depend on ℓ ?

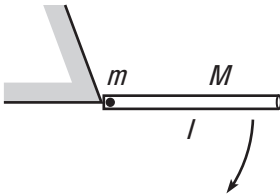


Fig. 1.5

1.3. Waves in a fluid *

How does the speed of waves in a fluid depend on its density, ρ , and “bulk modulus,” B (which has units of pressure, which is force per area)?

1.4. Vibrating star *

Consider a vibrating star, whose frequency ν depends (at most) on its radius R , mass density ρ , and Newton's gravitational constant G . How does ν depend on R , ρ , and G ?

1.5. Damping **

A particle with mass m and initial speed V is subject to a velocity-dependent damping force of the form bv^n .

- For $n = 0, 1, 2, \dots$, determine how the stopping time depends on m , V , and b .
- For $n = 0, 1, 2, \dots$, determine how the stopping distance depends on m , V , and b .

Be careful! See if your answers make sense. Dimensional analysis gives the answer only up to a numerical factor. This is a tricky problem, so don't let it discourage you from using dimensional analysis. Most applications of dimensional analysis are quite straightforward.

Section 1.3: Approximations, limiting cases

1.6. Projectile distance *

A person throws a ball (at an angle of her choosing, to achieve the maximum distance) with speed v from the edge of a cliff of height h . Assuming

that one of the following quantities is the maximum horizontal distance the ball can travel, which one is it? (Don't solve the problem from scratch, just check special cases.)

$$\frac{gh^2}{v^2}, \quad \frac{v^2}{g}, \quad \sqrt{\frac{v^2 h}{g}}, \quad \frac{v^2}{g} \sqrt{1 + \frac{2gh}{v^2}}, \quad \frac{v^2}{g} \left(1 + \frac{2gh}{v^2}\right), \quad \frac{v^2/g}{1 - \frac{2gh}{v^2}}.$$

Section 1.4: Solving differential equations numerically

1.7. Two masses, one swinging **

Two equal masses are connected by a string that hangs over two pulleys (of negligible size), as shown in Fig. 1.6. The left mass moves in a vertical line, but the right mass is free to swing back and forth in the plane of the pulleys. It can be shown (see Problem 6.4) that the equations of motion for r and θ (labeled in the figure) are

$$\begin{aligned} 2\ddot{r} &= r\dot{\theta}^2 - g(1 - \cos \theta), \\ \ddot{\theta} &= -\frac{2\dot{r}\dot{\theta}}{r} - \frac{g \sin \theta}{r}. \end{aligned} \quad (1.16)$$

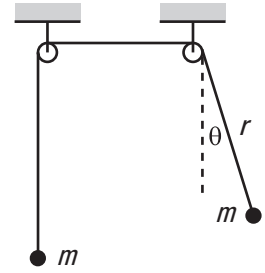


Fig. 1.6

Assume that both masses start out at rest, with the right mass making an initial angle of $10^\circ = \pi/18$ with the vertical. If the initial value of r is 1 m, how much time does it take for it to reach a length of 2 m? Write a program to solve this numerically. Use $g = 9.8 \text{ m/s}^2$.

1.6 Exercises

Section 1.2: Units, dimensional analysis

1.8. Pendulum on the moon

If a pendulum has a period of 3 s on the earth, what would its period be if it were placed on the moon? Use $g_M/g_E \approx 1/6$.

1.9. Escape velocity *

The *escape velocity* on the surface of a planet is given by

$$v = \sqrt{\frac{2GM}{R}}, \quad (1.17)$$

where M and R are the mass and radius of the planet, respectively, and G is Newton's gravitational constant. (The escape velocity is the velocity

needed to refute the “What goes up must come down” maxim, neglecting air resistance.)

- (a) Write v in terms of the average mass density ρ , instead of M .
- (b) Assuming that the average density of the earth is four times that of Jupiter, and that the radius of Jupiter is 11 times that of the earth, what is v_J/v_E ?

1.10. Downhill projectile *

A hill is sloped downward at an angle θ with respect to the horizontal. A projectile of mass m is fired with speed v_0 perpendicular to the hill. When it eventually lands on the hill, let its velocity make an angle β with respect to the horizontal. Which of the quantities θ , m , v_0 , and g does the angle β depend on?

1.11. Waves on a string *

How does the speed of waves on a string depend on its mass M , length L , and tension (that is, force) T ?

1.12. Vibrating water drop *

Consider a vibrating water drop, whose frequency ν depends on its radius R , mass density ρ , and surface tension S . The units of surface tension are (force)/(length). How does ν depend on R , ρ , and S ?

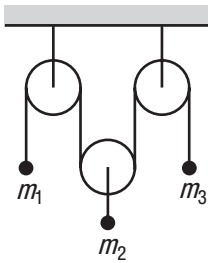


Fig. 1.7

Section 1.3: Approximations, limiting cases

1.13. Atwood's machine *

Consider the “Atwood's” machine shown in Fig. 1.7, consisting of three masses and three frictionless pulleys. It can be shown that the acceleration of m_1 is given by (just accept this):

$$a_1 = g \frac{3m_2m_3 - m_1(4m_3 + m_2)}{m_2m_3 + m_1(4m_3 + m_2)}, \quad (1.18)$$

with upward taken to be positive. Find a_1 in the following special cases:

- (a) $m_2 = 2m_1 = 2m_3$.
- (b) m_1 much larger than both m_2 and m_3 .
- (c) m_1 much smaller than both m_2 and m_3 .
- (d) $m_2 \gg m_1 = m_3$.
- (e) $m_1 = m_2 = m_3$.

1.14. Cone frustum *

A cone frustum has base radius b , top radius a , and height h , as shown in Fig. 1.8. Assuming that one of the following quantities is the volume

of the frustum, which one is it? (Don't solve the problem from scratch, just check special cases.)

$$\frac{\pi h}{3}(a^2 + b^2), \quad \frac{\pi h}{2}(a^2 + b^2), \quad \frac{\pi h}{3}(a^2 + ab + b^2),$$

$$\frac{\pi h}{3} \cdot \frac{a^4 + b^4}{a^2 + b^2}, \quad \pi hab.$$

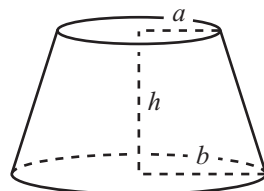


Fig. 1.8

1.15. Landing at the corner *

A ball is thrown at an angle θ up to the top of a cliff of height L , from a point a distance L from the base, as shown in Fig. 1.9. Assuming that one of the following quantities is the initial speed required to make the ball hit right at the edge of the cliff, which one is it? (Don't solve the problem from scratch, just check special cases.)

$$\sqrt{\frac{gL}{2(\tan \theta - 1)}}, \quad \frac{1}{\cos \theta} \sqrt{\frac{gL}{2(\tan \theta - 1)}}, \quad \frac{1}{\cos \theta} \sqrt{\frac{gL}{2(\tan \theta + 1)}},$$

$$\sqrt{\frac{gL \tan \theta}{2(\tan \theta + 1)}}.$$

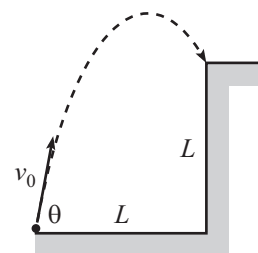


Fig. 1.9

1.16. Projectile with drag **

Consider a projectile subject to a drag force $\mathbf{F} = -m\alpha\mathbf{v}$. If it is fired with speed v_0 at an angle θ , it can be shown that the height as a function of time is given by (just accept this here; it's one of the tasks of Exercise 3.53)

$$y(t) = \frac{1}{\alpha} \left(v_0 \sin \theta + \frac{g}{\alpha} \right) \left(1 - e^{-\alpha t} \right) - \frac{gt}{\alpha}. \quad (1.19)$$

Show that this reduces to the usual projectile expression, $y(t) = (v_0 \sin \theta)t - gt^2/2$, in the limit of small α . What exactly is meant by "small α "?

Section 1.4: Solving differential equations numerically

1.17. Pendulum **

A pendulum of length ℓ is released from the horizontal position. It can be shown that the tangential $F = ma$ equation is (where θ is measured with respect to the vertical)

$$\ddot{\theta} = -\frac{g \sin \theta}{\ell}. \quad (1.20)$$

If $\ell = 1$ m, and $g = 9.8$ m/s², write a program to show that the time it takes the pendulum to swing down through the vertical position is

$t \approx 0.592$ s. This happens to be about 1.18 times the $(\pi/2)\sqrt{\ell/g} \approx 0.502$ s it would take the pendulum to swing down if it were released from very close to the vertical (this is $1/4$ of the standard period of $2\pi\sqrt{\ell/g}$ for a pendulum). It also happens to be about 1.31 times the $\sqrt{2\ell/g} \approx 0.452$ s it would take a mass to simply freefall a height ℓ .

1.18. Distance with damping **

A mass is subject to a damping force proportional to its velocity, which means that the equation of motion takes the form $\ddot{x} = -A\dot{x}$, where A is some constant. If the initial speed is 2 m/s, and if $A = 1 \text{ s}^{-1}$, how far has the mass traveled at 1 s? 10 s? 100 s? You should find that the distance approaches a limiting value.

Now assume that the mass is subject to a damping force proportional to the square of its velocity, which means that the equation of motion now takes the form $\ddot{x} = -A\dot{x}^2$, where A is some constant. If the initial speed is 2 m/s, and if $A = 1 \text{ m}^{-1}$, how far has the mass traveled at 1 s? 10 s? 100 s? How about some larger powers of 10? You should find that the distance keeps growing, but slowly like the log of t . (The results for these two forms of the damping are consistent with the results of Problem 1.5.)

1.7 Solutions

1.1. Escape velocity

It is tempting to use the same reasoning as in the low-orbit satellite example in Section 1.2. This reasoning gives the same result, $v = C\sqrt{gR} = C\sqrt{GM_E/R}$, where C is some number (it turns out that $C = \sqrt{2}$). Although this solution yields the correct answer, it isn't quite rigorous, in view of the footnote in the low-orbit satellite example. Because the particle isn't always at the same radius, the force changes, so it isn't obvious that we can absorb the M_E and G dependence into one quantity, g , as we did with the orbiting satellite. Let us therefore be more rigorous with the following reasoning.

The dimensionful quantities in the problem are $[m] = M$, the radius of the earth $[R] = L$, the mass of the earth $[M_E] = M$, and Newton's gravitational constant $[G] = L^3/MT^2$. These units for G follow from the gravitational force law, $F = Gm_1m_2/r^2$. If we use no information other than these given quantities, then there is no way to arrive at the speed of $C\sqrt{GM_E/R}$, because for all we know, there could be a factor of $(m/M_E)^7$ in the answer. This number is dimensionless, so it wouldn't mess up the units.

If we want to make any progress in this problem, we have to use the fact that the gravitational force takes the form of GM_Em/r^2 . This then implies (as was stated in the problem) that the acceleration is independent of m . And since the path of the particle is determined by its acceleration, we see that the answer can't depend on m . We are therefore left with the quantities G , R , and M_E , and you can show that the only combination of these quantities that gives the units of speed is $v = C\sqrt{GM_E/R}$.

1.2. Mass in a tube

The dimensionful quantities are $[g] = L/T^2$, $[\ell] = L$, $[m] = M$, and $[M] = M$. We want to produce a dimensionless number η . Since g is the only constant involving time, η cannot depend on g . This then implies that η cannot depend on ℓ , which is the only

length remaining. Therefore, η depends only on m and M (and furthermore only on the ratio m/M , since we want a dimensionless number). So the answer to the stated problem is, “No.”

It turns out that you have to solve the problem numerically if you actually want to find η (see Problem 8.5). Some results are: If $m \ll M$, then $\eta \approx 0.349$. If $m = M$, then $\eta \approx 0.378$. And if $m = 2M$, then $\eta \approx 0.410$.

1.3. Waves in a fluid

We want to make a speed, $[v] = L/T$, out of the quantities $[\rho] = M/L^3$, and $[B] = [F/A] = (ML/T^2)/(L^2) = M/(LT^2)$. We can play around with these quantities to find the combination that has the correct units, but let's do it the no-fail way. If $v \propto \rho^a B^b$, then we have

$$\frac{L}{T} = \left(\frac{M}{L^3}\right)^a \left(\frac{M}{LT^2}\right)^b. \quad (1.21)$$

Matching up the powers of the three kinds of units on each side of this equation gives

$$M : 0 = a + b, \quad L : 1 = -3a - b, \quad T : -1 = -2b. \quad (1.22)$$

The solution to this system of equations is $a = -1/2$ and $b = 1/2$. Therefore, our answer is $v \propto \sqrt{B/\rho}$. Fortunately, there was a solution to this system of three equations in two unknowns.

1.4. Vibrating star

We want to make a frequency, $[v] = 1/T$, out of the quantities $[R] = L$, $[\rho] = M/L^3$, and $[G] = L^3/(MT^2)$. These units for G follow from the gravitational force law, $F = Gm_1m_2/r^2$. As in the previous problem, we can play around with these quantities to find the combination that has the correct units, but let's do it the no-fail way. If $v \propto R^a \rho^b G^c$, then we have

$$\frac{1}{T} = L^a \left(\frac{M}{L^3}\right)^b \left(\frac{L^3}{MT^2}\right)^c. \quad (1.23)$$

Matching up the powers of the three kinds of units on each side of this equation gives

$$M : 0 = b - c, \quad L : 0 = a - 3b + 3c, \quad T : -1 = -2c. \quad (1.24)$$

The solution to this system of equations is $a = 0$, and $b = c = 1/2$. Therefore, our answer is $v \propto \sqrt{\rho G}$. So it turns out that there is no R dependence.

REMARK: Note the difference in the given quantities in this problem (R , ρ , and G) and the ones in Exercise 1.12 (R , ρ , and S). In this problem with the star, the mass is large enough so that we can ignore the surface tension, S . And in Exercise 1.12 with the drop, the mass is small enough so that we can ignore the gravitational force, and hence G . ♣

1.5. Damping

- (a) The constant b has units $[b] = [\text{Force}][v^{-n}] = (ML/T^2)(T^n/L^n)$. The other quantities are $[m] = M$ and $[V] = L/T$. There is also n , which is dimensionless. You can show that the only combination of these quantities that has units of T is

$$t = f(n) \frac{m}{bV^{n-1}}, \quad (1.25)$$

where $f(n)$ is a dimensionless function of n .

For $n = 0$, we have $t = f(0) mV/b$. This increases with m and V , and decreases with b , as it should.

For $n = 1$, we have $t = f(1) m/b$. So we *seem* to have $t \sim m/b$. This, however, cannot be correct, because t should definitely grow with V . A large initial speed V_1 requires some nonzero time to slow down to a smaller speed V_2 , after which time we simply have the same scenario with initial speed V_2 . So where did we go wrong? After all, dimensional analysis tells us that the answer *does* have to look like $t = f(1) m/b$, where $f(1)$ is a numerical factor. The resolution to this puzzle

is that $f(1)$ is infinite. If we worked out the problem using $F = ma$, we would encounter an integral that diverges. So for any V , we would find an infinite t .⁶

Similarly, for $n \geq 2$, there is at least one power of V in the denominator of t . This certainly cannot be correct, because t should not decrease with V . So $f(n)$ must likewise be infinite for all of these cases.

The moral of this exercise is that sometimes you have to be careful when using dimensional analysis. The numerical factor in front of your answer nearly always turns out to be of order 1, but in some strange cases it turns out to be 0 or ∞ .

REMARK: For $n \geq 1$, the expression in Eq. (1.25) still has relevance. For example, for $n = 2$, the $m/(Vb)$ expression is relevant if you want to know how long it takes to go from V to some final speed V_f . The answer involves $m/(V_f b)$, which diverges as $V_f \rightarrow 0$. ♣

(b) You can show that the only combination of the quantities that has units of L is

$$\ell = g(n) \frac{m}{bV^{n-2}}, \quad (1.26)$$

where $g(n)$ is a dimensionless function of n .

For $n = 0$, we have $\ell = g(0) mV^2/b$. This increases with V , as it should.

For $n = 1$, we have $\ell = g(1) mV/b$. This increases with V , as it should.

For $n = 2$ we have $\ell = g(2) m/b$. So we *seem* to have $\ell \sim m/b$. But as in part (a), this cannot be correct, because ℓ should definitely depend on V . A large initial speed V_1 requires some nonzero distance to slow down to a smaller speed V_2 , after which point we simply have the same scenario with initial speed V_2 . So, from the reasoning in part (a), the total distance is infinite for $n \geq 2$, because the function g is infinite.

REMARK: Note that for integral $n \neq 1$, t and ℓ are either both finite or both infinite. For $n = 1$, however, the total time is infinite, whereas the total distance is finite. This situation actually holds for $1 \leq n < 2$, if we want to consider fractional n . ♣

1.6. Projectile distance

All of the possible answers have the correct units, so we'll have to figure things out by looking at special cases. Let's look at each choice in turn:

$\frac{gh^2}{v^2}$: Incorrect, because the answer shouldn't be zero for $h = 0$. Also, it shouldn't grow with g . And even worse, it shouldn't be infinite for $v \rightarrow 0$.

$\frac{v^2}{g}$: Incorrect, because the answer should depend on h .

$\sqrt{\frac{v^2 h}{g}}$: Incorrect, because the answer shouldn't be zero for $h = 0$.

$\frac{v^2}{g} \sqrt{1 + \frac{2gh}{v^2}}$: Can't rule this out, and it happens to be the correct answer.

$\frac{v^2}{g} \left(1 + \frac{2gh}{v^2}\right)$: Incorrect, because the answer should be zero for $v \rightarrow 0$. But this expression goes to $2h$ for $v \rightarrow 0$.

$\frac{v^2/g}{1 - \frac{2gh}{v^2}}$: Incorrect, because the answer shouldn't be infinite for $v^2 = 2gh$.

⁶ The total time t is actually undefined, because the particle never comes to rest. But t does grow with V , in the sense that if t is defined to be the time to slow down to some certain small speed, then t grows with V .

1.7. Two masses, one swinging

As in Section 1.4, we'll write a Maple program. We'll let q stand for θ , and we'll use the notation where $q1$ stands for $\dot{\theta}$, and $q2$ stands for $\ddot{\theta}$. Likewise for r . We'll run the program for as long as $r < 2$. As soon as r exceeds 2, the program will stop and print the value of the time.

```

r:=1:                # initial r value
r1:=0:               # initial r velocity
q:=3.14/18:          # initial angle
q1:=0:               # initial angular velocity
e:=.001:             # small time interval
i:=0:                # i counts the number of time steps
while r<2 do         # run the program until r=2
i:=i+1:              # increase the counter by 1
r2:=(r*q1^2-9.8*(1-cos(q)))/2: # the first of the given eqs
r:=r+e*r1:           # how r changes, by definition of r1
r1:=r1+e*r2:         # how r1 changes, by definition of r2
q2:=-2*r1*q1/r-9.8*sin(q)/r: # the second of the given eqs
q:=q+e*q1:           # how q changes, by definition of q1
q1:=q1+e*q2:         # how q1 changes, by definition of q2
end do:              # the Maple command to stop the do loop
i*e;                 # print the value of the time

```

This yields a time of $t = 8.057$ s. If we instead use a time interval of 0.0001 s, we obtain $t = 8.1377$ s. And a time interval of 0.00001 s gives $t = 8.14591$ s. So the correct time must be somewhere around 8.15 s.

Chapter 3

Using $F = ma$

The general goal of classical mechanics is to determine what happens to a given set of objects in a given physical situation. In order to figure this out, we need to know what makes the objects move the way they do. There are two main ways of going about this task. The first one, which you are undoubtedly familiar with, involves Newton's laws. This is the subject of the present chapter. The second one, which is more advanced, is the *Lagrangian* method. This is the subject of Chapter 6. It should be noted that each of these methods is perfectly sufficient for solving any problem, and they both produce the same information in the end. But they are based on vastly different principles. We'll talk more about this in Chapter 6.

3.1 Newton's laws

In 1687 Newton published his three laws in his *Principia Mathematica*. These laws are fairly intuitive, although I suppose it's questionable to attach the adjective "intuitive" to a set of statements that weren't written down until a mere 300 years ago. At any rate, the laws may be stated as follows.

- **First law:** A body moves with constant velocity (which may be zero) unless acted on by a force.
- **Second law:** The time rate of change of the momentum of a body equals the force acting on the body.
- **Third law:** For every force on one body, there is an equal and opposite force on another body.

We could discuss for days on end the degree to which these statements are physical laws, and the degree to which they are definitions. Sir Arthur Eddington once made the unflattering remark that the first law essentially says that "every particle continues in its state of rest or uniform motion in a straight line except insofar as it doesn't." However, although the three laws might seem somewhat light on content at first glance, there's actually more to them than Eddington's comment implies. Let's look at each in turn.¹

¹ A disclaimer: This section represents my view on which parts of the laws are definitions and which parts have content. But you should take all of this with a grain of salt. For further reading, see Anderson (1990), Keller (1987), O'Sullivan (1980), and Eisenbud (1958).

First law

One thing this law does is give a definition of zero force. Another thing it does is give a definition of an *inertial frame*, which is defined simply as a frame of reference in which the first law holds; since the term “velocity” is used, we have to state what frame we’re measuring the velocity with respect to. The first law does *not* hold in an arbitrary frame. For example, it fails in the frame of a rotating turntable.² Intuitively, an inertial frame is one that moves with constant velocity. But this is ambiguous, because we have to say what the frame is moving with constant velocity *with respect to*. But all this aside, an inertial frame is defined as the special type of frame in which the first law holds.

So, what we now have are two intertwined definitions of “force” and “inertial frame.” Not much physical content here. But the important point is that the law holds for *all* particles. So if we have a frame in which one free particle moves with constant velocity, then *all* free particles move with constant velocity. This is a statement with content. We can’t have a bunch of free particles moving with constant velocity while another one is doing a fancy jig.

Second law

Momentum is defined³ to be $m\mathbf{v}$. If m is constant,⁴ then the second law says that

$$\mathbf{F} = m\mathbf{a}, \quad (3.1)$$

where $\mathbf{a} \equiv d\mathbf{v}/dt$. This law holds only in an inertial frame, which is defined by the first law.

For things moving free or at rest,
Observe what the first law does best.
It defines a key frame,
“Inertial” by name,
Where the second law then is expressed.

You might think that the second law merely gives a definition of force, but there is more to it than that. There is a tacit implication in the law that this “force” is something that has an existence that isn’t completely dependent on the particle whose “ m ” appears in the law (more on this in the third law below). A spring force, for example, doesn’t depend at all on the particle on which it acts. And the gravitational force, GMm/r^2 , depends partly on the particle and partly on something else (another mass).

² It’s possible to modify things so that Newton’s laws hold in such a frame, provided that we introduce the so-called “fictitious” forces. But we’ll save this discussion for Chapter 10.

³ We’re doing everything nonrelativistically here, of course. Chapter 12 gives the relativistic modification to the $m\mathbf{v}$ expression.

⁴ We’ll assume in this chapter that m is constant. But don’t worry, we’ll get plenty of practice with changing mass (in rockets and such) in Chapter 5.

If you feel like just making up definitions, then you can define a new quantity, $\mathbf{G} = m^2\mathbf{a}$. This is a perfectly legal thing to do; you can't really go wrong in making a definition (well, unless you've already defined the quantity to be something else). However, this definition is completely useless. You can define it for every particle in the world, and for any acceleration, but the point is that the definitions don't have anything to do with each other. There is simply no (uncontrived) quantity in this world that gives accelerations in the ratio of 4 to 1 when "acting" on masses m and $2m$. The quantity $\mathbf{G}$ has nothing to do with anything except the particle you defined it for. The main thing the second law says is that there does indeed exist a quantity $\mathbf{F}$ that gives the same $m\mathbf{a}$ when acting on different particles. The statement of the existence of such a thing is far more than a definition.

Along this same line, note that the second law says that $\mathbf{F} = m\mathbf{a}$, and not, for example, $\mathbf{F} = m\mathbf{v}$, or $\mathbf{F} = m d^3\mathbf{x}/dt^3$. In addition to being inconsistent with the real world, these expressions are inconsistent with the first law. $\mathbf{F} = m\mathbf{v}$ would say that a nonzero velocity requires a force, in contrast with the first law. And $\mathbf{F} = m d^3\mathbf{x}/dt^3$ would say that a particle moves with constant acceleration (instead of constant velocity) unless acted on by a force, also in contrast with the first law.

As with the first law, it is important to realize that the second law holds for *all* particles. In other words, if the same force (for example, the same spring stretched by the same amount) acts on two particles with masses m_1 and m_2 , then Eq. (3.1) says that their accelerations are related by

$$\frac{a_1}{a_2} = \frac{m_2}{m_1}. \quad (3.2)$$

This relation holds regardless of what the common force is. Therefore, once we've used one force to find the relative masses of two objects, then we know what the ratio of their a 's will be when they are subjected to any other force. Of course, we haven't really defined *mass* yet. But Eq. (3.2) gives an experimental method for determining an object's mass in terms of a standard (say, 1 kg) mass. All we have to do is compare its acceleration with that of the standard mass, when acted on by the same force.

Note that $\mathbf{F} = m\mathbf{a}$ is a vector equation, so it is really three equations in one. In Cartesian coordinates, it says that $F_x = ma_x$, $F_y = ma_y$, and $F_z = ma_z$.

Third law

One thing this law says is that if we have two isolated particles interacting through some force, then their accelerations are opposite in direction and inversely proportional to their masses. Equivalently, the third law essentially postulates that

the total momentum of an isolated system is conserved (that is, independent of time). To see this, consider two particles, each of which interacts only with the other particle and nothing else in the universe. Then we have

$$\begin{aligned}\frac{d\mathbf{p}_{\text{total}}}{dt} &= \frac{d\mathbf{p}_1}{dt} + \frac{d\mathbf{p}_2}{dt} \\ &= \mathbf{F}_1 + \mathbf{F}_2,\end{aligned}\tag{3.3}$$

where $\mathbf{F}_1$ and $\mathbf{F}_2$ are the forces acting on m_1 and m_2 , respectively. This demonstrates that momentum conservation (that is, $d\mathbf{p}_{\text{total}}/dt = 0$) is equivalent to Newton's third law (that is, $\mathbf{F}_1 = -\mathbf{F}_2$). Similar reasoning holds with more than two particles, but we'll save this more general case, along with many other aspects of momentum, for Chapter 5.

There isn't much left to be defined via this law, so this is a law of pure content. It can't be a definition, anyway, because it's actually not always valid. It holds for forces of the "pushing" and "pulling" type, but it fails for the magnetic force, for example. In that case, momentum is carried off in the electromagnetic field (so the total momentum of the particles *and* the field is conserved). But we won't deal with fields here. Just particles. So the third law will always hold in any situation we'll be concerned with.

The third law contains an extremely important piece of information. It says that we will never find a particle accelerating unless there's some other particle accelerating somewhere else. The other particle might be far away, as with the earth-sun system, but it's always out there somewhere. Note that if we were given only the second law, then it would be perfectly possible for a given particle to spontaneously accelerate with nothing else happening in the universe, as long as a similar particle with twice the mass accelerated with half the acceleration when placed in the same spot, etc. This would all be fine, as far as the second law goes. We would say that a force with a certain value is acting at the point, and everything would be consistent. But the third law says that this is simply not the way the world (at least the one we live in) works. In a sense, a force without a counterpart seems somewhat like magic, whereas a force with an equal and opposite counterpart has a "cause and effect" nature, which seems (and apparently is) more physical.

In the end, however, we shouldn't attach too much significance to Newton's laws, because although they were a remarkable intellectual achievement and work spectacularly for everyday physics, they are the laws of a theory that is only approximate. Newtonian physics is a limiting case of the more correct theories of relativity and quantum mechanics, which are in turn limiting cases of yet more correct theories. The way in which particles (or waves, or strings, or whatever) interact on the most fundamental level surely doesn't bear any resemblance to what we call forces.

3.2 Free-body diagrams

The law that allows us to be quantitative is the second law. Given a force, we can apply $\mathbf{F} = m\mathbf{a}$ to find the acceleration. And knowing the acceleration, we can determine the behavior of a given object (that is, the position and velocity), provided that we are given the initial position and velocity. This process sometimes takes a bit of work, but there are two basic types of situations that commonly arise.

- In many problems, all you are given is a physical situation (for example, a block resting on a plane, strings connecting masses, etc.), and it is up to you to find all the forces acting on all the objects, using $\mathbf{F} = m\mathbf{a}$. The forces generally point in various directions, so it's easy to lose track of them. It therefore proves useful to isolate the objects and draw all the forces acting on each of them. This is the subject of the present section.
- In other problems, you are *given* the force explicitly as a function of time, position, or velocity, and the task immediately becomes the mathematical one of solving the $F = ma \equiv m\ddot{x}$ equation (we'll just deal with one dimension here). These *differential equations* can be difficult (or impossible) to solve exactly. They are the subject of Section 3.3.

Let's consider here the first of these two types of scenarios, where we are presented with a physical situation and we must determine all the forces involved. The term *free-body diagram* is used to denote a diagram with all the forces drawn on a given object. After drawing such a diagram for each object in the setup, we simply write down all the $F = ma$ equations they imply. The result will be a system of linear equations in various unknown forces and accelerations, for which we can then solve. This procedure is best understood through an example.

Example (A plane and masses): Mass M_1 is held on a plane with inclination angle θ , and mass M_2 hangs over the side. The two masses are connected by a massless string which runs over a massless pulley (see Fig. 3.1). The coefficient of kinetic friction between M_1 and the plane is μ . M_1 is released from rest. Assuming that M_2 is sufficiently large so that M_1 gets pulled up the plane, what is the acceleration of the masses? What is the tension in the string?

Solution: The first thing to do is draw all the forces on the two masses. These are shown in Fig. 3.2. The forces on M_2 are gravity and the tension. The forces on M_1 are gravity, friction, the tension, and the normal force. Note that the friction force points down the plane, because we are assuming that M_1 moves up the plane.

Having drawn all the forces, we can now write down all the $F = ma$ equations. When dealing with M_1 , we could break things up into horizontal and vertical components, but it is much cleaner to use the components parallel and perpendicular to the

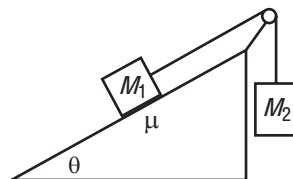


Fig. 3.1

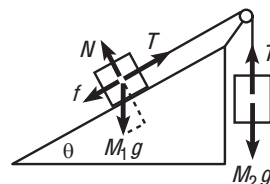


Fig. 3.2

plane.⁵ These two components of $\mathbf{F} = m\mathbf{a}$, along with the vertical $F = ma$ equation for M_2 , give

$$\begin{aligned} T - f - M_1 g \sin \theta &= M_1 a, \\ N - M_1 g \cos \theta &= 0, \\ M_2 g - T &= M_2 a, \end{aligned} \tag{3.4}$$

where we have used the fact that the two masses accelerate at the same rate (and we have defined the positive direction for M_2 to be downward). We have also used the fact that the tension is the same at both ends of the string, because otherwise there would be a net force on some part of the string which would then undergo infinite acceleration, because it is massless.

There are four unknowns in Eq. (3.4) (namely T , a , N , and f), but only three equations. Fortunately, we have a fourth equation: $f = \mu N$, because we are assuming that M_1 is in fact moving, so we can use the expression for kinetic friction. Using this in the second equation above gives $f = \mu M_1 g \cos \theta$. The first equation then becomes $T - \mu M_1 g \cos \theta - M_1 g \sin \theta = M_1 a$. Adding this to the third equation leaves us with only a , so we find

$$a = \frac{g(M_2 - \mu M_1 \cos \theta - M_1 \sin \theta)}{M_1 + M_2} \implies T = \frac{M_1 M_2 g(1 + \mu \cos \theta + \sin \theta)}{M_1 + M_2}. \tag{3.5}$$

Note that in order for M_1 to in fact accelerate upward (that is, $a > 0$), we must have $M_2 > M_1(\mu \cos \theta + \sin \theta)$. This is clear from looking at the forces along the plane.

REMARK: If we instead assume that M_1 is sufficiently large so that it slides down the plane, then the friction force points up the plane, and we find (as you can check),

$$a = \frac{g(M_2 + \mu M_1 \cos \theta - M_1 \sin \theta)}{M_1 + M_2}, \quad \text{and} \quad T = \frac{M_1 M_2 g(1 - \mu \cos \theta + \sin \theta)}{M_1 + M_2}. \tag{3.6}$$

In order for M_1 to in fact accelerate downward (that is, $a < 0$), we must have $M_2 < M_1(\sin \theta - \mu \cos \theta)$. Therefore, the range of M_2 for which the system doesn't accelerate (that is, it just sits there, assuming that it started at rest) is

$$M_1(\sin \theta - \mu \cos \theta) \leq M_2 \leq M_1(\sin \theta + \mu \cos \theta). \tag{3.7}$$

If μ is very small, then M_2 must essentially be equal to $M_1 \sin \theta$ if the system is to be static. Equation (3.7) also implies that if $\tan \theta \leq \mu$, then M_1 won't slide down, even if $M_2 = 0$. ♣

In problems like the one above, it's clear which things you should pick as the objects you're going to draw forces on. But in other problems, where there are

⁵ When dealing with inclined planes, it's usually the case that one of these two coordinate systems works much better than the other. Sometimes it isn't clear which one, but if things get messy with one system, you can always try the other.

various different subsystems you can choose, you must be careful to include all the relevant forces on a given subsystem. Which subsystems you want to pick depends on what quantities you’re trying to find. Consider the following example.

Example (Platform and pulley): A person stands on a platform-and-pulley system, as shown in Fig. 3.3. The masses of the platform, person, and pulley⁶ are M , m , and μ , respectively.⁷ The rope is massless. Let the person pull up on the rope so that she has acceleration a upward. (Assume that the platform is somehow constrained to stay level, perhaps by having the ends run along some rails.) Find the tension in the rope, the normal force between the person and the platform, and the tension in the rod connecting the pulley to the platform.

Solution: To find the tension in the rope, we simply want to let our subsystem be the whole system (except the ceiling). If we imagine putting the system in a black box (to emphasize the fact that we don’t care about any internal forces within the system), then the forces we see “protruding” from the box are the three weights (Mg , mg , and μg) downward, and the tension T upward. Applying $F = ma$ to the whole system gives

$$T - (M + m + \mu)g = (M + m + \mu)a \implies T = (M + m + \mu)(g + a). \quad (3.8)$$

To find the normal force N between the person and the platform, and also the tension f in the rod connecting the pulley to the platform, it is not sufficient to consider the system as a whole. This is true because these forces are internal forces to this system, so they won’t show up in any $F = ma$ equations (which involve only external forces to a system). So we must consider subsystems:

- Let’s apply $F = ma$ to the person. The forces acting on the person are gravity, the normal force from the platform, and the tension from the rope (pulling downward on her hand). So we have

$$N - T - mg = ma. \quad (3.9)$$

- Now apply $F = ma$ to the platform. The forces acting on the platform are gravity, the normal force from the person, and the force upward from the rod. So we have

$$f - N - Mg = Ma. \quad (3.10)$$

- Now apply $F = ma$ to the pulley. The forces acting on the pulley are gravity, the force downward from the rod, and *twice* the tension in the rope (because it pulls

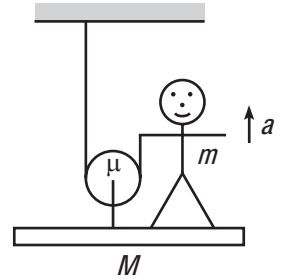


Fig. 3.3

⁶ Assume that the pulley’s mass is concentrated at its center, so that we don’t have to worry about any rotational dynamics (the subject of Chapter 8).

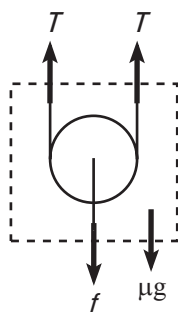
⁷ My apologies for using μ as a mass here, since it usually denotes a coefficient of friction. Alas, there are only so many symbols for “ m .”

up on both sides). So we have

$$2T - f - \mu g = \mu a. \quad (3.11)$$

Note that if we add up the three previous equations, we obtain the $F = ma$ equation in Eq. (3.8), as should be the case, because the whole system is the sum of the three above subsystems. Equations (3.9)–(3.11) are three equations in the three unknowns, T , N , and f . Their sum yields the T in (3.8), and then Eqs. (3.9) and (3.11) give, respectively, as you can show,

$$N = (M + 2m + \mu)(g + a), \quad \text{and} \quad f = (2M + 2m + \mu)(g + a). \quad (3.12)$$



REMARKS: You can also obtain these results by considering subsystems different from the ones we chose above. For example, you can choose the pulley-plus-platform subsystem, etc. But no matter how you choose to break up the system, you will need to produce three independent $F = ma$ equations in order to solve for the three unknowns, T , N , and f .

In problems like this one, it's easy to forget to include certain forces, such as the second T in Eq. (3.11). The safest thing to do is to always isolate each subsystem, draw a box around it, and then draw all the forces that “protrude” from the box. In other words, draw the free-body diagram. Figure 3.4 shows the free-body diagram for the subsystem consisting of only the pulley. ♣

Fig. 3.4

Another class of problems, similar to the above example, goes by the name of *Atwood's machines*. An Atwood's machine is the name used for any system consisting of a combination of masses, strings, and pulleys.⁸ In general, the pulleys and strings can have mass, but we'll deal only with massless ones in this chapter. As we'll see in the following example, there are two basic steps in solving an Atwood's problem: (1) write down all the $F = ma$ equations, and (2) relate the accelerations of the various masses by noting that the length of the string(s) doesn't change, a fact that we call “conservation of string.”

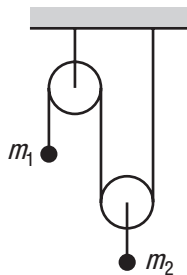


Fig. 3.5

Example (Atwood's machine): Consider the pulley system in Fig. 3.5, with masses m_1 and m_2 . The strings and pulleys are massless. What are the accelerations of the masses? What is the tension in the string?

Solution: The first thing to note is that the tension T is the same everywhere throughout the massless string, because otherwise there would be an infinite acceleration of some part of the string. It then follows that the tension in the short string connected to m_2 is $2T$. This is true because there must be zero net force on the massless right pulley, because otherwise it would have infinite acceleration. The $F = ma$

⁸ George Atwood (1746–1807) was a tutor at Cambridge University. He published the description of the first of his machines in Atwood (1784). For a history of Atwood's machines, see Greenslade (1985).

equations for the two masses are therefore (with upward taken to be positive)

$$\begin{aligned} T - m_1 g &= m_1 a_1, \\ 2T - m_2 g &= m_2 a_2. \end{aligned} \quad (3.13)$$

We now have two equations in the three unknowns, a_1 , a_2 , and T . So we need one more equation. This is the “conservation of string” fact, which relates a_1 and a_2 . If we imagine moving m_2 and the right pulley up a distance d , then a length $2d$ of string has disappeared from the two parts of the string touching the right pulley. This string has to go somewhere, so it ends up in the part of the string touching m_1 (see Fig. 3.6). Therefore, m_1 goes down by a distance $2d$. In other words, $y_1 = -2y_2$, where y_1 and y_2 are measured relative to the initial locations of the masses. Taking two time derivatives of this statement gives our desired relation between a_1 and a_2 ,

$$a_1 = -2a_2. \quad (3.14)$$

Combining this with Eq. (3.13), we can now solve for a_1 , a_2 , and T . The result is

$$a_1 = g \frac{2m_2 - 4m_1}{4m_1 + m_2}, \quad a_2 = g \frac{2m_1 - m_2}{4m_1 + m_2}, \quad T = \frac{3m_1 m_2 g}{4m_1 + m_2}. \quad (3.15)$$

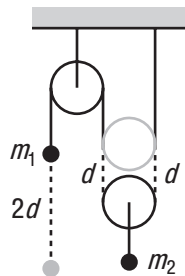


Fig. 3.6

REMARKS: There are all sorts of limits and special cases that we can check here. A couple are: (1) If $m_2 = 2m_1$, then Eq. (3.15) gives $a_1 = a_2 = 0$, and $T = m_1 g$. Everything is at rest. (2) If $m_2 \gg m_1$, then Eq. (3.15) gives $a_1 = 2g$, $a_2 = -g$, and $T = 3m_1 g$. In this case, m_2 is essentially in free fall, while m_1 gets yanked up with acceleration $2g$. The value of T is exactly what is needed to make the net force on m_1 equal to $m_1(2g)$, because $T - m_1 g = 3m_1 g - m_1 g = m_1(2g)$. You can check the case where $m_1 \gg m_2$.

For the more general case where there are N masses instead of two, the “conservation of string” statement is a single equation that relates all N accelerations. It is most easily obtained by imagining moving $N - 1$ of the masses, each by an arbitrary amount, and then seeing what happens to the last mass. Note that these arbitrary motions undoubtedly do *not* correspond to the actual motions of the masses. This is fine; the single “conservation of string” equation has nothing to do with the N $F = ma$ equations. The combination of all $N + 1$ equations is needed to constrain the motions down to a unique set. ♣

In the problems and exercises for this chapter, you will encounter some strange Atwood’s setups. But no matter how complicated they get, there are only two things you need to do to solve them, as mentioned above: write down the $F = ma$ equations for all the masses (which may involve relating the tensions in various strings), and then relate the accelerations of the masses, using “conservation of string.”

It may seem, with the angst it can bring,
That an Atwood’s machine’s a cruel thing.
But you just need to say
That F is ma ,
And use conservation of string!

3.3 Solving differential equations

Let's now consider the type of problem where we are *given* the force as a function of time, position, or velocity, and our task is to solve the $F = ma \equiv m\ddot{x}$ differential equation to find the position, $x(t)$, as a function of time.⁹ In what follows, we will develop a few techniques for solving differential equations. The ability to apply these techniques dramatically increases the number of systems we can understand.

It's also possible for the force F to be a function of higher derivatives of x , in addition to the quantities t , x , and $v \equiv \dot{x}$. But these cases don't arise much, so we won't worry about them. The $F = ma$ differential equation we want to solve is therefore (we'll just work in one dimension here)

$$m\ddot{x} = F(t, x, v). \quad (3.16)$$

In general, this equation cannot be solved exactly for $x(t)$.¹⁰ But for most of the problems we'll deal with, it can be solved. The problems we'll encounter will often fall into one of three special cases, namely, where F is a function of t only, or x only, or v only. In all of these cases, we must invoke the given initial conditions, $x_0 \equiv x(t_0)$ and $v_0 \equiv v(t_0)$, to obtain our final solutions. These initial conditions will appear in the limits of the integrals in the following discussion.¹¹

Note: You may want to just skim the following page and a half, and then refer back as needed. Don't try to memorize all the different steps. We present them only for completeness. The whole point here can basically be summarized by saying that sometimes you want to write $\ddot{x}$ as dv/dt , and sometimes you want to write it as $v dv/dx$ (see Eq. (3.20)). Then you "simply" have to separate variables and integrate. We'll go through the three special cases, and then we'll do some examples.

- F is a function of t only: $F = F(t)$.

Since $a = d^2x/dt^2$, we just need to integrate $F = ma$ twice to obtain $x(t)$. Let's do this in a very systematic way, to get used to the general procedure. First, write $F = ma$ as

$$m \frac{dv}{dt} = F(t). \quad (3.17)$$

⁹ In some setups, such as in Problem 3.11, the force isn't given, so you have to figure out what it is. But the main part of the problem is still solving the resulting differential equation.

¹⁰ You can always solve for $x(t)$ *numerically*, to any desired accuracy. This topic is discussed in Section 1.4.

¹¹ It is no coincidence that we need *two* initial conditions to completely specify the solution to our *second-order* (meaning the highest derivative of x that appears is the second one) $F = m\ddot{x}$ differential equation. It is a general result (which we'll just accept here) that the solution to an n th-order differential equation has n free parameters, which are determined by the initial conditions.

Then separate variables and integrate both sides to obtain¹²

$$m \int_{v_0}^{v(t)} dv' = \int_{t_0}^t F(t') dt'. \quad (3.18)$$

We have put primes on the integration variables so that we don't confuse them with the limits of integration, but in practice we usually don't bother with them. The integral of dv' is just v' , so Eq. (3.18) yields v as a function of t , that is, $v(t)$. We can then separate variables in $dx/dt \equiv v(t)$ and integrate to obtain

$$\int_{x_0}^{x(t)} dx' = \int_{t_0}^t v(t') dt'. \quad (3.19)$$

This yields x as a function of t , that is, $x(t)$. This procedure might seem like a cumbersome way to simply integrate something twice. That's because it is. But the technique proves more useful in the following case.

- F is a function of x only: $F = F(x)$.

We will use

$$a = \frac{dv}{dt} = \frac{dx}{dt} \frac{dv}{dx} = v \frac{dv}{dx} \quad (3.20)$$

to write $F = ma$ as

$$mv \frac{dv}{dx} = F(x). \quad (3.21)$$

Now separate variables and integrate both sides to obtain

$$m \int_{v_0}^{v(x)} v' dv' = \int_{x_0}^x F(x') dx'. \quad (3.22)$$

The integral of v' is $v'^2/2$, so the left-hand side involves the square of $v(x)$. Taking the square root, this gives v as a function of x , that is, $v(x)$. Separating variables in $dx/dt \equiv v(x)$ then yields

$$\int_{x_0}^{x(t)} \frac{dx'}{v(x')} = \int_{t_0}^t dt'. \quad (3.23)$$

Assuming that we can do the integral on the left-hand side, this equation gives t as a function of x . We can then (in principle) invert the result to obtain x as a function of t , that is, $x(t)$. The unfortunate thing about this case is that the integral in Eq. (3.23) might not be doable. And even if it is, it might not be possible to invert $t(x)$ to produce $x(t)$.

¹² If you haven't seen such a thing before, the act of multiplying both sides by the infinitesimal quantity dt might make you feel a bit uneasy. But it is in fact quite legal. If you wish, you can imagine working with the small (but not infinitesimal) quantities Δv and Δt , for which it is certainly legal to multiply both sides by Δt . Then you can take a discrete sum over many Δt intervals, and then finally take the limit $\Delta t \rightarrow 0$, which results in the integral in Eq. (3.18).

- F is a function of v only: $F = F(v)$.

Write $F = ma$ as

$$m \frac{dv}{dt} = F(v). \quad (3.24)$$

Separate variables and integrate both sides to obtain

$$m \int_{v_0}^{v(t)} \frac{dv'}{F(v')} = \int_{t_0}^t dt'. \quad (3.25)$$

Assuming that we can do this integral, it yields t as a function of v , and hence (in principle) v as a function of t , that is, $v(t)$. We can then integrate $dx/dt \equiv v(t)$ to obtain $x(t)$ from

$$\int_{x_0}^{x(t)} dx' = \int_{t_0}^t v(t') dt'. \quad (3.26)$$

Note: In this $F = F(v)$ case, if we want to find v as a function of x , $v(x)$, then we should write a as $v(dv/dx)$ and integrate

$$m \int_{v_0}^{v(x)} \frac{v' dv'}{F(v')} = \int_{x_0}^x dx'. \quad (3.27)$$

We can then obtain $x(t)$ from Eq. (3.23), if desired.

When dealing with the initial conditions, we have chosen to put them in the limits of integration above. If you wish, you can perform the integrals without any limits, and just tack on a constant of integration to your result. The constant is then determined by the initial conditions.

Again, as mentioned above, you do *not* have to memorize the above three procedures, because there are variations, depending on what you're given and what you want to solve for. All you have to remember is that $\ddot{x}$ can be written as either dv/dt or $v dv/dx$. One of these will get the job done (namely, the one that makes only two of the three variables, t , x , and v , appear in your differential equation). And then be prepared to separate variables and integrate as many times as needed.¹³

a is dv by dt .

Is this useful? There's no guarantee.

If it leads to "Oh, heck!"s,

Take dv by dx ,

And then write down its product with v .

¹³ We want only two of the variables to appear in the differential equation because the goal is to separate variables and integrate, and because equations have only two sides. If equations were triangles, it would be a different story.

Example (Gravitational force): A particle of mass m is subject to a constant force $F = -mg$. The particle starts at rest at height h . Because this constant force falls into all of the above three categories, we should be able to solve for $y(t)$ in two ways:

- (a) Find $y(t)$ by writing a as dv/dt .
- (b) Find $y(t)$ by writing a as $v\,dv/dy$.

Solution:

- (a) $F = ma$ gives $dv/dt = -g$. Multiplying by dt and integrating yields $v = -gt + A$, where A is a constant of integration.¹⁴ The initial condition $v(0) = 0$ gives $A = 0$. Therefore, $dy/dt = -gt$. Multiplying by dt and integrating yields $y = -gt^2/2 + B$. The initial condition $y(0) = h$ gives $B = h$. Therefore,

$$y = h - \frac{1}{2}gt^2. \quad (3.28)$$

- (b) $F = ma$ gives $v\,dv/dy = -g$. Separating variables and integrating yields $v^2/2 = -gy + C$. The initial condition $v(h) = 0$ gives $v^2/2 = -gy + gh$. Therefore, $v \equiv dy/dt = -\sqrt{2g(h-y)}$. We have chosen the negative square root because the particle is falling. Separating variables gives

$$\int \frac{dy}{\sqrt{h-y}} = -\sqrt{2g} \int dt. \quad (3.29)$$

This yields $2\sqrt{h-y} = \sqrt{2g}\,t$, where we have used the initial condition $y(0) = h$. Hence, $y = h - gt^2/2$, as in part (a). In part (b) here, we essentially derived conservation of energy, as we'll see in Chapter 5.

Example (Dropped ball): A beach ball is dropped from rest at height h . Assume that the drag force¹⁵ from the air takes the form of $F_d = -\beta v$. Find the velocity and height as a function of time.

Solution: For simplicity in future formulas, let's write the drag force as $F_d = -\beta v \equiv -m\alpha v$ (otherwise we'd have a bunch of $1/m$'s floating around). Taking upward to be the positive y direction, the force on the ball is

$$F = -mg - m\alpha v. \quad (3.30)$$

¹⁴ We'll do this example by adding on constants of integration which are then determined by the initial conditions. We'll do the following example by putting the initial conditions in the limits of integration.

¹⁵ The drag force is roughly proportional to v as long as the speed is fairly small (say, less than 10 m/s). For large speeds (say, greater than 100 m/s), the drag force is roughly proportional to v^2 . But these approximate cutoffs depend on various things, and in any event there is a messy transition region between the two cases.

Note that v is negative here, because the ball is falling, so the drag force points upward, as it should. Writing $F = m dv/dt$ and separating variables gives

$$\int_0^{v(t)} \frac{dv'}{g + \alpha v'} = - \int_0^t dt'. \quad (3.31)$$

Integration yields $\ln(1 + \alpha v/g) = -\alpha t$. Exponentiation then gives

$$v(t) = -\frac{g}{\alpha} (1 - e^{-\alpha t}). \quad (3.32)$$

Writing $dy/dt \equiv v(t)$, and then separating variables and integrating to obtain $y(t)$, yields

$$\int_h^{y(t)} dy' = -\frac{g}{\alpha} \int_0^t (1 - e^{-\alpha t'}) dt'. \quad (3.33)$$

Therefore,

$$y(t) = h - \frac{g}{\alpha} \left(t - \frac{1}{\alpha} (1 - e^{-\alpha t}) \right). \quad (3.34)$$

REMARKS:

1. Let's look at some limiting cases. If t is very small (more precisely, if $\alpha t \ll 1$), then we can use $e^{-x} \approx 1 - x + x^2/2$ to make approximations to leading order in t . You can show that Eq. (3.32) gives $v(t) \approx -gt$. This makes sense, because the drag force is negligible at the start, so the ball is essentially in freefall. And likewise you can show that Eq. (3.34) gives $y(t) \approx h - gt^2/2$, which is again the freefall result.

We can also look at large t . In this case, $e^{-\alpha t}$ is essentially equal to zero, so Eq. (3.32) gives $v(t) \approx -g/\alpha$. (This is the “terminal velocity.” Its value makes sense, because it is the velocity for which the total force, $-mg - m\alpha v$, vanishes.) And Eq. (3.34) gives $y(t) \approx h - (g/\alpha)t + g/\alpha^2$. Interestingly, we see that for large t , g/α^2 is the distance our ball lags behind another ball that started out already at the terminal velocity, $-g/\alpha$.

2. You might think that the velocity in Eq. (3.32) doesn't depend on m , since no m 's appear. However, there is an m hidden in α . The quantity α (which we introduced just to make our formulas look a little nicer) was defined by $F_d = -\beta v \equiv -m\alpha v$. But the quantity $\beta \equiv m\alpha$ is roughly proportional to the cross-sectional area, A , of the ball. Therefore, $\alpha \propto A/m$. Two balls of the same size, one made of lead and one made of styrofoam, have the same A but different m 's. So their α 's are different, and they fall at different rates.

If we have a solid ball with density ρ and radius r , then $\alpha \propto A/m \propto r^2/(\rho r^3) = 1/\rho r$. For large dense objects in a thin medium such as air, the quantity α is small, so the drag effects are not very noticeable over short times (because if we include the next term in the expansion for v , we obtain $v(t) \approx -gt + \alpha g t^2/2$). Large dense objects therefore all fall at roughly the same rate, with an acceleration essentially equal to g . But if the air were much thicker, then all the α 's would be larger, and maybe it would have taken Galileo a bit longer to come to his conclusions.

What would you have thought, Galileo,

If instead you dropped cows and did say, “Oh!

To lessen the sound
 Of the moos from the ground,
 They should fall not through air, but through mayo!”¹⁶ ♣

3.4 Projectile motion

Consider a ball thrown through the air, not necessarily vertically. We will neglect air resistance in the following discussion. Things get a bit more complicated when this is included, as Exercise 3.53 demonstrates.

Let x and y be the horizontal and vertical positions, respectively. The force in the x direction is $F_x = 0$, and the force in the y direction is $F_y = -mg$. So $\mathbf{F} = m\mathbf{a}$ gives

$$\ddot{x} = 0, \quad \text{and} \quad \ddot{y} = -g. \quad (3.35)$$

Note that these two equations are “decoupled.” That is, there is no mention of y in the equation for $\ddot{x}$, and vice versa. The motions in the x and y directions are therefore completely independent. The classic demonstration of the independence of the x and y motions is the following. Fire a bullet horizontally (or, preferably, just imagine firing a bullet horizontally), and at the same time drop a bullet from the height of the gun. Which bullet will hit the ground first? (Neglect air resistance, the curvature of the earth, etc.) The answer is that they will hit the ground at the same time, because the effect of gravity on the two y motions is exactly the same, independent of what is going on in the x direction.

If the initial position and velocity are (X, Y) and (V_x, V_y) , then we can easily integrate Eq. (3.35) to obtain

$$\dot{x}(t) = V_x, \quad \text{and} \quad \dot{y}(t) = V_y - gt. \quad (3.36)$$

Integrating again gives

$$x(t) = X + V_x t, \quad \text{and} \quad y(t) = Y + V_y t - \frac{1}{2}gt^2. \quad (3.37)$$

These equations for the speeds and positions are all you need to solve a projectile problem.

¹⁶ It’s actually much more likely that Galileo obtained his “all objects fall at the same rate in a vacuum” result by rolling balls down planes than by dropping balls from the Tower of Pisa; see Adler and Coulter (1978). So I suppose this limerick is relevant only in the approximation of the proverbial spherical cow.

Example (Throwing a ball):

- (a) For a given initial speed, at what inclination angle should a ball be thrown so that it travels the maximum horizontal distance by the time it returns to the ground? Assume that the ground is horizontal, and that the ball is released from ground level.
- (b) What is the optimal angle if the ground is sloped upward at an angle β (or downward, if β is negative)?

Solution:

- (a) Let the inclination angle be θ , and let the initial speed be V . Then the horizontal speed is always $V_x = V \cos \theta$, and the initial vertical speed is $V_y = V \sin \theta$. The first thing we need to do is find the time t in the air. We know that the vertical speed is zero at time $t/2$, because the ball is moving horizontally at the highest point. So the second of Eqs. (3.36) gives $V_y = g(t/2)$. Therefore, $t = 2V_y/g$.¹⁷ The first of Eqs. (3.37) tells us that the horizontal distance traveled is $d = V_x t$. Using $t = 2V_y/g$ in this gives

$$d = \frac{2V_x V_y}{g} = \frac{V^2 (2 \sin \theta \cos \theta)}{g} = \frac{V^2 \sin 2\theta}{g}. \quad (3.38)$$

The $\sin 2\theta$ factor has a maximum at $\theta = \pi/4$. The maximum horizontal distance traveled is then $d_{\max} = V^2/g$.

REMARK: For $\theta = \pi/4$, you can show that the maximum height achieved is $V^2/4g$. This is half the maximum height of $V^2/2g$ (as you can show) if the ball is thrown straight up. Note that any possible distance you might want to find in this problem must be proportional to V^2/g , by dimensional analysis. The only question is what the numerical factor is. ♣

- (b) As in part (a), the first thing we need to do is find the time t in the air. If the ground is sloped at an angle β , then the equation for the line of the ground is $y = (\tan \beta)x$. The path of the ball is given in terms of t by

$$x = (V \cos \theta)t, \quad \text{and} \quad y = (V \sin \theta)t - \frac{1}{2}gt^2, \quad (3.39)$$

where θ is the angle of the throw, as measured with respect to the horizontal (not the ground). We must solve for the t that makes $y = (\tan \beta)x$, because this gives the time when the path of the ball intersects the line of the ground. Using Eq. (3.39), we find that $y = (\tan \beta)x$ when

$$t = \frac{2V}{g}(\sin \theta - \tan \beta \cos \theta). \quad (3.40)$$

¹⁷ Alternatively, the time of flight can be found from the second of Eqs. (3.37), which says that the ball returns to the ground when $V_y t = gt^2/2$. We will have to use this second strategy in part (b), where the trajectory is not symmetric around the maximum.

(There is, of course, also the solution $t = 0$.) Plugging this into the expression for x in Eq. (3.39) gives

$$x = \frac{2V^2}{g} (\sin \theta \cos \theta - \tan \beta \cos^2 \theta). \quad (3.41)$$

We must now maximize this value for x , which is equivalent to maximizing the distance along the slope. Setting the derivative with respect to θ equal to zero, and using the double-angle formulas, $\sin 2\theta = 2 \sin \theta \cos \theta$ and $\cos 2\theta = \cos^2 \theta - \sin^2 \theta$, we find $\tan \beta = -\cot 2\theta$. This can be rewritten as $\tan \beta = -\tan(\pi/2 - 2\theta)$. Therefore, $\beta = -(\pi/2 - 2\theta)$, so we have

$$\theta = \frac{1}{2} \left(\beta + \frac{\pi}{2} \right). \quad (3.42)$$

In other words, the throwing angle should bisect the angle between the ground and the vertical.

REMARKS:

1. For $\beta \approx \pi/2$, we have $\theta \approx \pi/2$, as should be the case. For $\beta = 0$, we have $\theta = \pi/4$, as we found in part (a). And for $\beta \approx -\pi/2$, we have $\theta \approx 0$, which makes sense.
2. A quicker method of obtaining the time in Eq. (3.40) is the following. Consider the set of tilted axes parallel and perpendicular to the ground; let these be the x' and y' axes, respectively. The initial velocity in the y' direction is $V \sin(\theta - \beta)$, and the acceleration in this direction is $g \cos \beta$. The time in the air is twice the time it takes the ball to reach the maximum “height” above the ground (measured in the y' direction), which occurs when the velocity in the y' direction is instantaneously zero. The total time is therefore $2V \sin(\theta - \beta)/(g \cos \beta)$, which you can show is equivalent to the time in Eq. (3.40). Note that the $g \sin \beta$ acceleration in the x' direction is irrelevant in calculating this time. In the present example, using these tilted axes doesn’t save a huge amount of time, but in some situations (see Exercise 3.50) the tilted axes can save you a lot of grief.
3. An interesting fact about the motion of the ball in the maximum-distance case is that the initial and final velocities are perpendicular to each other. The demonstration of this is the task of Problem 3.16.
4. Substituting the value of θ from Eq. (3.42) into Eq. (3.41), you can show (after a bit of algebra) that the maximum distance traveled along the tilted ground is

$$d = \frac{x}{\cos \beta} = \frac{V^2/g}{1 + \sin \beta}. \quad (3.43)$$

Solving for V , we have $V^2 = g(d + d \sin \beta)$. This can be interpreted as saying that the minimum speed at which a ball must be thrown in order to pass over a wall of height h , at a distance L away on level ground, is given by $V^2 = g(\sqrt{L^2 + h^2} + h)$. This checks in the limits of $h \rightarrow 0$ and $L \rightarrow 0$.

5. A compilation of many other projectile results can be found in Buckmaster (1985). ♣

Along with the bullet example mentioned above, another classic example of the independence of the x and y motions is the “hunter and monkey” problem. In it, a hunter aims an arrow (a toy one, of course) at a monkey hanging from a branch in a tree. The monkey, thinking he’s being clever, tries to avoid the arrow by letting go of the branch right when he sees the arrow released. The unfortunate consequence of this action is that he in fact *will* get hit, because gravity acts on both him and the arrow in the same way; they both fall the same distance relative to where they would have been if there were no gravity. And the monkey *would* get hit in such a case, because the arrow is initially aimed at him. You can work this out in Exercise 3.44, in a more peaceful setting involving fruit.

If a monkey lets go of a tree,
The arrow will hit him, you see,
Because both heights are pared
By a half gt^2
From what they would be with no g .

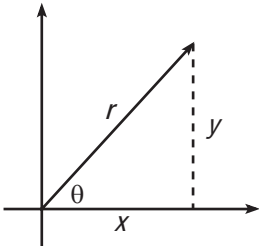


Fig. 3.7

3.5 Motion in a plane, polar coordinates

When dealing with problems where the motion lies in a plane, it is often convenient to work with polar coordinates, r and θ . These are related to the Cartesian coordinates by (see Fig. 3.7)

$$x = r \cos \theta, \quad \text{and} \quad y = r \sin \theta. \quad (3.44)$$

Depending on the problem, either Cartesian or polar coordinates are easier to use. It is usually clear from the setup which is better. For example, if the problem involves circular motion, then polar coordinates are a good bet. But to use polar coordinates, we need to know what Newton’s second law looks like when written in terms of them. Therefore, the goal of the present section is to determine what $\mathbf{F} = m\mathbf{a} \equiv m\ddot{\mathbf{r}}$ looks like when written in terms of polar coordinates.

At a given position $\mathbf{r}$ in the plane, the basis vectors in polar coordinates are $\hat{\mathbf{r}}$, which is a unit vector pointing in the radial direction; and $\hat{\boldsymbol{\theta}}$, which is a unit vector pointing in the counterclockwise tangential direction. In polar coordinates, a general vector may be written as

$$\mathbf{r} = r\hat{\mathbf{r}}. \quad (3.45)$$

Since the goal of this section is to find $\ddot{\mathbf{r}}$, we must, in view of Eq. (3.45), get a handle on the time derivative of $\hat{\mathbf{r}}$. And we’ll eventually need the derivative of $\hat{\boldsymbol{\theta}}$, too. In contrast with the fixed Cartesian basis vectors ($\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$), the polar basis vectors ($\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$) change as a point moves around in the plane. We can

find $\dot{\mathbf{r}}$ and $\dot{\boldsymbol{\theta}}$ in the following way. In terms of the Cartesian basis, Fig. 3.8 shows that

$$\begin{aligned}\hat{\mathbf{r}} &= \cos \theta \hat{\mathbf{x}} + \sin \theta \hat{\mathbf{y}}, \\ \hat{\boldsymbol{\theta}} &= -\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{y}}.\end{aligned}\tag{3.46}$$

Taking the time derivative of these equations gives

$$\begin{aligned}\dot{\hat{\mathbf{r}}} &= -\sin \theta \dot{\theta} \hat{\mathbf{x}} + \cos \theta \dot{\theta} \hat{\mathbf{y}}, \\ \dot{\hat{\boldsymbol{\theta}}} &= -\cos \theta \dot{\theta} \hat{\mathbf{x}} - \sin \theta \dot{\theta} \hat{\mathbf{y}}.\end{aligned}\tag{3.47}$$

Using Eq. (3.46), we arrive at the nice clean expressions,

$$\dot{\hat{\mathbf{r}}} = \dot{\theta} \hat{\boldsymbol{\theta}}, \quad \text{and} \quad \dot{\hat{\boldsymbol{\theta}}} = -\dot{\theta} \hat{\mathbf{r}}.\tag{3.48}$$

These relations are fairly evident if you look at what happens to the basis vectors as $\mathbf{r}$ moves a tiny distance in the tangential direction. Note that the basis vectors do not change as $\mathbf{r}$ moves in the radial direction. We can now start differentiating Eq. (3.45). One derivative gives (yes, the product rule works fine here)

$$\dot{\mathbf{r}} = \dot{r} \hat{\mathbf{r}} + r \dot{\hat{\mathbf{r}}} = \dot{r} \hat{\mathbf{r}} + r \dot{\theta} \hat{\boldsymbol{\theta}}.\tag{3.49}$$

This makes sense, because $\dot{r}$ is the velocity in the radial direction, and $r\dot{\theta}$ is the velocity in the tangential direction, often written as $r\omega$ (where $\omega \equiv \dot{\theta}$ is the angular velocity, or “angular frequency”).¹⁸ Differentiating Eq. (3.49) then gives

$$\begin{aligned}\ddot{\mathbf{r}} &= \ddot{r} \hat{\mathbf{r}} + \dot{r} \dot{\hat{\mathbf{r}}} + \dot{r} \dot{\theta} \hat{\boldsymbol{\theta}} + r \ddot{\theta} \hat{\boldsymbol{\theta}} + r \dot{\theta} \dot{\hat{\boldsymbol{\theta}}} \\ &= \ddot{r} \hat{\mathbf{r}} + \dot{r}(\dot{\theta} \hat{\boldsymbol{\theta}}) + \dot{r} \dot{\theta} \hat{\boldsymbol{\theta}} + r \ddot{\theta} \hat{\boldsymbol{\theta}} + r \dot{\theta}(-\dot{\theta} \hat{\mathbf{r}}) \\ &= (\ddot{r} - r\dot{\theta}^2) \hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \hat{\boldsymbol{\theta}}.\end{aligned}\tag{3.50}$$

Finally, equating $m\ddot{\mathbf{r}}$ with $\mathbf{F} \equiv F_r \hat{\mathbf{r}} + F_\theta \hat{\boldsymbol{\theta}}$ gives the radial and tangential forces as

$$\begin{aligned}F_r &= m(\ddot{r} - r\dot{\theta}^2), \\ F_\theta &= m(r\ddot{\theta} + 2\dot{r}\dot{\theta}).\end{aligned}\tag{3.51}$$

(See Exercise 3.67 for a slightly different derivation of these equations.) Let’s look at each of the four terms on the right-hand sides of Eqs. (3.51).

¹⁸ For $r\dot{\theta}$ to be the tangential velocity, we must measure θ in radians and not degrees. Then $r\dot{\theta}$ is by definition the position along the circumference, so $r\dot{\theta}$ is the velocity along the circumference.

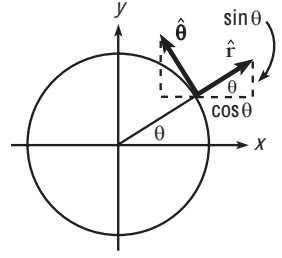


Fig. 3.8

- The $m\ddot{r}$ term is quite intuitive. For radial motion, it simply states that $F = ma$ along the radial direction.
- The $mr\ddot{\theta}$ term is also quite intuitive. For circular motion, it states that $F = ma$ along the tangential direction, because $r\ddot{\theta}$ is the second derivative of the distance $r\theta$ along the circumference.
- The $-mr\dot{\theta}^2$ term is also fairly clear. For circular motion, it says that the radial force is $-m(r\dot{\theta})^2/r = -mv^2/r$, which is the familiar force that causes the centripetal acceleration, v^2/r . See Problem 3.20 for an alternate (and quicker) derivation of this v^2/r result.
- The $2m\dot{r}\dot{\theta}$ term isn't so obvious. It is associated with the *Coriolis* force. There are various ways to look at this term. One is that it exists in order to keep angular momentum conserved. We'll have a great deal to say about the Coriolis force in Chapter 10.

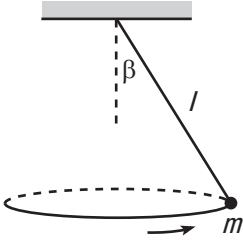


Fig. 3.9

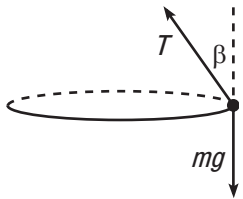


Fig. 3.10

Example (Circular pendulum): A mass hangs from a massless string of length ℓ . Conditions have been set up so that the mass swings around in a horizontal circle, with the string making a constant angle β with the vertical (see Fig. 3.9). What is the angular frequency, ω , of this motion?

Solution: The mass travels in a circle, so the horizontal radial force must be $F_r = mr\dot{\theta}^2 \equiv mr\omega^2$ (with $r = \ell \sin \beta$), directed radially inward. The forces on the mass are the tension in the string, T , and gravity, mg (see Fig. 3.10). There is no acceleration in the vertical direction, so $F = ma$ in the vertical and radial directions gives, respectively,

$$T \cos \beta - mg = 0, \quad (3.52)$$

$$T \sin \beta = m(\ell \sin \beta)\omega^2.$$

Solving for ω gives

$$\omega = \sqrt{\frac{g}{\ell \cos \beta}}. \quad (3.53)$$

If $\beta \approx 90^\circ$, then $\omega \rightarrow \infty$, which makes sense. And if $\beta \approx 0$, then $\omega \approx \sqrt{g/\ell}$, which happens to equal the frequency of a plane pendulum of length ℓ . The task of Exercise 3.60 is to explain why.



Fig. 3.11

3.6 Problems

Section 3.2: Free-body diagrams

3.1. Atwood's machine *

A massless pulley hangs from a fixed support. A massless string connecting two masses, m_1 and m_2 , hangs over the pulley (see Fig. 3.11). Find the acceleration of the masses and the tension in the string.

3.2. Double Atwood's machine **

A double Atwood's machine is shown in Fig. 3.12, with masses m_1 , m_2 , and m_3 . Find the accelerations of the masses.

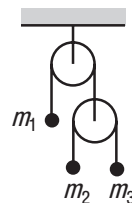


Fig. 3.12

3.3. Infinite Atwood's machine ***

Consider the infinite Atwood's machine shown in Fig. 3.13. A string passes over each pulley, with one end attached to a mass and the other end attached to another pulley. All the masses are equal to m , and all the pulleys and strings are massless. The masses are held fixed and then simultaneously released. What is the acceleration of the top mass? (You may define this infinite system as follows. Consider it to be made of N pulleys, with a nonzero mass replacing what would have been the $(N + 1)$ th pulley. Then take the limit as $N \rightarrow \infty$.)

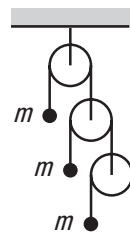


Fig. 3.13

3.4. Line of pulleys *

$N + 2$ equal masses hang from a system of pulleys, as shown in Fig. 3.14. What are the accelerations of all the masses?

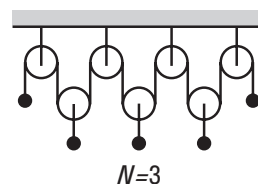


Fig. 3.14

3.5. Ring of pulleys **

Consider the system of pulleys shown in Fig. 3.15. The string (which is a loop with no ends) hangs over N fixed pulleys that circle around the underside of a ring. N masses, $m_1, m_2, \dots, m_N$, are attached to N pulleys that hang on the string. What are the accelerations of all the masses?

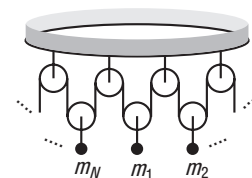


Fig. 3.15

3.6. Sliding down a plane **

- A block starts at rest and slides down a frictionless plane inclined at an angle θ . What should θ be so that the block travels a given horizontal distance in the minimum amount of time?
- Same question, but now let there be a coefficient of kinetic friction μ between the block and the plane.

3.7. Sliding sideways on a plane ***

A block is placed on a plane inclined at an angle θ . The coefficient of friction between the block and the plane is $\mu = \tan \theta$. The block is given a kick so that it initially moves with speed V horizontally along the plane (that is, in the direction perpendicular to the direction pointing straight down the plane). What is the speed of the block after a very long time?

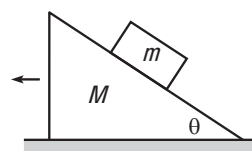


Fig. 3.16

3.8. Moving plane ***

A block of mass m is held motionless on a frictionless plane of mass M and angle of inclination θ (see Fig. 3.16). The plane rests on a frictionless horizontal surface. The block is released. What is the horizontal acceleration of the plane?

Section 3.3: Solving differential equations

3.9. Exponential force *

A particle of mass m is subject to a force $F(t) = ma_0 e^{-bt}$. The initial position and speed are zero. Find $x(t)$.

3.10. $-kx$ force **

A particle of mass m is subject to a force $F(x) = -kx$, with $k > 0$. The initial position is x_0 , and the initial speed is zero. Find $x(t)$.

3.11. Falling chain **

A chain with length ℓ is held stretched out on a frictionless horizontal table, with a length y_0 hanging down through a hole in the table. The chain is released. As a function of time, find the length that hangs down through the hole (don't bother with t after the chain loses contact with the table). Also, find the speed of the chain right when it loses contact with the table.¹⁹

3.12. Throwing a beach ball ***

A beach ball is thrown upward with initial speed v_0 . Assume that the drag force from the air is $F_d = -m\alpha v$. What is the speed of the ball, v_f , right before it hits the ground? (An implicit equation is sufficient.) Does the ball spend more time or less time in the air than it would if it were thrown in vacuum?

3.13. Balancing a pencil ***

Consider a pencil that stands upright on its tip and then falls over. Let's idealize the pencil as a mass m sitting at the end of a massless rod of length ℓ .²⁰

- (a) Assume that the pencil makes an initial (small) angle θ_0 with the vertical, and that its initial angular speed is ω_0 . The angle will eventually become large, but while it is small (so that $\sin \theta \approx \theta$), what is θ as a function of time?
- (b) You might think that it should be possible (theoretically, at least) to make the pencil balance for an arbitrarily long time, by making the initial θ_0 and ω_0 sufficiently small. However, it turns out that due to Heisenberg's uncertainty principle (which puts a constraint on how well we can know the position and momentum of

¹⁹ Assume that the hole is actually a short frictionless tube bent into a gradual right angle, so that the chain's horizontal momentum doesn't cause it to overshoot the hole. For a description of what happens in a similar problem when this constraint is removed, see Calkin (1989).

²⁰ It actually involves only a trivial modification to do the problem correctly using the moment of inertia and the torque. But the point-mass version is quite sufficient for the present purposes.

a particle), it is impossible to balance the pencil for more than a certain amount of time. The point is that you can't be sure that the pencil is initially both at the top *and* at rest. The goal of this problem is to be quantitative about this. The time limit is sure to surprise you.

Without getting into quantum mechanics, let's just say that the uncertainty principle says (up to factors of order 1) that $\Delta x \Delta p \geq \hbar$, where $\hbar = 1.05 \cdot 10^{-34}$ J s is Planck's constant. The implications of this are somewhat vague, but we'll just take it to mean that the initial conditions satisfy $(\ell \theta_0)(m \ell \omega_0) \geq \hbar$. With this constraint, your task is to find the maximum time it can take your $\theta(t)$ solution in part (a) to become of order 1. In other words, determine (roughly) the maximum time the pencil can balance. Assume $m = 0.01$ kg, and $\ell = 0.1$ m.

Section 3.4: Projectile motion

3.14. Maximum trajectory area *

A ball is thrown at speed v from zero height on level ground. At what angle should it be thrown so that the area under the trajectory is maximum?

3.15. Bouncing ball *

A ball is thrown straight upward so that it reaches a height h . It falls down and bounces repeatedly. After each bounce, it returns to a certain fraction f of its previous height. Find the total distance traveled, and also the total time, before it comes to rest. What is its average speed?

3.16. Perpendicular velocities **

In the maximum-distance case in part (b) of the example in Section 3.4, show that the initial and final velocities are perpendicular to each other.²¹

3.17. Throwing a ball from a cliff **

A ball is thrown with speed v from the edge of a cliff of height h . At what inclination angle should it be thrown so that it travels the maximum horizontal distance? What is this maximum distance? Assume that the ground below the cliff is horizontal.

²¹ You can grind through this problem and explicitly find the final angle, but there's a quicker way. This quicker method makes use of the conservation-of-energy statement that the difference in the squares of the initial and final speeds depends only on the change in height (the relation happens to be $v_i^2 - v_f^2 = 2gh$, but you don't need this actual expression). *Hint*: Consider the reverse path.

3.18. Redirected motion **

A ball is dropped from rest at height h above level ground, and it bounces off a surface at height y (with no loss in speed). The surface is inclined so that the ball bounces off at an angle θ with respect to the horizontal. What should y and θ be so that the ball travels the maximum horizontal distance by the time it hits the ground?

3.19. Maximum trajectory length ***

A ball is thrown at speed v from zero height on level ground. Let θ_0 be the angle at which the ball should be thrown so that the length of the trajectory is maximum. Show that θ_0 satisfies

$$\sin \theta_0 \ln \left(\frac{1 + \sin \theta_0}{\cos \theta_0} \right) = 1. \quad (3.54)$$

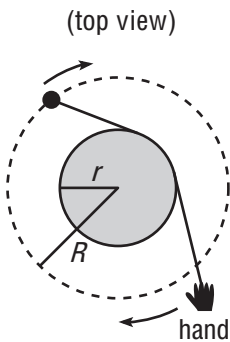
You can show numerically that $\theta_0 \approx 56.5^\circ$.

*Section 3.5: Motion in a plane, polar coordinates***3.20. Centripetal acceleration ***

Show that the magnitude of the acceleration of a particle moving in a circle at constant speed is v^2/r . Do this by drawing the position and velocity vectors at two nearby times, and then making use of some similar triangles.

3.21. Vertical acceleration **

A bead rests at the top of a fixed frictionless hoop of radius R that lies in a vertical plane. The bead is given a tiny push so that it slides down and around the hoop. At what points on the hoop is the bead's acceleration vertical?²² What is this vertical acceleration? *Note:* We haven't studied conservation of energy yet, but use the fact that the bead's speed after it has fallen a height h is given by $v = \sqrt{2gh}$.

**3.22. Circling around a pole ****

A mass, which is free to move on a horizontal frictionless surface, is attached to one end of a massless string that wraps partially around a frictionless vertical pole of radius r (see the top view in Fig. 3.17). You hold on to the other end of the string. At $t = 0$, the mass has speed v_0 in the tangential direction along the dotted circle of radius R shown. Your task is to pull on the string so that the mass keeps moving along the

Fig. 3.17

²² One such point is the bottom of the hoop. Another point is technically the top, where $a \approx 0$. Find the other two more interesting points (one on each side).

dotted circle. You are required to do this in such a way that the string remains in contact with the pole at all times. (You will have to move your hand around the pole, of course.) What is the speed of the mass as a function of time? There is a special value of the time; what is it and why is it special?

3.23. A force $F_\theta = m\dot{r}\dot{\theta}$ **

Consider a particle that feels an angular force only, of the form $F_\theta = m\dot{r}\dot{\theta}$. Show that $\dot{r} = \sqrt{A \ln r + B}$, where A and B are constants of integration, determined by the initial conditions. (There's nothing all that physical about this force. It simply makes the $F = ma$ equations solvable.)

3.24. Free particle ***

Consider a free particle in a plane. With Cartesian coordinates, it is easy to use $F = ma$ to show that the particle moves in a straight line. The task of this problem is to demonstrate this result in a much more cumbersome way, using polar coordinates and Eq. (3.51). More precisely, show that $\cos \theta = r_0/r$ for a free particle, where r_0 is the radius at closest approach to the origin, and θ is measured with respect to this radius.

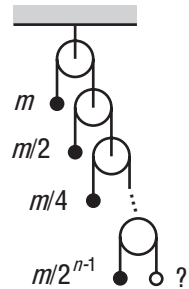


Fig. 3.18

3.7 Exercises

Section 3.2: Free-body diagrams

3.25. A peculiar Atwood's machine *

- The Atwood's machine in Fig. 3.18 consists of n masses, $m, m/2, m/4, \dots, m/2^{n-1}$. All the pulleys and strings are massless. Put a mass $m/2^{n-1}$ at the free end of the bottom string. What are the accelerations of all the masses?
- Remove the mass $m/2^{n-1}$ (which was arbitrarily small, for very large n) that was attached in part (a). What are the accelerations of all the masses, now that you've removed this infinitesimal piece?

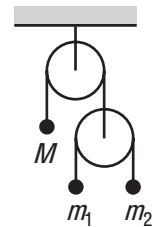


Fig. 3.19

3.26. Keeping the mass still *

In the Atwood's machine in Fig. 3.19, what should M be, in terms of m_1 and m_2 , so that it doesn't move?

3.27. Atwood's 1 *

Consider the Atwood's machine in Fig. 3.20. It consists of three pulleys, a short piece of string connecting one mass to the bottom pulley, and a continuous long piece of string that wraps twice around the bottom

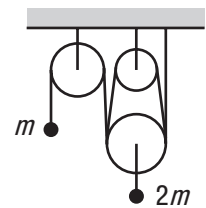


Fig. 3.20

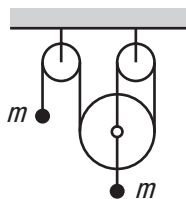


Fig. 3.21

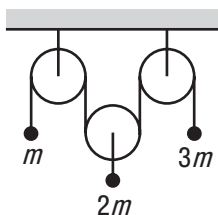


Fig. 3.22

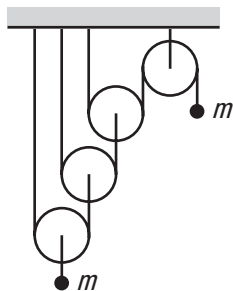


Fig. 3.23

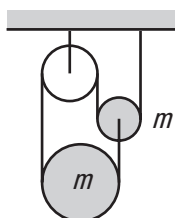


Fig. 3.24

side of the bottom pulley, and once around the top side of the top two pulleys. The two masses are m and $2m$. Assume that the parts of the string connecting the pulleys are essentially vertical. Find the accelerations of the masses.

3.28. Atwood's 2 *

Consider the Atwood's machine in Fig. 3.21, with two masses m . The axle of the bottom pulley has two string ends attached to it, as shown. Find the accelerations of the masses.

3.29. Atwood's 3 *

Consider the Atwood's machine in Fig. 3.22, with masses m , $2m$, and $3m$. Find the accelerations of the masses.

3.30. Atwood's 4 **

Consider the Atwood's machine in Fig. 3.23 (and also on the front cover). If the number of pulleys that have string passing beneath them is N instead of the 3 shown, find the accelerations of the masses.

3.31. Atwood's 5 **

Consider the Atwood's machine in Fig. 3.24. The two shaded pulleys have mass m , and the string slides *frictionlessly* along all of the pulleys (so you don't have to worry about any rotational motion). Find the accelerations of the two shaded pulleys.

3.32. Atwood's 6 **

Consider the Atwood's machine in Fig. 3.25. Find the accelerations of the masses. (This is a strange one.)

3.33. Accelerating plane **

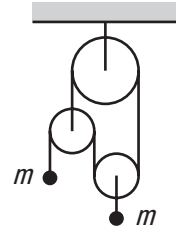
A block of mass m rests on a plane inclined at an angle θ . The coefficient of static friction between the block and the plane is μ . The plane is accelerated to the right with acceleration a (which may be negative); see Fig. 3.26. For what range of a does the block remain at rest with respect to the plane? In terms of μ , there are two special values of θ ; what are they, and why are they special?

3.34. Accelerating cylinders **

Three identical cylinders are arranged in a triangle as shown in Fig. 3.27, with the bottom two lying on the ground. The ground and the cylinders are frictionless. You apply a constant horizontal force (directed to the right) on the left cylinder. Let a be the acceleration you give to the system. For what range of a will all three cylinders remain in contact with each other?

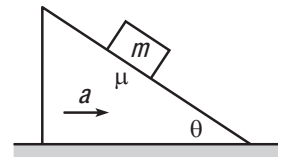
3.35. Leaving the sphere **

A small mass rests on top of a fixed sphere of radius R . The coefficient of friction is μ . The mass is given a sideways kick that produces an initial angular speed ω_0 . Let θ be the angle down from the top of the sphere. In terms of θ and its derivatives, what is the tangential $F = ma$ equation? Depending on the value of ω_0 , the mass either comes to rest on the sphere or flies off it. If $g = 10 \text{ m/s}^2$, $R = 1 \text{ m}$, and $\mu = 1$, write a program to numerically determine the minimum ω_0 for which the mass leaves the sphere. For this cutoff case, give the angle at which the mass loses contact, and describe (roughly) what the plot of $\dot{\theta}$ versus θ looks like. See Prior and Mele (2007) for the exact solution for $\dot{\theta}$ in terms of θ .

**Fig. 3.25****3.36. Comparing the times *****

A block of mass m is projected up along the surface of a plane inclined at an angle θ . The initial speed is v_0 , and the coefficients of static and kinetic friction are both equal to μ . The block reaches a highest point and then slides back down to the starting point.

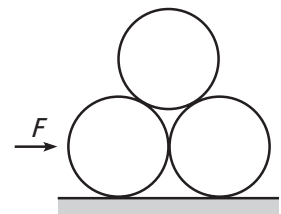
- Show that for the block to in fact slide back down instead of remaining at rest at the highest point, $\tan \theta$ must be greater than μ .
- Assuming that $\tan \theta > \mu$, is the total up and down time longer or shorter than the total time it would take if the plane were frictionless? Or does the answer depend on what θ and μ are?
- Assuming that $\tan \theta > \mu$, show that for a given θ , the value of μ that yields the minimum total time is given by $\mu \approx (0.397) \tan \theta$. (You will need to solve something numerically.) This minimum time turns out to be about 90% of the time it would take if the plane were frictionless.

**Fig. 3.26***Section 3.3: Solving differential equations***3.37. $-bv^2$ force ***

A particle of mass m is subject to a force $F(v) = -bv^2$. The initial position is zero, and the initial speed is v_0 . Find $x(t)$.

3.38. kx force **

A particle of mass m is subject to a force $F(x) = kx$, with $k > 0$. The initial position is x_0 , and the initial speed is zero. Find $x(t)$.

**Fig. 3.27***Section 3.4: Projectile motion***3.39. Equal distances ***

At what angle should a ball be thrown so that its maximum height equals the horizontal distance traveled?

3.40. Redirected motion *

A ball is dropped from rest at height h . At height y , it bounces off a surface with no loss in speed. The surface is inclined at 45° , so the ball bounces off horizontally. What should y be so that the ball travels the maximum horizontal distance? What is this maximum distance?

3.41. Throwing in the wind *

A ball is thrown horizontally to the right, from the top of a vertical cliff of height h . A wind blows horizontally to the left, and assume (simplistically) that the effect of the wind is to provide a constant force to the left, equal in magnitude to the weight of the ball. How fast should the ball be thrown so that it lands at the foot of the cliff?

3.42. Throwing in the wind again *

A ball is thrown eastward across level ground. A wind blows horizontally to the east, and assume (simplistically) that the effect of the wind is to provide a constant force to the east, equal in magnitude to the weight of the ball. At what angle θ should the ball be thrown so that it travels the maximum horizontal distance?

3.43. Increasing gravity *

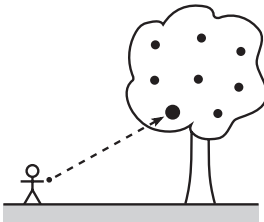
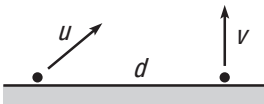
At $t = 0$ on the planet Gravitus Increasicus, a projectile is fired with speed v_0 at an angle θ above the horizontal. This planet is a strange one, in that the acceleration due to gravity increases linearly with time, starting with a value of zero when the projectile is fired. In other words, $g(t) = \beta t$, where β is a given constant. What horizontal distance does the projectile travel? What should θ be to maximize this distance?

3.44. Newton's apple *

Newton is tired of apples falling on his head, so he decides to throw a rock at one of the larger and more formidable-looking apples positioned directly above his favorite sitting spot. Forgetting all about his work on gravitation, he aims the rock directly at the apple (see Fig. 3.28). To his surprise, the apple falls from the tree just as he releases the rock. Show, by calculating the rock's height when it reaches the horizontal position of the apple, that the rock hits the apple.²³

3.45. Colliding projectiles *

Two balls are fired from ground level, a distance d apart. The right one is fired vertically with speed v (see Fig. 3.29). You wish to simultaneously

**Fig. 3.28****Fig. 3.29**

²³ This problem suggests a way in which William Tell and his son might survive their ordeal if they were plopped down, with no time to practice, on a planet with an unknown gravitational constant (provided that the son weren't too short or that g weren't too big).

fire the left one at the appropriate velocity $\mathbf{u}$ so that it collides with the right ball when they reach their highest point. What should $\mathbf{u}$ be (give the horizontal and vertical components)? Given d , what should v be so that the speed u is minimum?

3.46. Equal tilts *

A plane tilts down at an angle θ below the horizontal. On this plane, a projectile is fired with speed v at an angle θ above the horizontal, as shown in Fig. 3.30. What is the distance, d , along the plane that the projectile travels? What is d in the limit $\theta \rightarrow 90^\circ$? What θ yields the maximum *horizontal* distance?

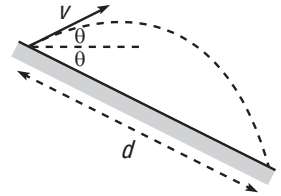


Fig. 3.30

3.47. Throwing at a wall *

You throw a ball with speed v_0 at a vertical wall, a distance ℓ away. At what angle should you throw the ball so that it hits the wall as high as possible? Assume $\ell < v_0^2/g$ (why?).

3.48. Firing a cannon **

A cannon, when aimed vertically, is observed to fire a ball to a maximum height of L . Another ball is then fired with this same speed, but with the cannon aimed up along a plane of length L , inclined at an angle θ , as shown in Fig. 3.31. What should θ be so that the ball travels the largest horizontal distance, d , by the time it returns to the height of the top of the plane?

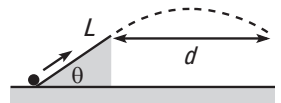


Fig. 3.31

3.49. Perpendicular and horizontal **

A plane is inclined at an angle θ below the horizontal. A person throws a ball with speed v_0 from the surface of the plane. How far down along the plane does the ball hit, if the person throws the ball (a) perpendicular to the plane? (b) horizontally?

3.50. Cart, ball, and plane **

A cart is held at rest on an inclined plane. A tube is positioned in the cart with its axis perpendicular to the plane. The cart is released, and at some later time a ball is fired from the tube. Will the ball eventually land back in the tube? *Hint:* Choose your coordinate system wisely.

3.51. Perpendicular to plane **

A hill is sloped downward at an angle β with respect to the horizontal. A projectile is fired with an initial velocity perpendicular to the hill. When it eventually lands on the hill, let its velocity make an angle θ with respect to the horizontal. What is θ ? What β yields the minimum value of θ ? What is this minimum θ ?

3.52. Increasing distance **

- (a) What is the maximum angle at which you can throw a ball so that its distance from you never decreases during its flight?
- (b) This maximum angle equals the minimum θ from Exercise 3.51. Explain why this is true. (It isn't necessary to have done that exercise.)

3.53. Projectile with drag ***

A ball is thrown with speed v_0 at an angle θ . Let the drag force from the air take the form $\mathbf{F}_d = -\beta\mathbf{v} \equiv -m\alpha\mathbf{v}$.

- (a) Find $x(t)$ and $y(t)$.
- (b) Assume that the drag coefficient takes the value that makes the magnitude of the initial drag force equal to the weight of the ball. If your goal is to have x be as large as possible when y achieves its maximum value (you don't care what this maximum value actually is), show that θ should satisfy $\sin \theta = (\sqrt{5} - 1)/2$, which just happens to be the inverse of the golden ratio.

*Section 3.5: Motion in a plane, polar coordinates***3.54. Low-orbit satellite**

What is the speed of a satellite whose orbit is just above the earth's surface? Give the numerical value.

3.55. Weight at the equator *

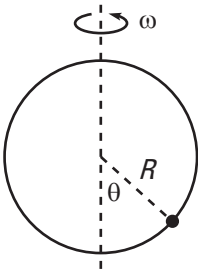
A person stands on a scale at the equator. If the earth somehow stopped spinning but kept its same shape, would the reading on the scale increase or decrease? By what fraction?

3.56. Banking an airplane *

An airplane flies at speed v in a horizontal circle of radius R . At what angle should the plane be banked so that you don't feel like you are getting flung to the side in your seat? At this angle, what is your apparent weight (that is, what is the normal force from the seat)?

3.57. Rotating hoop *

A bead lies on a frictionless hoop of radius R that rotates around a vertical diameter with constant angular frequency ω , as shown in Fig. 3.32. What should ω be so that the bead maintains the same position on the hoop, at an angle θ with respect to the vertical? There is a special value of ω ; what is it, and why is it special?

**Fig. 3.32**

3.58. **Swinging in circles** *

A large number of masses are attached by strings of various lengths to a point on the ceiling. All of the masses swing around in horizontal circles of various radii with the *same* frequency ω (one such circle is drawn in Fig. 3.33). If you take a picture (from the side) of the setup at an instant when all the masses lie in the plane of the paper (as shown for four masses), what does the “curve” formed by the masses look like?

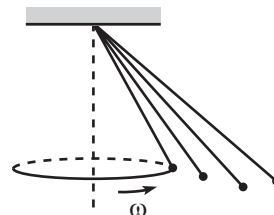


Fig. 3.33

3.59. **Swinging triangle** *

Two masses m are attached to two vertices of an equilateral triangle made of three massless rods of length ℓ . A pivot is located at the third vertex, and the triangle is free to swing back and forth in a vertical plane, as shown in Fig. 3.34. If it is initially released from rest when one of the rods is vertical (as shown), find the tensions in all three rods (and specify tension or compression), and also the accelerations of the masses, at the instant *right after* it is released.

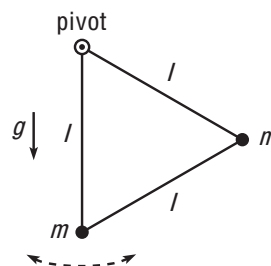


Fig. 3.34

3.60. **Circular and plane pendulums** *

Consider the circular pendulum in the example in Section 3.5. Let the x - y plane be the horizontal plane of the circle. For small β , what (approximately) is the F_x component of the force on the mass when it is at the position (x, y) on the circle?

Consider now a standard plane pendulum where the mass swings back and forth in the vertical plane containing the x axis. Let the maximum angle of the pendulum be the same small angle β .²⁴ What (approximately) is the F_x component of the force on the mass in terms of its x coordinate?

Your two results should be the same, which means that the x motions of the two systems are the same, because when they are each at their maximum x value ($\ell \sin \beta$), they have the same x speed (zero), and we just showed that they always have the same x acceleration, independent of any y motion. So the frequencies of the two systems must be equal. (We'll see in Section 4.2 that the frequency of a plane pendulum is $\sqrt{g/\ell}$, in agreement with this observation.)

3.61. **Rolling wheel** *

If you paint a dot on the rim of a rolling wheel, the coordinates of the dot may be written as²⁵

$$(x, y) = (R\theta + R \sin \theta, R + R \cos \theta). \quad (3.55)$$

²⁴ It's actually not required that this be the same angle, as long as it's small. See Problem 3.10.

²⁵ This follows from writing (x, y) as $(R\theta, R) + (R \sin \theta, R \cos \theta)$. The first term here is the position of the center of the wheel, and the second term is the position of the dot relative to the center, where θ is measured clockwise from the top.

The path of the dot is called a *cycloid*. Assume that the wheel is rolling at constant speed, which implies $\theta = \omega t$.

- (a) Find $\mathbf{v}(t)$ and $\mathbf{a}(t)$ of the dot.
- (b) At the instant the dot is at the top of the wheel, what is the radius of curvature of its path? The radius of curvature is defined to be the radius of the circle that matches up with the path locally at a given point. *Hint:* You know v and a .

3.62. Radius of curvature **

A projectile is fired at speed v_0 and angle θ . What is the radius of curvature (defined in Exercise 3.61) of the parabolic motion

- (a) at the top?
- (b) at the beginning?
- (c) At what angle should the projectile be fired so that the radius of curvature at the top equals half the maximum height, as shown in Fig. 3.35?

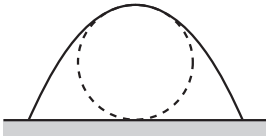


Fig. 3.35

3.63. Driving on tilted ground **

A driver encounters a large tilted parking lot, where the angle of the ground with respect to the horizontal is θ . The driver wishes to drive in a circle of radius R at constant speed. The coefficient of friction between the tires and the ground is μ .

- (a) What is the largest speed the driver can have if he wants to avoid slipping?
- (b) What is the largest speed the driver can have, assuming he is concerned only with whether or not he slips at one of the “side” points on the circle (that is, halfway between the top and bottom points; see Fig. 3.36)?

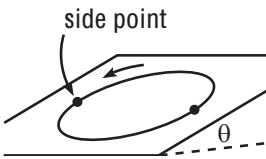


Fig. 3.36

3.64. Car on a banked track **

A car travels around a circular banked track of radius R . The angle of the bank is θ , and the coefficient of friction between the tires and the track is μ . For what range of speeds does the car not slip?

3.65. Horizontal acceleration **

A bead rests at the top of a fixed frictionless hoop of radius R that lies in a vertical plane. The bead is given a tiny push so that it slides down and around the hoop. At what points on the hoop is the bead’s acceleration horizontal? *Note:* We haven’t studied conservation of energy yet, but use the fact that the bead’s speed after it has fallen a height h is given by $v = \sqrt{2gh}$.

3.66. Maximum horizontal force **

A bead rests at the top of a fixed frictionless hoop of radius R that lies in a vertical plane. The bead is given a tiny push so that it slides down and around the hoop. Consider the horizontal component of the force from the hoop on the bead. At what points on the hoop does this component achieve a local maximum or minimum? As in Exercise 3.65, use $v = \sqrt{2gh}$.

3.67. Derivation of F_r and F_θ *

In Cartesian coordinates, a general vector takes the form,

$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} = r \cos \theta \hat{\mathbf{x}} + r \sin \theta \hat{\mathbf{y}}. \quad (3.56)$$

Derive Eq. (3.51) by taking two derivatives of this expression for $\mathbf{r}$, and then using Eq. (3.46) to show that the result can be written in the form of Eq. (3.50). Note that unlike $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$, the vectors $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ do not change with time.

3.68. A force $F_\theta = 3m\dot{r}\dot{\theta}$ **

Consider a particle that feels an angular force only, of the form $F_\theta = 3m\dot{r}\dot{\theta}$. Show that $\dot{r} = \pm\sqrt{Ar^4 + B}$, where A and B are constants of integration, determined by the initial conditions. Also, show that if the particle starts with $\dot{\theta} \neq 0$ and $\dot{r} > 0$, it reaches $r = \infty$ in a finite time. (As in Problem 3.23, there's nothing all that physical about this force. It simply makes the $F = ma$ equations solvable.)

3.69. A force $F_\theta = 2m\dot{r}\dot{\theta}$ **

Consider a particle that feels an angular force only, of the form $F_\theta = 2m\dot{r}\dot{\theta}$. Show that $r = Ae^\theta + Be^{-\theta}$, where A and B are constants of integration, determined by the initial conditions. (This force is actually a physical one. If you put a bead on a stick and swing the stick around one end at a constant rate, then the normal force from the stick happens to be $2m\dot{r}\dot{\theta}$.²⁶)

3.70. Stopping on a cone **

When viewed from the side, the cone in Fig. 3.37 subtends an angle 2θ at its tip. A block of mass m is connected to the tip by a massless string and moves in a horizontal circle of radius R around the surface. If the

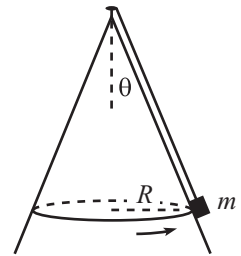


Fig. 3.37

²⁶ Depending on what is meant by “physical,” the forces in Exercise 3.68 and Problem 3.23 might also be considered to be physical. They correspond to putting a bead on a stick and swinging the stick around with angular speeds proportional to the bead's r or $1/r$, respectively (as is evident from the values of $\dot{\theta}$ in the solutions). This can also be deduced from $\tau = dL/dt$, but we won't get to torque until Chapter 8.

initial speed is v_0 , and if the coefficient of kinetic friction between the block and the cone is μ , how much time does it take the block to stop? (The answer is a little messy, but there are some limits you can check that will make you feel better about it.)

3.71. Motorcycle circle ***

A motorcyclist wishes to travel in a circle of radius R on level ground. The coefficient of friction between the tires and the ground is μ . The motorcycle starts at rest. What is the minimum distance it must travel in order to achieve its maximum allowable speed, that is, the speed above which it will skid out of the circular path?²⁷ Solve this in two ways:

- Write down the radial and tangential $F = ma$ equations (you'll want to write a as $v dv/dx$), and then demand that the magnitude of the friction force equals μmg in the optimal case. Take it from there.
- Let the friction force make an angle $\beta(t)$ with respect to the tangential direction. Write down the radial and tangential $F = ma$ equations (you'll want to write a as dv/dt), and then take the derivative of the radial equation. Take it from there (this is the slick way).

3.8 Solutions

3.1. Atwood's machine

Let T be the tension in the string, and let a be the acceleration of m_1 (with upward taken to be positive). Then $-a$ is the acceleration of m_2 . So the $F = ma$ equations are

$$T - m_1g = m_1a, \quad \text{and} \quad T - m_2g = m_2(-a). \quad (3.57)$$

Solving these two equations for a and T gives

$$a = \frac{(m_2 - m_1)g}{m_2 + m_1}, \quad \text{and} \quad T = \frac{2m_1m_2g}{m_2 + m_1}. \quad (3.58)$$

REMARKS: As a double-check, a has the correct limits when $m_2 \gg m_1$, $m_1 \gg m_2$, and $m_2 = m_1$ (namely $a \approx g$, $a \approx -g$, and $a = 0$, respectively). As far as T goes, if $m_1 = m_2 \equiv m$, then $T = mg$, as it should. And if $m_1 \ll m_2$, then $T \approx 2m_1g$. This is correct, because it makes the net upward force on m_1 equal to m_1g , which means that its acceleration is g upward, which is consistent with the fact that m_2 is essentially in free fall. ♣

3.2. Double Atwood's machine

Let the tension in the lower string be T . Then the tension in the upper string is $2T$ (by balancing the forces on the bottom pulley). The three $F = ma$ equations are therefore (with all the a 's taken to be positive upward)

$$2T - m_1g = m_1a_1, \quad T - m_2g = m_2a_2, \quad T - m_3g = m_3a_3. \quad (3.59)$$

²⁷ This problem can be traced to an old edition of the Russian magazine *Kvant*.

And conservation of string says that the acceleration of m_1 is

$$a_1 = -\left(\frac{a_2 + a_3}{2}\right). \quad (3.60)$$

This follows from the fact that the average position of m_2 and m_3 moves the same distance as the bottom pulley, which in turn moves the same distance (but in the opposite direction) as m_1 . We now have four equations in the four unknowns, a_1 , a_2 , a_3 , and T . With a little work, we can solve for the accelerations,

$$\begin{aligned} a_1 &= g \frac{4m_2m_3 - m_1(m_2 + m_3)}{4m_2m_3 + m_1(m_2 + m_3)}, \\ a_2 &= -g \frac{4m_2m_3 + m_1(m_2 - 3m_3)}{4m_2m_3 + m_1(m_2 + m_3)}, \\ a_3 &= -g \frac{4m_2m_3 + m_1(m_3 - 3m_2)}{4m_2m_3 + m_1(m_2 + m_3)}. \end{aligned} \quad (3.61)$$

REMARKS: There are many limits we can check here. A couple are: (1) If $m_2 = m_3 = m_1/2$, then all the a 's are zero, which is correct. (2) If m_3 is much less than both m_1 and m_2 , then $a_1 = -g$, $a_2 = -g$, and $a_3 = 3g$. To understand this $3g$, convince yourself that if m_1 and m_2 go down by d , then m_3 goes up by $3d$.

Note that a_1 can be written as

$$a_1 = g \left(\frac{4m_2m_3}{m_2 + m_3} - m_1 \right) / \left(\frac{4m_2m_3}{m_2 + m_3} + m_1 \right). \quad (3.62)$$

In view of the result for a in Eq. (3.58) in Problem 3.1, we see that as far as m_1 is concerned here, the m_2, m_3 pulley system acts just like a mass of $4m_2m_3/(m_2 + m_3)$. This has the expected properties of equaling zero when either m_2 or m_3 is zero, and equaling $2m$ if $m_2 = m_3 \equiv m$. ♣

3.3. Infinite Atwood's machine

FIRST SOLUTION: If the strength of gravity on the earth were multiplied by a factor η , then the tension in all of the strings in the Atwood's machine would likewise be multiplied by η . This is true because the only way to produce a quantity with the units of tension (that is, force) is to multiply a mass by g . Conversely, if we put the Atwood's machine on another planet and discover that all of the tensions are multiplied by η , then we know that the gravity there must be ηg .

Let the tension in the string above the first pulley be T . Then the tension in the string above the second pulley is $T/2$ (because the pulley is massless). Let the downward acceleration of the second pulley be a_2 . Then the second pulley effectively lives in a world where gravity has strength $g - a_2$. Consider the subsystem of all the pulleys except the top one. This infinite subsystem is identical to the original infinite system of all the pulleys. Therefore, by the arguments in the above paragraph, we must have

$$\frac{T}{g} = \frac{T/2}{g - a_2}, \quad (3.63)$$

which gives $a_2 = g/2$. But a_2 is also the acceleration of the top mass, so our answer is $g/2$.

REMARKS: You can show that the relative acceleration of the second and third pulleys is $g/4$, and that of the third and fourth is $g/8$, etc. The acceleration of a mass far down in the system therefore equals $g(1/2 + 1/4 + 1/8 + \dots) = g$, which makes intuitive sense.

Note that $T = 0$ also makes Eq. (3.63) true. But this corresponds to putting a mass of zero at the end of a finite pulley system (see the following solution). ♣

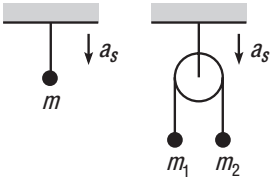


Fig. 3.38

SECOND SOLUTION: Consider the following auxiliary problem.

Problem: Two setups are shown in Fig. 3.38. The first contains a hanging mass m . The second contains two masses, m_1 and m_2 , hanging over a pulley. Let both supports have acceleration a_s downward. What should m be, in terms of m_1 and m_2 , so that the tension T in the top string is the same in both cases?

Answer: In the first case, we have

$$mg - T = ma_s. \quad (3.64)$$

In the second case, let a be the acceleration of m_2 relative to the support (with downward taken to be positive). Then we have

$$m_1g - \frac{T}{2} = m_1(a_s - a), \quad \text{and} \quad m_2g - \frac{T}{2} = m_2(a_s + a). \quad (3.65)$$

If we define $g' \equiv g - a_s$, then we may write the above three equations as

$$mg' = T, \quad m_1g' - \frac{T}{2} = -m_1a, \quad m_2g' - \frac{T}{2} = m_2a. \quad (3.66)$$

Eliminating a from the last two of these equations gives $T = 4m_1m_2g'/(m_1 + m_2)$. Using this value of T in the first equation then gives

$$m = \frac{4m_1m_2}{m_1 + m_2}. \quad (3.67)$$

Note that the value of a_s is irrelevant. We effectively have a fixed support in a world where the acceleration due to gravity is g' (see Eq. (3.66)), and the desired m can't depend on g' , by dimensional analysis. This auxiliary problem shows that for any a_s the two-mass system in the second case can equivalently be treated like a mass m , given by Eq. (3.67), as far as the upper string is concerned. ■

Now let's look at our infinite Atwood's machine. Assume that the system has N pulleys, where $N \rightarrow \infty$. Let the bottom mass be x . Then the auxiliary problem shows that the bottom two masses, m and x , can be treated like an effective mass $f(x)$, where

$$f(x) = \frac{4mx}{m+x} = \frac{4x}{1+(x/m)}. \quad (3.68)$$

We can then treat the combination of the mass $f(x)$ and the next m as an effective mass $f(f(x))$. These iterations can be repeated, until we finally have a mass m and a mass $f^{(N-1)}(x)$ hanging over the top pulley. So we must determine the behavior of $f^N(x)$, as $N \rightarrow \infty$. This behavior is clear if we look at the plot of $f(x)$ in Fig. 3.39. We see that $x = 3m$ is a fixed point of $f(x)$. That is, $f(3m) = 3m$. This plot shows that no matter what x we start with, the iterations approach $3m$ (unless we start at $x = 0$, in which case we remain there). These iterations are shown graphically by the directed lines in the plot. After reaching the value $f(x)$ on the curve, the line moves horizontally to the x value of $f(x)$, and then vertically to the value $f(f(x))$ on the curve, and so on. Therefore, since $f^N(x) \rightarrow 3m$ as $N \rightarrow \infty$, our infinite Atwood's machine is equivalent to (as far as the top mass is concerned) just two masses, m and $3m$. You can then quickly show that the acceleration of the top mass is $g/2$. Note that as far as the support is concerned, the whole apparatus is equivalent to a mass $3m$. So $3mg$ is the upward force exerted by the support.

3.4. Line of pulleys

Let m be the common mass, and let T be the tension in the string. Let a be the acceleration of the end masses, and let a' be the acceleration of the other N masses, with upward taken to be positive. These N accelerations are indeed all equal, because the same net force acts on all of the internal N masses, namely $2T$ upwards and mg downwards. The $F = ma$ equations for the end and internal masses are, respectively,

$$T - mg = ma, \quad \text{and} \quad 2T - mg = ma'. \quad (3.69)$$

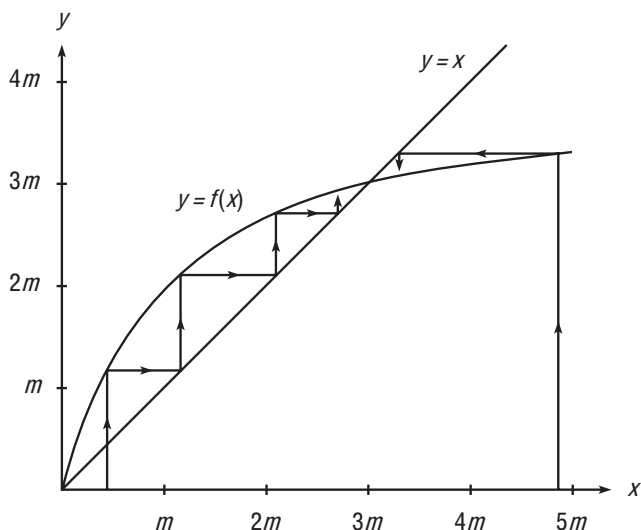


Fig. 3.39 Problem 3.3, second solution

But the string has fixed length. Therefore,

$$N(2a') + a + a = 0. \quad (3.70)$$

The “2” here comes from the fact that if one of the inside masses goes up by a distance d , then a length $2d$ of string has disappeared and must therefore appear somewhere else (namely, in the outer two segments). Eliminating T from Eq. (3.69) gives $a' = 2a + g$. Combining this with Eq. (3.70) then gives

$$a = -\frac{Ng}{2N+1}, \quad \text{and} \quad a' = \frac{g}{2N+1}. \quad (3.71)$$

REMARKS: For $N = 1$, we have $a = -g/3$ and $a' = g/3$. For larger N , a increases in magnitude and approaches $-g/2$ as $N \rightarrow \infty$, and a' decreases in magnitude and approaches zero as $N \rightarrow \infty$. The signs of a and a' in Eq. (3.71) may be surprising. You might think that if, say, $N = 100$, then these 100 masses will “win” out over the two end masses, so that the N masses will fall. But this is not correct, because there are many ($2N$, in fact) tensions acting up on the N masses. They do *not* behave like a mass Nm hanging below one pulley. In fact, two masses of $m/2$ on the ends will balance any number N of masses m in the interior (with the help of the upward forces from the top row of pulleys). ♣

3.5. Ring of pulleys

Let T be the tension in the string. Then $F = ma$ for m_i gives

$$2T - m_i g = m_i a_i, \quad (3.72)$$

with upward taken to be positive. The a_i ’s are related by the fact that the string has fixed length, which implies that the sum of the displacements of all the masses is zero. In other words,

$$a_1 + a_2 + \cdots + a_N = 0. \quad (3.73)$$

If we divide Eq. (3.72) by m_i , and then add the N such equations together and use Eq. (3.73), we find that T is given by

$$2T \left(\frac{1}{m_1} + \frac{1}{m_2} + \cdots + \frac{1}{m_N} \right) - Ng = 0. \quad (3.74)$$

Therefore,

$$T = \frac{NMg}{2}, \quad \text{where} \quad \frac{1}{M} \equiv \frac{1}{m_1} + \frac{1}{m_2} + \cdots + \frac{1}{m_N} \quad (3.75)$$

is the so-called *reduced mass* of the system. Substituting this value for T into (3.72) gives

$$a_i = g \left(\frac{NM}{m_i} - 1 \right). \quad (3.76)$$

REMARK: A few special cases are: If all the masses are equal, then all the $a_i = 0$. If $m_k = 0$ (and all the others are not zero), then $a_k = (N-1)g$, and all the other $a_i = -g$. If $N-1$ of the masses are equal and much smaller than the remaining one, m_k , then $m_k \approx -g$, and all the other $a_i \approx g/(N-1)$. ♣

3.6. Sliding down a plane

- The component of gravity along the plane is $g \sin \theta$. The acceleration in the horizontal direction is therefore $a_x = (g \sin \theta) \cos \theta$. Our goal is to maximize a_x . By taking the derivative, or by noting that $\sin \theta \cos \theta = (\sin 2\theta)/2$, we obtain $\theta = \pi/4$. The maximum a_x is then $g/2$.
- The normal force from the plane is $mg \cos \theta$, so the kinetic friction force is $\mu mg \cos \theta$. The acceleration along the plane is therefore $g(\sin \theta - \mu \cos \theta)$, and so the acceleration in the horizontal direction is $a_x = g(\sin \theta - \mu \cos \theta) \cos \theta$. We want to maximize this. Setting the derivative equal to zero gives

$$\begin{aligned} (\cos^2 \theta - \sin^2 \theta) + 2\mu \sin \theta \cos \theta &= 0 \implies \cos 2\theta + \mu \sin 2\theta = 0 \\ \implies \tan 2\theta &= -\frac{1}{\mu}. \end{aligned} \quad (3.77)$$

For $\mu \rightarrow 0$, this gives the $\pi/4$ result in part (a). For $\mu \rightarrow \infty$, we obtain $\theta \approx \pi/2$, which makes sense.

REMARKS: The time to travel a horizontal distance d is obtained from $a_x t^2/2 = d$. In part (a), this gives a minimum time of $2\sqrt{d/g}$. In part (b), you can show that the maximum a_x is $(g/2)(\sqrt{1+\mu^2} - \mu)$, which then leads to a minimum time of $2\sqrt{d/g}(\sqrt{1+\mu^2} + \mu)^{1/2}$. This has the correct $\mu \rightarrow 0$ limit, and it behaves like $2\sqrt{2\mu d/g}$ for $\mu \rightarrow \infty$. ♣

3.7. Sliding sideways on a plane

The normal force from the plane is $N = mg \cos \theta$. Therefore, the friction force on the block is $\mu N = (\tan \theta)(mg \cos \theta) = mg \sin \theta$. This force acts in the direction opposite to the motion. The block also feels the gravitational force of $mg \sin \theta$ pointing down the plane.

Because the magnitudes of the friction force and the gravitational force along the plane are equal, the acceleration along the direction of motion equals the negative of the acceleration in the direction down the plane. Therefore, in a small increment of time, the speed that the block loses along its direction of motion exactly equals the speed that it gains in the direction down the plane. Letting v be the total speed of the

block, and letting v_y be the component of the velocity in the direction down the plane, we therefore have

$$v + v_y = C, \quad (3.78)$$

where C is a constant. C is given by its initial value, which is $V + 0 = V$. The final value of C is $V_f + V_f = 2V_f$ (where V_f is the final speed of the block), because the block is essentially moving straight down the plane after a very long time. Therefore,

$$2V_f = V \implies V_f = \frac{V}{2}. \quad (3.79)$$

3.8. Moving plane

Let N be the normal force between the block and the plane. Note that we *cannot* assume that $N = mg \cos \theta$, because the plane recoils. We can see that $N = mg \cos \theta$ is in fact incorrect, because in the limiting case where $M = 0$, we have no normal force at all.

The various $F = ma$ equations (vertical and horizontal for the block, and horizontal for the plane) are

$$\begin{aligned} mg - N \cos \theta &= ma_y, \\ N \sin \theta &= ma_x, \\ N \sin \theta &= MA_x, \end{aligned} \quad (3.80)$$

where we have chosen the positive directions for a_y , a_x , and A_x to be downward, rightward, and leftward, respectively. There are four unknowns here: a_x , a_y , A_x , and N , so we need one more equation. This fourth equation is the constraint that the block remains in contact with the plane. The horizontal distance between the block and its starting point on the plane is $(a_x + A_x)t^2/2$, and the vertical distance is $a_y t^2/2$. The ratio of these distances must equal $\tan \theta$ if the block is to remain on the plane (imagine looking at things in the frame of the plane). Therefore, we must have

$$\frac{a_y}{a_x + A_x} = \tan \theta. \quad (3.81)$$

Using Eq. (3.80) to solve for a_y , a_x , and A_x in terms of N , and then plugging the results into Eq. (3.81), gives

$$\frac{g - \frac{N}{m} \cos \theta}{\frac{N}{m} \sin \theta + \frac{N}{M} \sin \theta} = \tan \theta \implies N = g \left(\sin \theta \tan \theta \left(\frac{1}{m} + \frac{1}{M} \right) + \frac{\cos \theta}{m} \right)^{-1}. \quad (3.82)$$

(In the limit $M \rightarrow \infty$, this reduces to $N = mg \cos \theta$, as it should.) Having found N , the third of Eqs. (3.80) gives A_x , which may be written as

$$A_x = \frac{N \sin \theta}{M} = \frac{mg \sin \theta \cos \theta}{M + m \sin^2 \theta}. \quad (3.83)$$

REMARKS:

1. For given M and m , you can show that the angle θ_0 that maximizes A_x is $\tan \theta_0 = \sqrt{M/(M+m)}$. If $M \ll m$, then $\theta_0 \approx 0$; this makes sense, because the plane gets squeezed out very fast. If $M \gg m$, then $\theta_0 \approx \pi/4$; this is consistent with the $\pi/4$ result from Problem 3.6(a).
2. In the limit $M \ll m$, Eq. (3.83) gives $A_x \approx g/\tan \theta$. This makes sense, because m falls essentially straight down with acceleration g , and the plane gets squeezed out to the left.
3. In the limit $M \gg m$, Eq. (3.83) gives $A_x \approx g(m/M) \sin \theta \cos \theta$. The correctness of this is more transparent if we instead look at $a_x = (M/m)A_x \approx g \sin \theta \cos \theta$. Since the plane is essentially at rest in this limit, this value of a_x implies that the acceleration of m along the plane equals $a_x/\cos \theta \approx g \sin \theta$, as expected. ♣

3.9. Exponential force

$F = ma$ gives $\ddot{x} = a_0 e^{-bt}$. Integrating this with respect to time gives $v(t) = -a_0 e^{-bt}/b + A$. Integrating again gives $x(t) = a_0 e^{-bt}/b^2 + At + B$. The initial condition $v(0) = 0$ gives $-a_0/b + A = 0 \implies A = a_0/b$. And the initial condition $x(0) = 0$ gives $a_0/b^2 + B = 0 \implies B = -a_0/b^2$. Therefore,

$$x(t) = a_0 \left(\frac{e^{-bt}}{b^2} + \frac{t}{b} - \frac{1}{b^2} \right). \quad (3.84)$$

For $t \rightarrow \infty$ (more precisely, for $bt \rightarrow \infty$), v approaches a_0/b , and x approaches $a_0(t/b - 1/b^2)$. We see that the particle eventually lags a distance a_0/b^2 behind another particle that starts at the same position but moves with the constant speed $v = a_0/b$. For $t \approx 0$ (more precisely, for $bt \approx 0$), we can expand e^{-bt} in its Taylor series to obtain $x(t) \approx a_0 t^2/2$. This makes sense, because the exponential factor in the force is essentially equal to 1, so we essentially have a constant force with constant acceleration.

3.10. $-kx$ force

This is simply a Hooke's-law spring force, which we'll see much more of in Chapter 4. $F = ma$ gives $-kx = m dv/dx$. Separating variables and integrating yields

$$-\int_{x_0}^x kx \, dx = \int_0^v mv \, dv \implies \frac{1}{2}kx_0^2 - \frac{1}{2}kx^2 = \frac{1}{2}mv^2. \quad (3.85)$$

Solving for $v \equiv dx/dt$ and then separating variables and integrating again gives

$$\int_{x_0}^x \frac{dx}{\sqrt{x_0^2 - x^2}} = \pm \int_0^t \sqrt{\frac{k}{m}} \, dt. \quad (3.86)$$

You can look up this integral, or you can solve it with a trig substitution. Letting $x = x_0 \cos \theta$ gives $dx = -x_0 \sin \theta \, d\theta$, and so we have

$$\int_0^\theta \frac{-x_0 \sin \theta \, d\theta}{x_0 \sin \theta} = \pm \sqrt{\frac{k}{m}} t \implies \theta = \mp \sqrt{\frac{k}{m}} t. \quad (3.87)$$

From the definition of θ , the solution for $x(t)$ is therefore

$$x(t) = x_0 \cos \left(\sqrt{\frac{k}{m}} t \right). \quad (3.88)$$

We see that the particle oscillates back and forth sinusoidally. It completes a full oscillation when the argument of the cosine increases by 2π . So the period of the motion is $T = 2\pi\sqrt{m/k}$, which interestingly is independent of x_0 . It increases with m and decreases with k , as expected.

3.11. Falling chain

Let the density of the chain be ρ , and let $y(t)$ be the length hanging down through the hole at time t . Then the total mass is $\rho\ell$, and the mass hanging below the hole is ρy . The net downward force on the chain is $(\rho y)g$, so $F = ma$ gives

$$\rho g y = (\rho\ell)\ddot{y} \implies \ddot{y} = \frac{g}{\ell} y. \quad (3.89)$$

At this point, there are two ways we can proceed:

FIRST METHOD: Since we have a function whose second derivative is proportional to itself, a good bet for the solution is an exponential function. And indeed, a quick check shows that the solution is

$$y(t) = Ae^{\alpha t} + Be^{-\alpha t}, \quad \text{where } \alpha \equiv \sqrt{\frac{g}{\ell}}. \quad (3.90)$$

Taking the derivative of this to obtain $\dot{y}(t)$, and using the given information that $\dot{y}(0) = 0$, we find $A = B$. Using $y(0) = y_0$, we then find $A = B = y_0/2$. So the length that hangs below the hole is

$$y(t) = \frac{y_0}{2} (e^{\alpha t} + e^{-\alpha t}) \equiv y_0 \cosh(\alpha t). \quad (3.91)$$

And the speed is

$$\dot{y}(t) = \frac{\alpha y_0}{2} (e^{\alpha t} - e^{-\alpha t}) \equiv \alpha y_0 \sinh(\alpha t). \quad (3.92)$$

The time T that satisfies $y(T) = \ell$ is given by $\ell = y_0 \cosh(\alpha T)$. Using $\sinh x = \sqrt{\cosh^2 x - 1}$, we find that the speed of the chain right when it loses contact with the table is

$$\dot{y}(T) = \alpha y_0 \sinh(\alpha T) = \alpha \sqrt{\ell^2 - y_0^2} \equiv \sqrt{g\ell} \sqrt{1 - \eta_0^2}, \quad (3.93)$$

where $\eta_0 \equiv y_0/\ell$ is the initial fraction hanging below the hole. If $\eta_0 \approx 0$, then the speed at time T is $\sqrt{g\ell}$ (this quickly follows from conservation of energy, which is the subject of Chapter 5). Also, you can show that Eq. (3.91) implies that T goes to infinity logarithmically as $\eta_0 \rightarrow 0$.

SECOND METHOD: Write $\ddot{y}$ as $v dv/dy$ in Eq. (3.89), and then separate variables and integrate to obtain

$$\int_0^v v dv = \alpha^2 \int_{y_0}^y y dy \implies v^2 = \alpha^2 (y^2 - y_0^2), \quad (3.94)$$

where $\alpha \equiv \sqrt{g/\ell}$. Now write v as dy/dt and separate variables again to obtain

$$\int_{y_0}^y \frac{dy}{\sqrt{y^2 - y_0^2}} = \alpha \int_0^t dt. \quad (3.95)$$

The integral on the left-hand side is $\cosh^{-1}(y/y_0)$, so we arrive at $y(t) = y_0 \cosh(\alpha t)$, in agreement with Eq. (3.91). The solution then proceeds as above. However, an easier way to obtain the final speed with this method is to simply use the result for v in Eq. (3.94). This tells us that the speed of the chain when it leaves the table (that is, when $y = \ell$) is $v = \alpha \sqrt{\ell^2 - y_0^2}$, in agreement with Eq. (3.93).

3.12. Throwing a beach ball

On both the way up and the way down, the total force on the ball is

$$F = -mg - m\alpha v. \quad (3.96)$$

On the way up, v is positive, so the drag force points downward, as it should. And on the way down, v is negative, so the drag force points upward. Our strategy for finding v_f will be to produce two different expressions for the maximum height h , and then equate them. We'll find these two expressions by considering the upward and then the downward motion of the ball. In doing so, we will need to write the acceleration of the ball as $a = v dv/dy$. For the upward motion, $F = ma$ gives

$$-mg - m\alpha v = m v \frac{dv}{dy} \implies \int_0^h dy = - \int_{v_0}^0 \frac{v dv}{g + \alpha v}. \quad (3.97)$$

where we have taken advantage of the fact that the speed of the ball at the top is zero. Writing $v/(g + \alpha v)$ as $[1 - g/(g + \alpha v)]/\alpha$, the integral yields

$$h = \frac{v_0}{\alpha} - \frac{g}{\alpha^2} \ln \left(1 + \frac{\alpha v_0}{g} \right). \quad (3.98)$$

Now consider the downward motion. Let v_f be the final speed, which is a positive quantity. The final velocity is then the negative quantity, $-v_f$. Using $F = ma$, we obtain

$$\int_h^0 dy = - \int_0^{-v_f} \frac{v dv}{g + \alpha v}. \quad (3.99)$$

Performing the integration (or just replacing the v_0 in Eq. (3.98) with $-v_f$) gives

$$h = -\frac{v_f}{\alpha} - \frac{g}{\alpha^2} \ln \left(1 - \frac{\alpha v_f}{g} \right). \quad (3.100)$$

Equating the expressions for h in Eqs. (3.98) and (3.100) gives an implicit equation for v_f in terms of v_0 ,

$$v_0 + v_f = \frac{g}{\alpha} \ln \left(\frac{g + \alpha v_0}{g - \alpha v_f} \right). \quad (3.101)$$

REMARKS: In the limit of small α (more precisely, in the limit $\alpha v_0/g \ll 1$), we can use $\ln(1+x) = x - x^2/2 + \dots$ to obtain approximate values for h in Eqs. (3.98) and (3.100). The results are, as expected,

$$h \approx \frac{v_0^2}{2g}, \quad \text{and} \quad h \approx \frac{v_f^2}{2g}. \quad (3.102)$$

We can also make approximations for large α (or large $\alpha v_0/g$). In this limit, the log term in Eq. (3.98) is negligible, so we obtain $h \approx v_0/\alpha$. And Eq. (3.100) gives $v_f \approx g/\alpha$, because the argument of the log must be very small in order to give a very large negative number, which is needed to produce a positive h on the left-hand side. There is no way to relate v_f and h in this limit, because the ball quickly reaches the terminal velocity of $-g/\alpha$ (which is the velocity that makes the net force equal to zero), independent of h . ♣

Let's now find the times it takes for the ball to go up and to go down. We'll present two methods for doing this.

FIRST METHOD: Let T_1 be the time for the upward path. If we write the acceleration of the ball as $a = dv/dt$, then $F = ma$ gives $-mg - m\alpha v = m dv/dt$. Separating variables and integrating yields

$$\int_0^{T_1} dt = - \int_{v_0}^0 \frac{dv}{g + \alpha v} \implies T_1 = \frac{1}{\alpha} \ln \left(1 + \frac{\alpha v_0}{g} \right). \quad (3.103)$$

In a similar manner, we find that the time T_2 for the downward path is

$$T_2 = -\frac{1}{\alpha} \ln \left(1 - \frac{\alpha v_f}{g} \right). \quad (3.104)$$

Therefore,

$$T_1 + T_2 = \frac{1}{\alpha} \ln \left(\frac{g + \alpha v_0}{g - \alpha v_f} \right) = \frac{v_0 + v_f}{g}, \quad (3.105)$$

where we have used Eq. (3.101). This result is shorter than the time in vacuum (namely $2v_0/g$) because $v_f < v_0$.

SECOND METHOD: The very simple form of Eq. (3.105) suggests that there is a cleaner way of deriving it. And indeed, if we integrate $m dv/dt = -mg - m\alpha v$ with respect to time on the way up, we obtain $-v_0 = -gT_1 - \alpha h$ (because $\int v dt = h$). Likewise, if we integrate $m dv/dt = -mg - m\alpha v$ with respect to time on the way down, we obtain $-v_f = -gT_2 + \alpha h$ (because $\int v dt = -h$). Adding these two results gives Eq. (3.105). Note that this procedure works only because the drag force is proportional to v .

REMARK: The fact that the time here is shorter than the time in vacuum isn't obvious. On one hand, the ball doesn't travel as high in air as it would in vacuum, so you might think that $T_1 + T_2 < 2v_0/g$. But on the other hand, the ball moves slower in air on the way down, so you might think that $T_1 + T_2 > 2v_0/g$. It isn't obvious which effect wins, without doing a calculation.²⁸ For any α , you can use Eq. (3.103) to show that $T_1 < v_0/g$. But T_2 is harder to get a handle on, because it is given in terms of v_f . But in the limit of large α , the ball quickly reaches terminal velocity, so we have $T_2 \approx h/v_f$. Using the results from the previous remark, this becomes $T_2 \approx (v_0/\alpha)/(g/\alpha) = v_0/g$. Interestingly, this equals the downward (and upward) time for a ball thrown in vacuum. ♣

3.13. Balancing a pencil

- (a) The component of gravity in the tangential direction is $mg \sin \theta \approx mg\theta$. Therefore, the tangential $F = ma$ equation is $mg\theta = m\ell\ddot{\theta}$, which may be written as $\ddot{\theta} = (g/\ell)\theta$. The general solution to this equation is²⁹

$$\theta(t) = Ae^{t/\tau} + Be^{-t/\tau}, \quad \text{where } \tau \equiv \sqrt{\ell/g}. \quad (3.106)$$

The constants A and B are found from the initial conditions,

$$\begin{aligned} \theta(0) = \theta_0 &\implies A + B = \theta_0, \\ \dot{\theta}(0) = \omega_0 &\implies (A - B)/\tau = \omega_0. \end{aligned} \quad (3.107)$$

Solving for A and B , and then plugging the results into Eq. (3.106) gives

$$\theta(t) = \frac{1}{2}(\theta_0 + \omega_0\tau)e^{t/\tau} + \frac{1}{2}(\theta_0 - \omega_0\tau)e^{-t/\tau}. \quad (3.108)$$

- (b) The constants A and B will turn out to be small (they will each be of order $\sqrt{\hbar}$). Therefore, by the time the positive exponential has increased enough to make θ of order 1, the negative exponential will have become negligible. We will therefore ignore this term from here on. In other words,

$$\theta(t) \approx \frac{1}{2}(\theta_0 + \omega_0\tau)e^{t/\tau}. \quad (3.109)$$

The goal is to keep θ small for as long as possible. Hence, we want to minimize the coefficient of the exponential, subject to the uncertainty-principle constraint, $(\ell\theta_0)(m\ell\omega_0) \geq \hbar$. This constraint gives $\omega_0 \geq \hbar/(m\ell^2\theta_0)$. Therefore,

$$\theta(t) \geq \frac{1}{2} \left(\theta_0 + \frac{\hbar\tau}{m\ell^2\theta_0} \right) e^{t/\tau}. \quad (3.110)$$

Taking the derivative with respect to θ_0 to minimize the coefficient, we find that the minimum value occurs at $\theta_0 = \sqrt{\hbar\tau}/m\ell^2$. Substituting this back into Eq. (3.110) gives

$$\theta(t) \geq \sqrt{\frac{\hbar\tau}{m\ell^2}} e^{t/\tau}. \quad (3.111)$$

Setting $\theta \approx 1$, and then solving for t gives (using $\tau \equiv \sqrt{\ell/g}$)

$$t \leq \frac{1}{4} \sqrt{\frac{\ell}{g}} \ln \left(\frac{m^2 \ell^3 g}{\hbar^2} \right). \quad (3.112)$$

²⁸ For a similar setup where it again isn't obvious (for good reason) which effect wins, see Exercise 3.36.

²⁹ If you want, you can derive this by separating variables and integrating. The solution is essentially the same as in the second method presented in the solution to Problem 3.11.

With the given values, $m = 0.01$ kg and $\ell = 0.1$ m, along with $g = 10$ m/s² and $\hbar = 1.06 \cdot 10^{-34}$ J s, we obtain

$$t \leq \frac{1}{4} (0.1 \text{ s}) \ln(9 \cdot 10^{61}) \approx 3.5 \text{ s.} \quad (3.113)$$

No matter how clever you are, and no matter how much money you spend on the newest cutting-edge pencil balancing equipment, you can never get a pencil to balance for more than about four seconds.

REMARKS:

1. The smallness of this answer is quite amazing. It is remarkable that a quantum effect on a macroscopic object can produce an everyday value for a time scale. Basically, the point is that the fast exponential growth of θ (which gives rise to the log in the final result for t) wins out over the smallness of $\hbar$, and produces a result for t of order 1. When push comes to shove, exponential effects always win.
2. The above value for t depends strongly on ℓ and g , through the $\sqrt{\ell/g}$ term. But the dependence on m , ℓ , and g in the log term is very weak, because $\hbar$ is so small. If m is increased by a factor of 1000, for example, the result for t increases by only about 10%. This implies that any factors of order 1 that we neglected throughout this problem are completely irrelevant. They will appear in the argument of the log term, and will therefore have negligible effect.
3. Note that dimensional analysis, which is generally a very powerful tool, won't get you too far in this problem. The quantity $\sqrt{\ell/g}$ has dimensions of time, and the quantity $\eta \equiv m^2 \ell^3 g / \hbar^2$ is dimensionless (it is the only such quantity), so the balancing time must take the form,

$$t \approx \sqrt{\frac{\ell}{g}} f(\eta), \quad (3.114)$$

where f is some function. If the leading term in f were a power (even, for example, a square root), then t would essentially be infinite ($t \approx 10^{30}$ s $\approx 10^{22}$ years for the square root). But f in fact turns out to be a log (which you can't know without solving the problem), which completely cancels out the smallness of $\hbar$, reducing an essentially infinite time down to a few seconds. ♣

3.14. Maximum trajectory area

Let θ be the angle at which the ball is thrown. Then the coordinates are given by $x = (v \cos \theta)t$ and $y = (v \sin \theta)t - gt^2/2$. The total time in the air is $2(v \sin \theta)/g$, so the area under the trajectory, $A = \int y dx$, is

$$\int_0^{x_{\max}} y dx = \int_0^{2v \sin \theta / g} \left((v \sin \theta)t - \frac{gt^2}{2} \right) (v \cos \theta dt) = \frac{2v^4}{3g^2} \sin^3 \theta \cos \theta. \quad (3.115)$$

Taking the derivative of this, we find that the maximum occurs when $\tan \theta = \sqrt{3}$, that is, when $\theta = 60^\circ$. The maximum area is then $A_{\max} = \sqrt{3}v^4/8g^2$. Note that by dimensional analysis we know that the area, which has dimensions of distance squared, must be proportional to v^4/g^2 .

3.15. Bouncing ball

The ball travels $2h$ during the first up-and-down journey. It travels $2hf$ during the second, then $2hf^2$ during the third, and so on. Therefore, the total distance traveled is

$$D = 2h(1 + f + f^2 + f^3 + \cdots) = \frac{2h}{1-f}. \quad (3.116)$$

The time it takes to fall down during the first up-and-down is obtained from $h = gt^2/2$. Therefore, the time for the first up-and-down equals $2t = 2\sqrt{2h/g}$. Likewise, the time for the second up-and-down equals $2\sqrt{2(hf)/g}$. Each successive up-and-down time decreases by a factor of $\sqrt{f}$, so the total time is

$$T = 2\sqrt{\frac{2h}{g}}(1 + f^{1/2} + f^1 + f^{3/2} + \dots) = 2\sqrt{\frac{2h}{g}} \cdot \frac{1}{1 - \sqrt{f}}. \quad (3.117)$$

The average speed is therefore

$$\frac{D}{T} = \frac{\sqrt{gh/2}}{1 + \sqrt{f}}. \quad (3.118)$$

REMARK: The average speed for $f \approx 1$ is roughly half of the average speed for $f \approx 0$. This may seem counterintuitive, because in the $f \approx 0$ case the ball slows down far more quickly than in the $f \approx 1$ case. But the $f \approx 0$ case consists of essentially only one bounce, and the average speed for that one bounce is the largest of any bounce. Both D and T are smaller for $f \approx 0$ than for $f \approx 1$, but T is smaller by a larger factor. ♣

3.16. Perpendicular velocities

In the maximum-distance case, let v_i be the initial speed of the ball, and let v_f be the final speed right before it hits the plane (so $v_f = \sqrt{v_i^2 - 2gh}$, where h is the final height of the ball). Let the parabolic path be labeled P , and let the beginning and ending points be A and B , as shown in Fig. 3.40.

Consider the question, “Given an initial speed v_f , at what inclination angle should a ball be thrown *down* the plane from point B so that it travels the maximum distance?” The answer is that it should be thrown along the same path P , tracing out the path backward. This is certainly a possible physical trajectory (reversing time still leads to a solution to $F = ma$ for projectile motion), and we claim that it does indeed yield the maximum distance. This can be seen in the following way.

Assume (in search of a contradiction) that with initial speed v_f , the maximum distance down the plane is obtained via some other path P' that lands farther down the plane, as shown in Fig. 3.40. Then if we decrease the initial speed v_f by an appropriate amount, we can have the ball land at point A via some path P'' (not shown, lest the figure get too cluttered). From the conservation-of-energy result, the final speed at A in this case is less than v_i . But if we simply reverse the motion along path P'' , we see that we can get from A to B by using an initial speed that is less than v_i . So if we then increase the speed up to v_i , we can hit a point above B on the plane via some other path P''' , contradicting the fact that B was the endpoint of the maximum-distance trajectory starting at A with initial speed v_i . This contradiction shows that the maximum distance down the plane, starting at B with speed v_f , must in fact be attained via the path P .

Now, from the example in Section 3.4, we know that in the maximum-distance case, the throwing angle bisects the angle between the ground and the vertical. We therefore have the situation shown in Fig. 3.41, because the same path gives the maximum distance for both the upward and downward throws. Since $2\alpha + 2\gamma = 180^\circ$, we see that $\alpha + \gamma = 90^\circ$, as we wanted to show.

3.17. Throwing a ball from a cliff

Let the inclination angle be θ . Then the horizontal speed is $v_x = v \cos \theta$, and the initial vertical speed is $v_y = v \sin \theta$. The time it takes for the ball to hit the ground is given by $h + (v \sin \theta)t - gt^2/2 = 0$. Therefore,

$$t = \frac{v}{g} \left(\sin \theta + \sqrt{\sin^2 \theta + \beta} \right), \quad \text{where } \beta \equiv \frac{2gh}{v^2}. \quad (3.119)$$

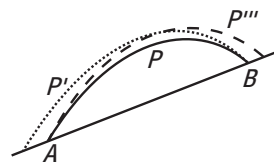


Fig. 3.40

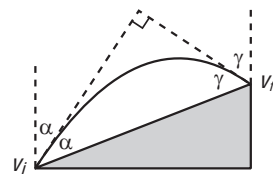


Fig. 3.41

(The “−” solution for t from the quadratic formula corresponds to the ball being thrown backward down through the cliff.) The horizontal distance traveled is $d = (v \cos \theta)t$, which gives

$$d = \frac{v^2}{g} \cos \theta \left(\sin \theta + \sqrt{\sin^2 \theta + \beta} \right). \quad (3.120)$$

We want to maximize this function of θ . Taking the derivative, multiplying through by $\sqrt{\sin^2 \theta + \beta}$, and setting the result equal to zero, gives

$$(\cos^2 \theta - \sin^2 \theta) \sqrt{\sin^2 \theta + \beta} = \sin \theta (\beta - (\cos^2 \theta - \sin^2 \theta)). \quad (3.121)$$

Using $\cos^2 \theta = 1 - \sin^2 \theta$, and then squaring and simplifying this equation, gives an optimal angle of

$$\sin \theta_{\max} = \frac{1}{\sqrt{2 + \beta}} \equiv \frac{1}{\sqrt{2 + 2gh/v^2}}. \quad (3.122)$$

Plugging this into Eq. (3.120), and simplifying, gives a maximum distance of

$$d_{\max} = \frac{v^2}{g} \sqrt{1 + \beta} \equiv \frac{v^2}{g} \sqrt{1 + \frac{2gh}{v^2}}. \quad (3.123)$$

If $h = 0$, then $\theta_{\max} = \pi/4$ and $d_{\max} = v^2/g$, in agreement with the example in Section 3.4. If $h \rightarrow \infty$ or $v \rightarrow 0$, then $\theta_{\max} \approx 0$, which makes sense.

REMARK: If we make use of conservation of energy (discussed in Chapter 5), it turns out that the final speed of the ball when it hits the ground is $v_f = \sqrt{v^2 + 2gh}$. The maximum distance in Eq. (3.123) can therefore be written as (with $v_i \equiv v$ being the initial speed)

$$d_{\max} = \frac{v_i v_f}{g}. \quad (3.124)$$

Note that this is symmetric in v_i and v_f , as it must be, because we could imagine the trajectory running backward. Also, it equals zero if v_i is zero, and it reduces to v^2/g on level ground, as it should. We can also write the angle θ in Eq. (3.122) in terms of v_f (instead of h). You can show that the result is $\tan \theta = v_i/v_f$. This implies that the initial and final velocities are perpendicular to each other, because running the trajectory backward interchanges v_i and v_f , which means that the product of the tangents of the two angles equals 1. This is basically the same result as in Problem 3.16. ♣

3.18. Redirected motion

FIRST SOLUTION: We will use the results of Problem 3.17, namely Eqs. (3.123) and (3.122), which say that an object projected from height y at speed v travels a maximum horizontal distance of

$$d_{\max} = \frac{v^2}{g} \sqrt{1 + \frac{2gy}{v^2}}, \quad (3.125)$$

and the optimal angle yielding this distance is

$$\sin \theta = \frac{1}{\sqrt{2 + 2gy/v^2}}. \quad (3.126)$$

In the problem at hand, the object is dropped from height h , so the kinematic relation $v_f^2 = v_i^2 + 2ad$ gives the speed at height y as $v = \sqrt{2g(h-y)}$. Plugging this into Eq. (3.125) tells us that the maximum horizontal distance, as a function of y , is

$$d_{\max}(y) = 2\sqrt{h(h-y)}. \quad (3.127)$$

This is maximum when $y = 0$ (assuming we can't have negative y), in which case the distance is $d_{\max} = 2h$. Equation (3.126) gives the associated optimal angle as $\theta = 45^\circ$.

SECOND SOLUTION: Assume that the greatest distance d_0 is obtained when the surface is at $y = y_0$. We will show that y_0 must be 0. We will do this by assuming $y_0 \neq 0$ and explicitly constructing a situation that yields a greater distance.

Let P be the point where the ball finally hits the ground after it bounces off the surface at height y_0 . Consider a second scenario where a ball is dropped from height h directly above P . The speed of this ball at P will be the same as the speed of the original ball at P . This follows from conservation of energy. (Or, since we haven't covered energy yet, you can use the kinematic relation $v_f^2 = v_i^2 + 2ad$ for the y speeds of the two balls; apply it in two steps for the first ball.)

Now imagine putting a surface at P at the appropriate angle so that the second ball bounces off in the direction from which the first ball came. Then the second ball will travel backward along the parabolic trajectory of the first one. But this means that after the second ball gets to the location of the platform at y_0 (which we have now removed), it will cover more horizontal distance before it hits the ground. We have therefore constructed a setup in which the ball travels farther horizontally than in our proposed maximal case. So the optimal setup must have $y_0 = 0$, in which case the example in Section 3.4 says that the optimal angle is $\theta = 45^\circ$. If we want the ball to go even farther, we can simply dig a (wide enough) hole in the ground and have the ball bounce from the bottom of the hole.

3.19. Maximum trajectory length

Let θ be the angle at which the ball is thrown. Then the coordinates are given by $x = (v \cos \theta)t$ and $y = (v \sin \theta)t - gt^2/2$. The ball reaches its maximum height at $t = v \sin \theta/g$, so the length of the trajectory is

$$\begin{aligned} L &= 2 \int_0^{v \sin \theta/g} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= 2 \int_0^{v \sin \theta/g} \sqrt{(v \cos \theta)^2 + (v \sin \theta - gt)^2} dt \\ &= 2v \cos \theta \int_0^{v \sin \theta/g} \sqrt{1 + \left(\tan \theta - \frac{gt}{v \cos \theta}\right)^2} dt. \end{aligned} \quad (3.128)$$

Letting $z \equiv \tan \theta - gt/v \cos \theta$, we obtain

$$L = -\frac{2v^2 \cos^2 \theta}{g} \int_{\tan \theta}^0 \sqrt{1 + z^2} dz. \quad (3.129)$$

We can look up this integral, or we can derive it by making a $z \equiv \sinh \alpha$ substitution. The result is

$$\begin{aligned} L &= \frac{2v^2 \cos^2 \theta}{g} \cdot \frac{1}{2} \left(z\sqrt{1+z^2} + \ln(z + \sqrt{1+z^2}) \right) \Big|_0^{\tan \theta} \\ &= \frac{v^2}{g} \left(\sin \theta + \cos^2 \theta \ln \left(\frac{\sin \theta + 1}{\cos \theta} \right) \right). \end{aligned} \quad (3.130)$$

As an intermediate check, you can verify that $L = 0$ when $\theta = 0$, and $L = v^2/g$ when $\theta = 90^\circ$. Taking the derivative of Eq. (3.130) to find the maximum, we obtain

$$\begin{aligned} 0 &= \cos \theta - 2 \cos \theta \sin \theta \ln \left(\frac{1 + \sin \theta}{\cos \theta} \right) \\ &\quad + \cos^2 \theta \left(\frac{\cos \theta}{1 + \sin \theta} \right) \frac{\cos^2 \theta + (1 + \sin \theta) \sin \theta}{\cos^2 \theta}. \end{aligned} \quad (3.131)$$

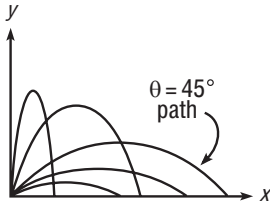


Fig. 3.42

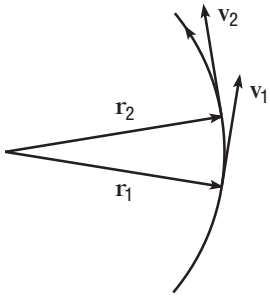


Fig. 3.43

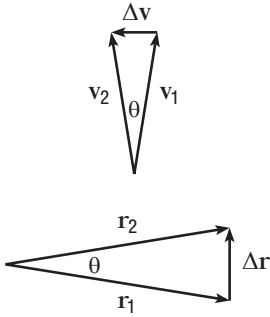


Fig. 3.44

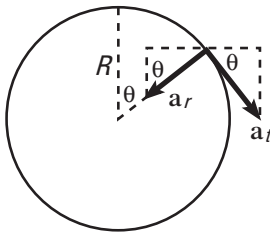


Fig. 3.45

This reduces to

$$1 = \sin \theta \ln \left(\frac{1 + \sin \theta}{\cos \theta} \right). \quad (3.132)$$

You can show numerically that the solution for θ is $\theta_0 \approx 56.5^\circ$.

REMARK: A few possible trajectories are shown in Fig. 3.42. Using the standard result that $\theta = 45^\circ$ provides the maximum *horizontal* distance, it follows from the figure that the θ_0 that yields the maximum trajectory length must satisfy $\theta_0 \geq 45^\circ$. The exact angle, however, requires the above detailed calculation. ♣

3.20. Centripetal acceleration

The position and velocity vectors at two nearby times are shown in Fig. 3.43. Their differences, $\Delta \mathbf{r} \equiv \mathbf{r}_2 - \mathbf{r}_1$ and $\Delta \mathbf{v} \equiv \mathbf{v}_2 - \mathbf{v}_1$, are shown in Fig. 3.44. The angle between the $\mathbf{v}$'s is the same as the angle between the $\mathbf{r}$'s, because each $\mathbf{v}$ makes a right angle with the corresponding $\mathbf{r}$. Therefore, the triangles in Fig. 3.44 are similar, so we have

$$\frac{|\Delta \mathbf{v}|}{v} = \frac{|\Delta \mathbf{r}|}{r}, \quad (3.133)$$

where $r \equiv |\mathbf{r}|$ and $v \equiv |\mathbf{v}|$. Dividing through by Δt gives

$$\frac{1}{v} \left| \frac{\Delta \mathbf{v}}{\Delta t} \right| = \frac{1}{r} \left| \frac{\Delta \mathbf{r}}{\Delta t} \right| \implies \frac{|\mathbf{a}|}{v} = \frac{|\mathbf{v}|}{r} \implies a = \frac{v^2}{r}. \quad (3.134)$$

We have assumed that Δt is infinitesimal here, which allows us to get rid of the Δ 's in favor of instantaneous quantities.

3.21. Vertical acceleration

Let θ be the angle down from the top of the hoop. The tangential acceleration is $a_t = g \sin \theta$, and the radial acceleration is $a_r = v^2/R = 2gh/R$. But the height fallen is $h = R - R \cos \theta$, so we have

$$a_r = \frac{2gR(1 - \cos \theta)}{R} = 2g(1 - \cos \theta). \quad (3.135)$$

We want the total acceleration to be vertical, which means that we want the horizontal components of $\mathbf{a}_t$ and $\mathbf{a}_r$ in Fig. 3.45 to cancel. That is, $a_t \cos \theta = a_r \sin \theta$. This gives

$$(g \sin \theta) \cos \theta = 2g(1 - \cos \theta) \sin \theta \implies \sin \theta = 0 \text{ or } \cos \theta = 2/3. \quad (3.136)$$

The $\sin \theta = 0$ root corresponds to the top and bottom of the hoop ($\theta = 0$ and $\theta = \pi$). So we want the $\cos \theta = 2/3 \implies \theta \approx \pm 48.2^\circ$ root. The vertical acceleration is the sum of the vertical components of $\mathbf{a}_t$ and $\mathbf{a}_r$, so

$$\begin{aligned} a_y &= a_t \sin \theta + a_r \cos \theta = (g \sin \theta) \sin \theta + 2g(1 - \cos \theta) \cos \theta \\ &= g(\sin^2 \theta + 2 \cos \theta - 2 \cos^2 \theta). \end{aligned} \quad (3.137)$$

Using $\cos \theta = 2/3$, and hence $\sin \theta = \sqrt{5}/3$, we have

$$a_y = g \left(\frac{5}{9} + 2 \cdot \frac{2}{3} - 2 \cdot \frac{4}{9} \right) = g. \quad (3.138)$$

REMARK: The reason for this nice answer is the following. If there is no horizontal acceleration, then the normal force from the hoop must have no horizontal component. In other words, $N \sin \theta = 0$. Therefore, either $\sin \theta = 0$ (which gives the top and bottom solutions of $\theta = 0$ and $\theta = \pi$), or $N = 0$, which means that there is no normal force, so the bead feels only gravity, so it's in freefall with $a_y = g$.

If we want, we can use this $N = 0$ requirement as the starting point for a second solution. Using the a_r from above, the radial $F = ma$ equation is $mg \cos \theta - N = 2mg(1 - \cos \theta)$, with positive N defined to point outward. Setting $N = 0$ gives the desired result, $\cos \theta = 2/3$. ♣

3.22. Circling around a pole

Let F be the tension in the string. At the mass, the angle between the string and the radius of the dotted circle is $\theta = \sin^{-1}(r/R)$. In terms of θ , the radial and tangential $F = ma$ equations are

$$F \cos \theta = \frac{mv^2}{R}, \quad \text{and} \quad F \sin \theta = m\dot{v}. \quad (3.139)$$

Dividing these two equations gives $\tan \theta = (R\dot{v})/v^2$. Separating variables and integrating gives

$$\begin{aligned} \int_{v_0}^v \frac{dv}{v^2} &= \frac{\tan \theta}{R} \int_0^t dt \implies \frac{1}{v_0} - \frac{1}{v} = \frac{(\tan \theta)t}{R} \\ \implies v(t) &= \left(\frac{1}{v_0} - \frac{(\tan \theta)t}{R} \right)^{-1}. \end{aligned} \quad (3.140)$$

The speed v becomes infinite when

$$t = T \equiv \frac{R}{v_0 \tan \theta}. \quad (3.141)$$

This means that you can keep the mass moving in the desired circle only up to time T . After that, it is impossible. (Of course, it will become impossible, for all practical purposes, long before v becomes infinite.) The total distance, $d = \int v dt$, is infinite, because this integral diverges (barely, like a log) as t approaches T .

3.23. A force $F_\theta = m\dot{r}\dot{\theta}$

With the given force, Eq. (3.51) becomes

$$0 = m(\ddot{r} - r\dot{\theta}^2), \quad \text{and} \quad m\dot{r}\dot{\theta} = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}). \quad (3.142)$$

The second of these equations gives $-\dot{r}\dot{\theta} = r\ddot{\theta}$. Therefore,

$$\int \frac{\ddot{\theta}}{\dot{\theta}} dt = - \int \frac{\dot{r}}{r} dt \implies \ln \dot{\theta} = -\ln r + C \implies \dot{\theta} = \frac{D}{r}, \quad (3.143)$$

where $D = e^C$ is a constant of integration, determined by the initial conditions. Substituting this value of $\dot{\theta}$ into the first of Eqs. (3.142), and then multiplying through by $\dot{r}$ and integrating, gives

$$\dot{r} = r \left(\frac{D}{r} \right)^2 \implies \int \ddot{r} \dot{r} dt = D^2 \int \frac{\dot{r}}{r} dt \implies \frac{\dot{r}^2}{2} = D^2 \ln r + E. \quad (3.144)$$

Therefore,

$$\dot{r} = \sqrt{A \ln r + B}, \quad (3.145)$$

where $A \equiv 2D^2$ and $B \equiv 2E$.

3.24. Free particle

For zero force, Eq. (3.51) gives

$$\ddot{r} = r\dot{\theta}^2, \quad \text{and} \quad r\ddot{\theta} = -2\dot{r}\dot{\theta}. \quad (3.146)$$

Separating variables in the second equation and integrating yields

$$\int \frac{\ddot{\theta}}{\dot{\theta}} dt = - \int \frac{2\dot{r}}{r} dt \implies \ln \dot{\theta} = -2 \ln r + C \implies \dot{\theta} = \frac{D}{r^2}, \quad (3.147)$$

where $D = e^C$ is a constant of integration, determined by the initial conditions.³⁰ Substituting this value of $\dot{\theta}$ into the first of Eqs. (3.146), and then multiplying through by $\dot{r}$ and integrating, gives

$$\ddot{r} = r \left(\frac{D}{r^2} \right)^2 \implies \int \ddot{r} dt = D^2 \int \frac{\dot{r}}{r^3} dt \implies \frac{\dot{r}^2}{2} = -\frac{D^2}{2r^2} + E. \quad (3.148)$$

We want $\dot{r} = 0$ when $r = r_0$, which implies that $E = D^2/2r_0^2$. Therefore,

$$\dot{r} = V \sqrt{1 - \frac{r_0^2}{r^2}}, \quad (3.149)$$

where $V \equiv D/r_0$. Separating variables and integrating gives

$$\int \frac{r \dot{r} dt}{\sqrt{r^2 - r_0^2}} = \int V dt \implies \sqrt{r^2 - r_0^2} = Vt \implies r = \sqrt{r_0^2 + (Vt)^2}, \quad (3.150)$$

where the constant of integration is zero, because we have chosen $t = 0$ to correspond to $r = r_0$. Plugging this value for r into the $\dot{\theta} = D/r^2 \equiv Vr_0/r^2$ result in Eq. (3.147) gives

$$\int d\theta = \int \frac{Vr_0 dt}{r_0^2 + (Vt)^2} \implies \theta = \tan^{-1} \left(\frac{Vt}{r_0} \right) \implies \cos \theta = \frac{r_0}{\sqrt{r_0^2 + (Vt)^2}}. \quad (3.151)$$

Finally, combining this with the result for r in Eq. (3.150) gives $\cos \theta = r_0/r$, as desired.

³⁰ The statement that $r^2\dot{\theta}$ is constant is simply the statement of conservation of angular momentum, because $r^2\dot{\theta} = r(r\dot{\theta}) = rv_\theta$, where v_θ is the tangential velocity. More on this in Chapters 7 and 8.

1. Newton's Laws of Motion

“So few went to hear him, and fewer understood him, that oftimes he did, for want of hearers, read to the walls. He usually stayed about half an hour; when he had no auditors he commonly returned in a quarter of that time.”

Appraisal of a Cambridge lecturer in classical mechanics, circa 1690

1.1 Introduction

The fundamental principles of classical mechanics were laid down by Galileo and Newton in the 16th and 17th centuries. In 1686, Newton wrote the *Principia* where he gave us three laws of motion, one law of gravity and pretended he didn't know calculus. Probably the single greatest scientific achievement in history, you might think this pretty much wraps it up for classical mechanics. And, in a sense, it does. Given a collection of particles, acted upon by a collection of forces, you have to draw a nice diagram, with the particles as points and the forces as arrows. The forces are then added up and Newton's famous “ $F = ma$ ” is employed to figure out where the particle's velocities are heading next. All you need is enough patience and a big enough computer and you're done.

From a modern perspective this is a little unsatisfactory on several levels: it's messy and inelegant; it's hard to deal with problems that involve extended objects rather than point particles; it obscures certain features of dynamics so that concepts such as chaos theory took over 200 years to discover; and it's not at all clear what the relationship is between Newton's classical laws and quantum physics.

The purpose of this course is to resolve these issues by presenting new perspectives on Newton's ideas. We shall describe the advances that took place during the 150 years after Newton when the laws of motion were reformulated using more powerful techniques and ideas developed by some of the giants of mathematical physics: people such as Euler, Lagrange, Hamilton and Jacobi. This will give us an immediate practical advantage, allowing us to solve certain complicated problems with relative ease (the strange motion of spinning tops is a good example). But, perhaps more importantly, it will provide an elegant viewpoint from which we'll see the profound basic principles which underlie Newton's familiar laws of motion. We shall prise open “ $F = ma$ ” to reveal the structures and symmetries that lie beneath.

Moreover, the formalisms that we'll develop here are the basis for *all* of fundamental modern physics. Every theory of Nature, from electromagnetism and general relativity, to the standard model of particle physics and more speculative pursuits such as string theory, is best described in the language we shall develop in this course. The new formalisms that we'll see here also provide the bridge between the classical world and the quantum world.

There are phenomena in Nature for which these formalisms are not particularly useful. Systems which are dissipative, for example, are not so well suited to these new techniques. But if you want to understand the dynamics of planets and stars and galaxies as they orbit and spin, or you want to understand what's happening at the LHC where protons are collided at unprecedented energies, or you want to know how electrons meld together in solids to form new states of matter, then the foundations that we'll lay in in this course are a must.

1.2 Newtonian Mechanics: A Single Particle

In the rest of this section, we'll take a flying tour through the basic ideas of classical mechanics handed down to us by Newton. We'll start with a single particle.

A *particle* is defined to be an object of insignificant size. e.g. an electron, a tennis ball or a planet. Obviously the validity of this statement depends on the context: to first approximation, the earth can be treated as a particle when computing its orbit around the sun. But if you want to understand its spin, it must be treated as an extended object.

The motion of a particle of mass m at the position $\mathbf{r}$ is governed by *Newton's Second Law* $\mathbf{F} = m\mathbf{a}$ or, more precisely,

$$\mathbf{F}(\mathbf{r}, \dot{\mathbf{r}}) = \dot{\mathbf{p}} \tag{1.1}$$

where $\mathbf{F}$ is the force which, in general, can depend on both the position $\mathbf{r}$ as well as the velocity $\dot{\mathbf{r}}$ (for example, friction forces depend on $\dot{\mathbf{r}}$) and $\mathbf{p} = m\dot{\mathbf{r}}$ is the momentum. Both $\mathbf{F}$ and $\mathbf{p}$ are 3-vectors which we denote by the bold font. Equation (1.1) reduces to $\mathbf{F} = m\mathbf{a}$ if $\dot{m} = 0$. But if $m = m(t)$ (e.g. in rocket science) then the form with $\dot{\mathbf{p}}$ is correct.

General theorems governing differential equations guarantee that if we are given $\mathbf{r}$ and $\dot{\mathbf{r}}$ at an initial time $t = t_0$, we can integrate equation (1.1) to determine $\mathbf{r}(t)$ for all t (as long as $\mathbf{F}$ remains finite). This is the goal of classical dynamics.

Equation (1.1) is not quite correct as stated: we must add the caveat that it holds only in an *inertial frame*. This is defined to be a frame in which a free particle with $\dot{m} = 0$ travels in a straight line,

$$\mathbf{r} = \mathbf{r}_0 + \mathbf{v}t \quad (1.2)$$

Newton's first law is the statement that such frames exist.

An inertial frame is not unique. In fact, there are an infinite number of inertial frames. Let S be an inertial frame. Then there are 10 linearly independent transformations $S \rightarrow S'$ such that S' is also an inertial frame (i.e. if (1.2) holds in S , then it also holds in S'). These are

- 3 Rotations: $\mathbf{r}' = O\mathbf{r}$ where O is a 3×3 orthogonal matrix.
- 3 Translations: $\mathbf{r}' = \mathbf{r} + \mathbf{c}$ for a constant vector $\mathbf{c}$.
- 3 Boosts: $\mathbf{r}' = \mathbf{r} + \mathbf{u}t$ for a constant velocity $\mathbf{u}$.
- 1 Time Translation: $t' = t + c$ for a constant real number c

If motion is uniform in S , it will also be uniform in S' . These transformations make up the *Galilean Group* under which Newton's laws are invariant. They will be important in section 2.4 where we will see that these symmetries of space and time are the underlying reason for conservation laws. As a parenthetical remark, recall from special relativity that Einstein's laws of motion are invariant under Lorentz transformations which, together with translations, make up the Poincaré group. We can recover the Galilean group from the Poincaré group by taking the speed of light to infinity.

1.2.1 Angular Momentum

We define the *angular momentum* $\mathbf{L}$ of a particle and the *torque* $\boldsymbol{\tau}$ acting upon it as

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \quad , \quad \boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} \quad (1.3)$$

Note that, unlike linear momentum $\mathbf{p}$, both $\mathbf{L}$ and $\boldsymbol{\tau}$ depend on where we take the origin: we measure angular momentum with respect to a particular point. Let us cross both sides of equation (1.1) with $\mathbf{r}$. Using the fact that $\dot{\mathbf{r}}$ is parallel to $\mathbf{p}$, we can write $\frac{d}{dt}(\mathbf{r} \times \mathbf{p}) = \mathbf{r} \times \dot{\mathbf{p}}$. Then we get a version of Newton's second law that holds for angular momentum:

$$\boldsymbol{\tau} = \dot{\mathbf{L}} \quad (1.4)$$

1.2.2 Conservation Laws

From (1.1) and (1.4), two important conservation laws follow immediately.

- If $\mathbf{F} = 0$ then $\mathbf{p}$ is constant throughout the motion
- If $\boldsymbol{\tau} = 0$ then $\mathbf{L}$ is constant throughout the motion

Notice that $\boldsymbol{\tau} = 0$ does not require $\mathbf{F} = 0$, but only $\mathbf{r} \times \mathbf{F} = 0$. This means that $\mathbf{F}$ must be parallel to $\mathbf{r}$. This is the definition of a *central force*. An example is given by the gravitational force between the earth and the sun: the earth's angular momentum about the sun is constant. As written above in terms of forces and torques, these conservation laws appear trivial. In section 2.4, we'll see how they arise as a property of the symmetry of space as encoded in the Galilean group.

1.2.3 Energy

Let's now recall the definitions of energy. We firstly define the *kinetic energy* T as

$$T = \frac{1}{2}m \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} \quad (1.5)$$

Suppose from now on that the mass is constant. We can compute the change of kinetic energy with time: $\frac{dT}{dt} = \dot{\mathbf{p}} \cdot \dot{\mathbf{r}} = \mathbf{F} \cdot \dot{\mathbf{r}}$. If the particle travels from position $\mathbf{r}_1$ at time t_1 to position $\mathbf{r}_2$ at time t_2 then this change in kinetic energy is given by

$$T(t_2) - T(t_1) = \int_{t_1}^{t_2} \frac{dT}{dt} dt = \int_{t_1}^{t_2} \mathbf{F} \cdot \dot{\mathbf{r}} dt = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} \quad (1.6)$$

where the final expression involving the integral of the force over the path is called the *work done* by the force. So we see that the work done is equal to the change in kinetic energy. From now on we will mostly focus on a very special type of force known as a *conservative* force. Such a force depends only on position $\mathbf{r}$ rather than velocity $\dot{\mathbf{r}}$ and is such that the work done is independent of the path taken. In particular, for a closed path, the work done vanishes.

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0 \quad \Leftrightarrow \quad \nabla \times \mathbf{F} = 0 \quad (1.7)$$

It is a deep property of flat space $\mathbf{R}^3$ that this property implies we may write the force as

$$\mathbf{F} = -\nabla V(\mathbf{r}) \quad (1.8)$$

for some *potential* $V(\mathbf{r})$. Systems which admit a potential of this form include gravitational, electrostatic and interatomic forces. When we have a conservative force, we

necessarily have a conservation law for energy. To see this, return to equation (1.6) which now reads

$$T(t_2) - T(t_1) = - \int_{\mathbf{r}_1}^{\mathbf{r}_2} \nabla V \cdot d\mathbf{r} = -V(t_2) + V(t_1) \quad (1.9)$$

or, rearranging things,

$$T(t_1) + V(t_1) = T(t_2) + V(t_2) \equiv E \quad (1.10)$$

So $E = T + V$ is also a constant of motion. It is the energy. When the energy is considered to be a function of position $\mathbf{r}$ and momentum $\mathbf{p}$ it is referred to as the *Hamiltonian* H . In section 4 we will be seeing much more of the Hamiltonian.

1.2.4 Examples

- Example 1: The Simple Harmonic Oscillator

This is a one-dimensional system with a force proportional to the distance x to the origin: $F(x) = -kx$. This force arises from a potential $V = \frac{1}{2}kx^2$. Since $F \neq 0$, momentum is not conserved (the object oscillates backwards and forwards) and, since the system lives in only one dimension, angular momentum is not defined. But energy $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ is conserved.

- Example 2: The Damped Simple Harmonic Oscillator

We now include a friction term so that $F(x, \dot{x}) = -kx - \gamma\dot{x}$. Since F is not conservative, energy is not conserved. This system loses energy until it comes to rest.

- Example 3: Particle Moving Under Gravity

Consider a particle of mass m moving in 3 dimensions under the gravitational pull of a much larger particle of mass M . The force is $\mathbf{F} = -(GMm/r^2)\hat{\mathbf{r}}$ which arises from the potential $V = -GMm/r$. Again, the linear momentum $\mathbf{p}$ of the smaller particle is not conserved, but the force is both central and conservative, ensuring the particle's total energy E and the angular momentum $\mathbf{L}$ are conserved.

1.3 Newtonian Mechanics: Many Particles

It's easy to generalise the above discussion to many particles: we simply add an index to everything in sight! Let particle i have mass m_i and position $\mathbf{r}_i$ where $i = 1, \dots, N$ is the number of particles. Newton's law now reads

$$\mathbf{F}_i = \dot{\mathbf{p}}_i \quad (1.11)$$

where $\mathbf{F}_i$ is the force on the i^{th} particle. The subtlety is that forces can now be working between particles. In general, we can decompose the force in the following way:

$$\mathbf{F}_i = \sum_{j \neq i} \mathbf{F}_{ij} + \mathbf{F}_i^{\text{ext}} \quad (1.12)$$

where $\mathbf{F}_{ij}$ is the force acting on the i^{th} particle due to the j^{th} particle, while $\mathbf{F}_i^{\text{ext}}$ is the external force on the i^{th} particle. We now sum over all N particles

$$\begin{aligned} \sum_i \mathbf{F}_i &= \sum_{i,j \text{ with } j \neq i} \mathbf{F}_{ij} + \sum_i \mathbf{F}_i^{\text{ext}} \\ &= \sum_{i < j} (\mathbf{F}_{ij} + \mathbf{F}_{ji}) + \sum_i \mathbf{F}_i^{\text{ext}} \end{aligned} \quad (1.13)$$

where, in the second line, we've re-written the sum to be over all pairs $i < j$. At this stage we make use of *Newton's third law of motion*: every action has an equal and opposite reaction. Or, in other words, $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$. We see that the first term vanishes and we are left simply with

$$\sum_i \mathbf{F}_i = \mathbf{F}^{\text{ext}} \quad (1.14)$$

where we've defined the total external force to be $\mathbf{F}^{\text{ext}} = \sum_i \mathbf{F}_i^{\text{ext}}$. We now define the total mass of the system $M = \sum_i m_i$ as well as the *centre of mass* $\mathbf{R}$

$$\mathbf{R} = \frac{\sum_i m_i \mathbf{r}_i}{M} \quad (1.15)$$

Then using (1.11), and summing over all particles, we arrive at the simple formula,

$$\mathbf{F}^{\text{ext}} = M \ddot{\mathbf{R}} \quad (1.16)$$

which is identical to that of a single particle. This is an important formula. It tells that the centre of mass of a system of particles acts just as if all the mass were concentrated there. In other words, it doesn't matter if you throw a tennis ball or a very lively cat: the center of mass of each traces the same path.

1.3.1 Momentum Revisited

The *total momentum* is defined to be $\mathbf{P} = \sum_i \mathbf{p}_i$ and, from the formulae above, it is simple to derive $\dot{\mathbf{P}} = \mathbf{F}^{\text{ext}}$. So we find the conservation law of total linear momentum for a system of many particles: $\mathbf{P}$ is constant if $\mathbf{F}^{\text{ext}}$ vanishes.

Similarly, we define *total angular momentum* to be $\mathbf{L} = \sum_i \mathbf{L}_i$. Now let's see what happens when we compute the time derivative.

$$\begin{aligned}\dot{\mathbf{L}} &= \sum_i \mathbf{r}_i \times \dot{\mathbf{p}}_i \\ &= \sum_i \mathbf{r}_i \times \left(\sum_{j \neq i} \mathbf{F}_{ij} + \mathbf{F}_i^{\text{ext}} \right)\end{aligned}\tag{1.17}$$

$$= \sum_{i,j \text{ with } i \neq j} \mathbf{r}_i \times \mathbf{F}_{ji} + \sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{ext}}\tag{1.18}$$

The last term in this expression is the definition of *total external torque*: $\boldsymbol{\tau}^{\text{ext}} = \sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{ext}}$. But what are we going to do with the first term on the right hand side? Ideally we would like it to vanish! Let's look at the circumstances under which this will happen. We can again rewrite it as a sum over pairs $i < j$ to get

$$\sum_{i < j} (\mathbf{r}_i - \mathbf{r}_j) \times \mathbf{F}_{ij}\tag{1.19}$$

which will vanish if and only if the force $\mathbf{F}_{ij}$ is parallel to the line joining to two particles $(\mathbf{r}_i - \mathbf{r}_j)$. This is the strong form of Newton's third law. If this is true, then we have a statement about the conservation of total angular momentum, namely $\mathbf{L}$ is constant if $\boldsymbol{\tau}^{\text{ext}} = 0$.

Most forces do indeed obey both forms of Newton's third law: $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$ and $\mathbf{F}_{ij}$ is parallel to $(\mathbf{r}_i - \mathbf{r}_j)$. For example, gravitational and electrostatic forces have this property. And the total momentum and angular momentum are both conserved in these systems. But some forces don't have these properties! The most famous example is the Lorentz force on two moving particles with electric charge Q . This is given by,

$$\mathbf{F}_{ij} = Q\mathbf{v}_i \times \mathbf{B}_j\tag{1.20}$$

where $\mathbf{v}_i$ is the velocity of the i^{th} particle and $\mathbf{B}_j$ is the magnetic field generated by the j^{th} particle. Consider two particles crossing each other in a "T" as shown in the diagram. The force on particle 1 from particle 2 vanishes. Meanwhile, the force on particle 2 from particle 1 is non-zero, and in the direction

$$\mathbf{F}_{21} \sim \uparrow \times \otimes \sim \leftarrow\tag{1.21}$$

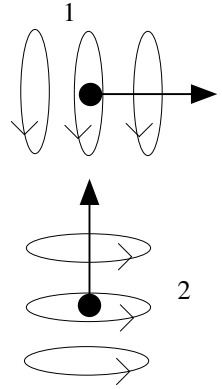


Figure 1: The magnetic field for two particles.

Does this mean that conservation of total linear and angular momentum is violated? Thankfully, no! We need to realise that the electromagnetic field itself carries angular momentum which restores the conservation law. Once we realise this, it becomes a rather cheap counterexample to Newton's third law, little different from an underwater swimmer who can appear to violate Newton's third law if we don't take into account the momentum of the water.

1.3.2 Energy Revisited

The total kinetic energy of a system of many particles is $T = \frac{1}{2} \sum_i m_i \dot{\mathbf{r}}_i^2$. Let us decompose the position vector $\mathbf{r}_i$ as

$$\mathbf{r}_i = \mathbf{R} + \tilde{\mathbf{r}}_i \quad (1.22)$$

where $\tilde{\mathbf{r}}_i$ is the distance from the centre of mass to the particle i . Then we can write the total kinetic energy as

$$T = \frac{1}{2} M \dot{\mathbf{R}}^2 + \frac{1}{2} \sum_i m_i \dot{\tilde{\mathbf{r}}}_i^2 \quad (1.23)$$

Which shows us that the kinetic energy splits up into the kinetic energy of the centre of mass, together with an *internal energy* describing how the system is moving around its centre of mass. As for a single particle, we may calculate the change in the total kinetic energy,

$$T(t_2) - T(t_1) = \sum_i \int \mathbf{F}_i^{\text{ext}} \cdot d\mathbf{r}_i + \sum_{i \neq j} \int \mathbf{F}_{ij} \cdot d\mathbf{r}_i \quad (1.24)$$

Like before, we need to consider conservative forces to get energy conservation. But now we need both

- Conservative external forces: $\mathbf{F}_i^{\text{ext}} = -\nabla_i V_i(\mathbf{r}_1, \dots, \mathbf{r}_N)$
- Conservative internal forces: $\mathbf{F}_{ij} = -\nabla_i V_{ij}(\mathbf{r}_1, \dots, \mathbf{r}_N)$

where $\nabla_i \equiv \partial/\partial \mathbf{r}_i$. To get Newton's third law $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$ together with the requirement that this is parallel to $(\mathbf{r}_i - \mathbf{r}_j)$, we should take the internal potentials to satisfy $V_{ij} = V_{ji}$ with

$$V_{ij}(\mathbf{r}_1, \dots, \mathbf{r}_N) = V_{ij}(|\mathbf{r}_i - \mathbf{r}_j|) \quad (1.25)$$

so that V_{ij} depends only on the distance between the i^{th} and j^{th} particles. We also insist on a restriction for the external forces, $V_i(\mathbf{r}_1, \dots, \mathbf{r}_N) = V_i(\mathbf{r}_i)$, so that the force on particle i does not depend on the positions of the other particles. Then, following the steps we took in the single particle case, we can define the *total potential energy* $V = \sum_i V_i + \sum_{i < j} V_{ij}$ and we can show that $H = T + V$ is conserved.

1.3.3 An Example

Let us return to the case of gravitational attraction between two bodies but, unlike in Section 1.2.4, now including both particles. We have $T = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2$. The potential is $V = -Gm_1m_2/|\mathbf{r}_1 - \mathbf{r}_2|$. This system has total linear momentum and total angular momentum conserved, as well as the total energy $H = T + V$.